

Supplementary material

False-Alarm Calibration Transport Across Engines in Full-Flight Aero-Engine Anomaly Detection: A Multi-Subset Audit of Context Conditioning and Calibration-Fleet Coverage

Jinghang Mei

Part A. Multi-subset extension

Contents. Part A reports the frozen multi-subset extension (Sections 3–4 of the article): cohort eligibility, engine roles, locks and numerical checks, every detector run’s phase FPRs and transport metrics, per-engine errors, bootstrap intervals, target sensitivity, calibration-fleet composition, alarm persistence, matched-burden delay, abnormal-state alarm rates, the reference reproduction gates, the post hoc checks of Section 4.9 and the final focused validation of Section 4.8. Part B reproduces, unchanged, the supplement of the pre-extension study: the frozen DS02 discovery and one-shot DS03 confirmation, the post-confirmation analyses and the adversarial validation, which Section 4.1 of the article summarizes.

Conventions. FPRs are healthy ($hs = 1$) row-level false-positive rates; pp = percentage points. Runs: PCA (one run), Isolation Forest (IF), past-only LSTM and CVAE (seeds s0–s2). Arms: P pooled; C retrospective phase-conditioned (uses complete-flight phases); Q continuous-context quantile regression (adversarial control). Families: F1 = DS01, F2 = DS04, F3 = DS05–DS07, F4 = DS08a, F5 = DS08c; DS02 and DS03 are references and decide no hypothesis. Machine-readable values: results/extension/ and paper/mssp_extended/evidence/ (with SHA-256 sums); every number in the article is traced in docs/extension/EXTENSION_EVIDENCE_LEDGER.csv.

Amendments (docs/extension/AMENDMENTS.md; each preceded every affected outcome). (1) Two classification rules (original-story class and H2 verdict) were completed so that every outcome configuration has a class; no threshold changed. (2) After a stop rule fired on a non-finite CVAE training loss (DS01), the encoder log-variance was bounded and gradients were clipped at norm 1.0 for every subset and seed; the DS01 lock was rerun from scratch. (3) Healthy and post-onset audit rows are scored in separate calls, as in the frozen stages, after the first DS03 reference attempt failed a reproduction gate at float32 rounding level; the reference work was rerun and passed before any new-subset audit.

A.1 Cohort, roles, locks and training

Table S1: Candidate subsets, metadata-only eligibility audit (C1–C9 of the subset selection lock; y = pass, n = fail). DS02 and DS03 fail C9 by construction (opened by the frozen study) and are references. DS08d fails C1 (the official file is 32 bytes shorter than its HDF5 superblock declares) and was not opened.

Subset	Role	Dev engines (unit:class)	Test engines (unit:class)	Healthy flights dev / test	Post-onset flights (test)	Min. test engine rows per phase	C1–C9
DS01	primary cohort	1:1 2:3 3:2 4:1 5:3 6:2	7:1 8:2 9:1 10:3	158 / 111	230	31,432	yyyyyyyyy
DS02	REFERENCE	2:3 5:3 10:3 16:3 18:3 20:3	11:3 14:1 15:2	95 / 76	126	12,241	yyyyyyyyn
DS03	REFERENCE	1:1 2:2 3:2 4:2 5:1 6:3 7:2 8:3 9:1	10:3 11:3 12:1 13:3 14:1 15:2	237 / 148	290	33,331	yyyyyyyyn
DS04	primary cohort	1:2 2:3 3:2 4:3 5:3 6:3	7:2 8:2 9:2 10:3	105 / 79	265	47,345	yyyyyyyyy
DS05	primary cohort	1:2 2:3 3:2 4:1 5:1 6:3	7:2 8:1 9:3 10:1	153 / 112	215	33,438	yyyyyyyyy
DS06	primary cohort	1:2 2:3 3:2 4:1 5:1 6:3	7:2 8:1 9:3 10:1	141 / 100	222	29,592	yyyyyyyyy
DS07	primary cohort	1:2 2:3 3:2 4:1 5:1 6:3	7:2 8:1 9:3 10:1	154 / 113	231	34,148	yyyyyyyyy
DS08a	primary cohort	1:1 2:3 3:1 4:2 5:2 6:3 7:2 8:2 9:1	10:1 11:2 12:3 13:3 14:3 15:1	197 / 122	261	24,255	yyyyyyyyy
DS08c	primary cohort	1:3 2:3 3:3 4:3 5:3 6:2	7:2 8:2 9:2 10:2	100 / 86	151	48,940	yyyyyyyyy
DS08d	EXCLUDED	–	–	–	–	–	C1 fails; not opened

Source: docs/extension/subset_metadata_audit.csv (labels, variable names and healthy-row altitude only).

Table S2: Engine roles (R-ALLOC for new subsets; frozen roles for DS02 and DS03).

Subset	Family	Fit (class)	Validation	Calibration (class)	Audit (class)	Composition N
DS02	ref.	2 (3), 5 (3), 10 (3), 16 (3)	16 (3)	18 (3), 20 (3)	11 (3), 14 (1), 15 (2)	not eligible
DS03	ref.	1 (1), 2 (2), 3 (2), 5 (1), 6 (3), 7 (2), 9 (1)	9 (1)	4 (2), 8 (3)	10 (3), 11 (3), 12 (1), 13 (3), 14 (1), 15 (2)	102,000
DS01	F1	1 (1), 2 (3), 3 (2)	3 (2)	4 (1), 5 (3), 6 (2)	7 (1), 8 (2), 9 (1), 10 (3)	78,000
DS04	F2	1 (2), 2 (3), 4 (3)	4 (3)	3 (2), 5 (3), 6 (3)	7 (2), 8 (2), 9 (2), 10 (3)	96,000
DS05	F3	1 (2), 2 (3), 4 (1)	4 (1)	3 (2), 5 (1), 6 (3)	7 (2), 8 (1), 9 (3), 10 (1)	78,000
DS06	F3	1 (2), 2 (3), 4 (1)	4 (1)	3 (2), 5 (1), 6 (3)	7 (2), 8 (1), 9 (3), 10 (1)	66,000
DS07	F3	1 (2), 2 (3), 4 (1)	4 (1)	3 (2), 5 (1), 6 (3)	7 (2), 8 (1), 9 (3), 10 (1)	78,000
DS08a	F4	1 (1), 2 (3), 4 (2)	4 (2)	3 (1), 5 (2), 6 (3), 7 (2), 8 (2), 9 (1)	10 (1), 11 (2), 12 (3), 13 (3), 14 (3), 15 (1)	60,000
DS08c	F5	1 (3), 2 (3), 6 (2)	6 (2)	3 (3), 4 (3), 5 (3)	7 (2), 8 (2), 9 (2), 10 (2)	not eligible

Source: results/extension/(subset)/lock/calibration_lock.json.

Table S3: Calibration locks, compute and CVAE numerical checks. The float64 check rescored calibration rows in double precision; the variance-bound share is the fraction of decoder outputs within 1% of the lower log-variance bound. For the references DS02 and DS03 the audit time is the whole reference rerun (CVAE training, lock and audit; the frozen residual models are refitted or loaded).

Subset	Calibration rows / flights	Lock (s)	Audit (s)	Audit rows	CVAE float64 check: min Spearman	CVAE exceedance changes	CVAE variance-bound share (seeds 0/1/2)
DS01	648,562 / 79	3,732	126	2,735,232	1.000000	0	73.5% / 42.0% / 25.5%
DS04	649,001 / 53	4,627	169	3,602,561	1.000000	0	0.1% / 11.4% / 26.0%
DS05	600,857 / 77	3,336	113	2,562,046	1.000000	2	67.5% / 1.7% / 46.7%
DS06	559,506 / 71	3,029	116	2,522,447	1.000000	2	0.0% / 31.8% / 37.2%
DS07	612,770 / 78	3,681	126	2,869,786	1.000000	1	0.7% / 19.4% / 11.9%
DS08a	984,555 / 131	10,496	196	3,722,997	1.000000	2	0.0% / 1.7% / 30.6%
DS08c	692,465 / 47	4,766	118	2,117,819	1.000000	1	16.0% / 0.2% / 0.2%
DS02	386,643 / 32	–	2,215	1,253,743	1.000000	2	45.5% / 11.9% / 0.0%
DS03	458,105 / 40	–	3,266	4,251,560	1.000000	0	88.2% / 84.1% / 9.6%

Source: results/extension/summary/numerical_audit_locks.csv.

Table S4: Epoch selection and refit of the LSTM and CVAE (selection on the validation engine, refit on the full fit pool). DS02/DS03 LSTM checkpoints are frozen and not retrained.

Subset	Detector	Selection epochs run	Selected epoch	Hit 600	Best validation loss	Refit epochs	Refit final loss	Refit finite
DS01	CVAE s0	81	75	no	-30.16	75	-44.62	yes
DS01	CVAE s1	58	52	no	-32.12	52	-39.84	yes
DS01	CVAE s2	42	36	no	-36.99	36	-36.66	yes
DS01	LSTM s0	180	174	no	138.2	174	0.05613	yes
DS01	LSTM s1	130	124	no	138.2	124	0.07002	yes

Subset	Detector	Selection epochs run	Selected epoch	Hit 600	Best validation loss	Refit epochs	Refit final loss	Refit finite
DS01	LSTM s2	150	144	no	138.2	144	0.06771	yes
DS04	CVAE s0	22	16	no	-34.07	16	-20.23	yes
DS04	CVAE s1	35	29	no	-37.4	29	-34.24	yes
DS04	CVAE s2	39	33	no	-36.36	33	-34.3	yes
DS04	LSTM s0	598	592	no	0.1726	592	0.1658	yes
DS04	LSTM s1	584	582	no	0.1699	582	0.1618	yes
DS04	LSTM s2	352	349	no	0.1876	349	0.1808	yes
DS05	CVAE s0	72	66	no	-40.8	66	-41.89	yes
DS05	CVAE s1	24	18	no	-30.79	18	-26.11	yes
DS05	CVAE s2	55	49	no	-40.14	49	-39.41	yes
DS05	LSTM s0	370	369	no	0.07237	369	0.03416	yes
DS05	LSTM s1	341	337	no	0.07487	337	0.03637	yes
DS05	LSTM s2	414	411	no	0.06801	411	0.03142	yes
DS06	CVAE s0	20	14	no	-27.3	14	-19.74	yes
DS06	CVAE s1	45	39	no	-36.21	39	-38.38	yes
DS06	CVAE s2	46	40	no	-35.35	40	-40.64	yes
DS06	LSTM s0	473	468	no	0.07893	468	0.03078	yes
DS06	LSTM s1	510	508	no	0.07897	508	0.02874	yes
DS06	LSTM s2	447	446	no	0.07853	446	0.03265	yes
DS07	CVAE s0	41	35	no	-35.43	35	-35.39	yes
DS07	CVAE s1	39	33	no	-35.87	33	-38.18	yes
DS07	CVAE s2	29	23	no	-33.78	23	-34.66	yes
DS07	LSTM s0	365	365	no	0.09256	365	0.03516	yes
DS07	LSTM s1	355	352	no	0.08867	352	0.0365	yes
DS07	LSTM s2	417	417	no	0.0843	417	0.03184	yes
DS08a	CVAE s0	33	27	no	-35.72	27	-35.55	yes
DS08a	CVAE s1	30	24	no	-34.36	24	-34.85	yes
DS08a	CVAE s2	52	46	no	-39.7	46	-38.66	yes
DS08a	LSTM s0	535	532	no	0.0911	532	0.02193	yes
DS08a	LSTM s1	380	374	no	0.1369	374	0.0317	yes
DS08a	LSTM s2	321	315	no	0.1571	315	0.03802	yes
DS08c	CVAE s0	36	30	no	-29.71	30	-30.66	yes
DS08c	CVAE s1	24	18	no	-7.642	18	-18.64	yes
DS08c	CVAE s2	21	15	no	-16.24	15	-26.25	yes
DS08c	LSTM s0	257	251	no	233.6	251	0.03831	yes
DS08c	LSTM s1	134	128	no	233.8	128	0.06229	yes
DS08c	LSTM s2	210	204	no	233.7	204	0.04536	yes
DS02	CVAE s0	35	29	no	-40.65	29	-40.27	yes
DS02	CVAE s1	24	18	no	-36.17	18	-38.54	yes
DS02	CVAE s2	16	10	no	-35.22	10	-32.37	yes
DS03	CVAE s0	60	54	no	-46.75	54	-47.02	yes
DS03	CVAE s1	48	42	no	-46.79	42	-45.89	yes
DS03	CVAE s2	21	15	no	-41.88	15	-38.66	yes

Source: results/extension/summary/numerical_audit_training.csv. Losses are per-row negative log-likelihoods (CVAE) or mean squared errors (LSTM) on the standardized scale.

A.2 Calibration transport (1% target)

Table S5: Row-pooled phase FPRs and transport metrics for every subset, detector run and arm at the 1% target (P pooled, C retrospective phase-conditioned, Q continuous context).

Subset	Run	Arm	Climb FPR (%)	Cruise FPR (%)	Descent FPR (%)	Pooled A1 (pp)	Pooled A2 (pp)	ME-A2 (pp)	WE-A2 (pp)	EW-A2 (pp)	Worst engine
DS02	PCA	P	0.12	0.43	3.46	3.33	2.46	2.81	3.97	2.81	14
DS02	PCA	C	1.40	1.33	1.08	0.32	0.40	1.66	4.31	1.40	14
DS02	PCA	Q	0.70	2.12	1.27	1.42	1.12	1.60	2.20	0.91	14
DS02	IF s0	P	0.29	0.57	2.34	2.05	1.34	1.38	1.51	1.38	14
DS02	IF s0	C	1.65	1.85	0.92	0.92	0.85	2.31	6.42	2.12	14
DS02	IF s0	Q	0.49	2.24	2.14	1.75	1.24	2.41	3.63	1.48	15
DS02	IF s1	P	0.25	0.53	2.67	2.42	1.67	1.82	2.29	1.82	14
DS02	IF s1	C	1.54	1.53	0.98	0.57	0.54	2.19	5.96	1.93	14
DS02	IF s1	Q	0.42	1.34	1.94	1.52	0.94	1.31	1.58	0.95	14
DS02	IF s2	P	0.27	0.58	2.46	2.19	1.46	1.52	1.72	1.52	14
DS02	IF s2	C	1.69	1.71	0.93	0.79	0.71	2.37	6.59	2.19	14
DS02	IF s2	Q	0.56	2.46	2.11	1.89	1.46	2.52	4.15	1.69	15
DS02	LSTM s0	P	0.45	0.71	2.08	1.63	1.08	1.03	1.36	1.02	11
DS02	LSTM s0	C	1.39	1.11	1.08	0.31	0.39	1.66	4.42	1.42	14
DS02	LSTM s0	Q	0.72	1.55	1.29	0.83	0.55	0.84	1.30	0.71	14
DS02	LSTM s1	P	0.47	0.73	2.10	1.63	1.10	1.06	1.52	0.98	11
DS02	LSTM s1	C	1.24	1.28	0.96	0.32	0.28	1.42	3.74	1.14	14
DS02	LSTM s1	Q	0.56	1.29	0.97	0.73	0.44	0.54	0.64	0.33	15
DS02	LSTM s2	P	0.54	0.75	2.06	1.52	1.06	1.13	1.38	0.96	11
DS02	LSTM s2	C	1.46	1.07	1.00	0.46	0.46	1.77	4.65	1.53	14
DS02	LSTM s2	Q	0.82	2.71	3.14	2.32	2.14	3.28	4.25	1.96	11
DS02	CVAE s0	P	0.75	1.12	2.17	1.42	1.17	1.41	2.79	1.41	14
DS02	CVAE s0	C	1.97	1.45	1.42	0.55	0.97	2.12	4.65	1.93	14
DS02	CVAE s0	Q	0.86	1.73	1.94	1.08	0.94	1.68	2.66	1.16	14
DS02	CVAE s1	P	1.41	1.66	2.84	1.42	1.84	2.86	6.63	2.64	14
DS02	CVAE s1	C	1.90	2.65	1.74	0.91	1.65	2.96	7.37	2.65	14
DS02	CVAE s1	Q	1.24	2.05	2.42	1.18	1.42	2.39	4.64	1.88	14
DS02	CVAE s2	P	1.64	1.15	2.83	1.68	1.83	2.69	6.29	2.63	14
DS02	CVAE s2	C	1.55	2.78	1.95	1.22	1.78	3.06	8.35	2.97	14
DS02	CVAE s2	Q	1.23	1.84	1.96	0.73	0.96	1.81	3.43	1.35	14
DS03	PCA	P	0.88	0.91	1.98	1.10	0.98	1.02	1.45	0.92	10
DS03	PCA	C	1.86	1.79	1.23	0.63	0.86	1.03	1.52	0.80	10
DS03	PCA	Q	2.66	1.80	1.41	1.25	1.66	1.84	7.57	1.33	10
DS03	IF s0	P	0.42	1.09	2.17	1.75	1.17	1.21	2.11	1.11	10
DS03	IF s0	C	1.66	1.39	1.27	0.39	0.66	0.90	2.37	0.62	10
DS03	IF s0	Q	1.96	1.61	1.70	0.35	0.96	1.81	6.31	0.77	10
DS03	IF s1	P	0.40	1.06	2.13	1.73	1.13	1.19	2.16	1.07	10
DS03	IF s1	C	1.66	1.37	1.27	0.40	0.66	0.86	2.06	0.62	10
DS03	IF s1	Q	1.87	1.50	1.51	0.37	0.87	1.74	6.27	0.63	10
DS03	IF s2	P	0.41	1.16	2.13	1.72	1.13	1.20	2.23	1.07	10
DS03	IF s2	C	1.74	1.42	1.27	0.48	0.74	0.95	2.56	0.69	10
DS03	IF s2	Q	1.90	1.46	1.59	0.44	0.90	1.80	6.44	0.65	10
DS03	LSTM s0	P	0.76	1.80	2.23	1.47	1.23	1.22	2.70	1.14	10
DS03	LSTM s0	C	1.78	1.83	1.49	0.34	0.83	1.22	2.04	0.86	12
DS03	LSTM s0	Q	1.51	1.12	1.69	0.57	0.69	1.18	3.67	0.64	10
DS03	LSTM s1	P	0.72	1.89	2.22	1.50	1.22	1.22	2.91	1.13	10
DS03	LSTM s1	C	1.84	1.76	1.50	0.35	0.84	1.20	2.17	0.94	12
DS03	LSTM s1	Q	1.72	1.19	1.71	0.53	0.72	1.38	4.91	0.67	10
DS03	LSTM s2	P	0.77	1.98	2.19	1.42	1.19	1.30	3.08	1.09	10
DS03	LSTM s2	C	1.80	1.72	1.53	0.27	0.80	1.24	2.02	0.87	12
DS03	LSTM s2	Q	1.49	1.06	1.63	0.57	0.63	1.18	3.59	0.58	10
DS03	CVAE s0	P	0.50	2.71	1.95	2.22	1.71	1.87	3.83	1.19	10
DS03	CVAE s0	C	2.17	2.32	1.33	0.99	1.32	1.83	4.73	1.01	10
DS03	CVAE s0	Q	2.63	3.13	1.95	1.18	2.13	2.10	8.47	1.60	10
DS03	CVAE s1	P	0.92	1.18	1.65	0.74	0.65	1.03	1.64	0.55	11
DS03	CVAE s1	C	1.77	1.06	1.28	0.71	0.77	1.23	3.06	0.92	12
DS03	CVAE s1	Q	1.28	1.15	1.68	0.52	0.68	0.93	1.62	0.60	11
DS03	CVAE s2	P	0.69	1.17	2.05	1.37	1.05	1.00	1.74	0.99	11
DS03	CVAE s2	C	1.30	1.97	1.26	0.71	0.97	1.10	1.53	0.92	12
DS03	CVAE s2	Q	0.86	1.52	1.32	0.67	0.52	0.69	1.06	0.52	11
DS01	PCA	P	1.15	0.62	1.99	1.37	0.99	0.98	1.50	0.98	8
DS01	PCA	C	2.08	1.08	1.16	0.99	1.08	1.11	1.31	1.09	9
DS01	PCA	Q	2.29	1.76	1.51	0.78	1.29	1.28	2.57	0.98	8
DS01	IF s0	P	0.85	0.44	1.95	1.51	0.95	0.97	1.57	0.97	7
DS01	IF s0	C	1.18	1.10	0.97	0.21	0.18	0.47	0.96	0.32	7
DS01	IF s0	Q	0.82	0.97	1.14	0.32	0.18	0.63	0.73	0.27	9
DS01	IF s1	P	0.80	0.44	1.89	1.45	0.89	0.91	1.31	0.91	7
DS01	IF s1	C	1.19	1.17	0.89	0.30	0.19	0.43	0.96	0.31	7

Subset	Run	Arm	Climb FPR (%)	Cruise FPR (%)	Descent FPR (%)	Pooled A1 (pp)	Pooled A2 (pp)	ME-A2 (pp)	WE-A2 (pp)	EW-A2 (pp)	Worst engine
DS01	IF s1	Q	0.78	0.98	1.09	0.32	0.22	0.63	0.71	0.32	9
DS01	IF s2	P	0.84	0.45	1.93	1.48	0.93	0.95	1.49	0.95	7
DS01	IF s2	C	1.19	0.98	0.91	0.28	0.19	0.44	0.98	0.32	7
DS01	IF s2	Q	0.83	0.98	1.10	0.27	0.17	0.65	0.70	0.27	10
DS01	LSTM s0	P	1.05	0.67	1.46	0.79	0.46	0.54	0.68	0.46	7
DS01	LSTM s0	C	1.13	1.38	0.92	0.46	0.38	0.53	0.75	0.38	7
DS01	LSTM s0	Q	1.05	1.33	1.37	0.32	0.37	0.75	1.16	0.42	10
DS01	LSTM s1	P	1.07	0.70	1.51	0.82	0.51	0.58	0.77	0.51	7
DS01	LSTM s1	C	1.14	1.38	0.96	0.43	0.38	0.52	0.90	0.41	7
DS01	LSTM s1	Q	1.17	1.35	1.40	0.23	0.40	0.85	1.50	0.45	10
DS01	LSTM s2	P	1.08	0.70	1.49	0.79	0.49	0.58	0.72	0.48	7
DS01	LSTM s2	C	1.19	1.35	0.93	0.42	0.35	0.56	0.93	0.37	7
DS01	LSTM s2	Q	1.14	1.32	1.47	0.33	0.47	0.78	1.45	0.41	10
DS01	CVAE s0	P	0.73	1.10	1.74	1.02	0.74	0.84	1.53	0.69	10
DS01	CVAE s0	C	1.21	1.59	1.09	0.49	0.59	0.64	1.03	0.52	8
DS01	CVAE s0	Q	1.18	1.67	1.26	0.49	0.67	0.93	1.62	0.57	8
DS01	CVAE s1	P	0.80	1.10	1.85	1.05	0.85	0.91	1.67	0.81	10
DS01	CVAE s1	C	1.25	1.87	1.18	0.70	0.87	0.87	1.23	0.86	8
DS01	CVAE s1	Q	1.52	1.61	1.44	0.18	0.61	0.93	1.67	0.56	8
DS01	CVAE s2	P	1.01	0.60	1.78	1.18	0.78	0.79	1.40	0.78	8
DS01	CVAE s2	C	1.16	1.60	0.98	0.61	0.60	0.69	0.97	0.68	9
DS01	CVAE s2	Q	1.34	1.30	1.13	0.21	0.34	0.64	0.81	0.31	8
DS04	PCA	P	0.60	0.73	1.22	0.61	0.40	0.40	0.47	0.40	10
DS04	PCA	C	1.16	0.65	0.98	0.51	0.35	0.35	0.39	0.35	7
DS04	PCA	Q	1.13	0.94	1.33	0.40	0.33	0.33	0.41	0.33	8
DS04	IF s0	P	0.14	0.22	1.60	1.46	0.86	0.94	1.21	0.86	10
DS04	IF s0	C	0.97	0.41	0.81	0.56	0.59	0.59	0.65	0.59	9
DS04	IF s0	Q	0.85	0.55	1.05	0.50	0.45	0.46	0.58	0.45	7
DS04	IF s1	P	0.13	0.22	1.53	1.41	0.87	0.96	1.28	0.87	10
DS04	IF s1	C	0.90	0.39	0.79	0.50	0.61	0.61	0.67	0.61	9
DS04	IF s1	Q	0.84	0.51	1.03	0.52	0.49	0.49	0.62	0.49	7
DS04	IF s2	P	0.14	0.22	1.56	1.42	0.86	0.93	1.23	0.86	10
DS04	IF s2	C	0.96	0.40	0.81	0.56	0.60	0.60	0.66	0.60	9
DS04	IF s2	Q	0.90	0.51	1.06	0.56	0.49	0.51	0.66	0.49	7
DS04	LSTM s0	P	0.31	0.49	1.38	1.07	0.69	0.71	0.75	0.69	10
DS04	LSTM s0	C	1.21	0.48	0.92	0.73	0.52	0.52	0.53	0.52	7
DS04	LSTM s0	Q	1.14	0.82	1.14	0.32	0.18	0.27	0.45	0.16	10
DS04	LSTM s1	P	0.36	0.52	1.27	0.92	0.64	0.64	0.69	0.64	10
DS04	LSTM s1	C	1.14	0.48	0.91	0.66	0.52	0.52	0.54	0.52	7
DS04	LSTM s1	Q	1.14	0.88	1.18	0.30	0.18	0.26	0.42	0.19	10
DS04	LSTM s2	P	0.29	0.48	1.34	1.05	0.71	0.71	0.78	0.71	10
DS04	LSTM s2	C	1.15	0.43	0.93	0.72	0.57	0.57	0.59	0.57	7
DS04	LSTM s2	Q	1.22	0.77	1.17	0.46	0.23	0.34	0.46	0.22	8
DS04	CVAE s0	P	0.31	0.16	0.64	0.48	0.84	0.85	0.93	0.85	8
DS04	CVAE s0	C	1.76	0.00	0.86	1.76	1.00	1.11	1.37	1.00	7
DS04	CVAE s0	Q	0.96	1.01	1.09	0.13	0.09	0.72	1.19	0.09	7
DS04	CVAE s1	P	1.12	0.21	0.54	0.91	0.79	1.15	2.11	0.79	7
DS04	CVAE s1	C	2.43	0.00	0.72	2.42	1.43	2.00	5.01	1.42	7
DS04	CVAE s1	Q	1.79	1.17	1.15	0.65	0.79	1.44	4.14	0.78	7
DS04	CVAE s2	P	0.43	0.27	0.61	0.33	0.73	0.77	0.82	0.73	10
DS04	CVAE s2	C	1.71	0.01	0.72	1.70	0.99	1.19	1.77	0.99	7
DS04	CVAE s2	Q	0.86	0.93	1.07	0.21	0.14	0.76	1.15	0.14	7
DS05	PCA	P	0.71	1.96	3.64	2.93	2.64	2.44	4.95	2.44	9
DS05	PCA	C	5.72	2.71	1.89	3.83	4.72	4.27	6.57	4.27	7
DS05	PCA	Q	17.41	7.74	3.75	13.66	16.41	14.05	24.17	13.75	7
DS05	IF s0	P	0.49	0.48	1.18	0.70	0.52	0.72	0.90	0.54	9
DS05	IF s0	C	1.01	0.46	0.96	0.55	0.54	0.74	1.31	0.56	8
DS05	IF s0	Q	3.20	2.25	1.69	1.51	2.20	2.60	3.90	2.02	10
DS05	IF s1	P	0.43	0.46	1.12	0.69	0.57	0.73	0.91	0.57	9
DS05	IF s1	C	1.05	0.40	1.02	0.65	0.60	0.79	1.36	0.62	8
DS05	IF s1	Q	3.03	2.49	1.91	1.12	2.03	2.81	4.28	1.91	10
DS05	IF s2	P	0.48	0.54	1.38	0.90	0.52	0.74	0.90	0.48	9
DS05	IF s2	C	1.07	0.54	1.02	0.54	0.46	0.67	1.29	0.48	8
DS05	IF s2	Q	2.75	2.45	2.06	0.69	1.75	2.07	2.97	1.47	9
DS05	LSTM s0	P	0.59	1.15	1.61	1.01	0.61	0.61	0.97	0.57	7
DS05	LSTM s0	C	1.12	1.47	1.14	0.35	0.47	0.78	1.20	0.34	8
DS05	LSTM s0	Q	3.06	1.70	1.54	1.53	2.06	2.27	3.96	1.43	9
DS05	LSTM s1	P	0.65	1.15	1.56	0.91	0.56	0.59	0.97	0.53	7
DS05	LSTM s1	C	1.12	1.47	1.19	0.35	0.47	0.83	1.46	0.37	8
DS05	LSTM s1	Q	2.86	1.69	1.59	1.27	1.86	2.15	3.38	1.30	9
DS05	LSTM s2	P	0.85	1.41	1.66	0.82	0.66	0.66	1.03	0.62	9
DS05	LSTM s2	C	1.37	1.84	1.22	0.62	0.84	1.15	1.43	0.64	9
DS05	LSTM s2	Q	4.81	2.18	1.61	3.20	3.81	3.75	6.79	2.80	9
DS05	CVAE s0	P	0.33	1.03	1.65	1.32	0.67	0.95	1.67	0.67	7

Subset	Run	Arm	Climb FPR (%)	Cruise FPR (%)	Descent FPR (%)	Pooled A1 (pp)	Pooled A2 (pp)	ME-A2 (pp)	WE-A2 (pp)	EW-A2 (pp)	Worst engine
DS05	CVAE s0	C	0.78	1.90	0.95	1.12	0.90	0.87	1.31	0.82	7
DS05	CVAE s0	Q	0.66	1.27	1.18	0.61	0.34	0.51	0.75	0.37	10
DS05	CVAE s1	P	1.09	1.18	1.26	0.17	0.26	0.82	1.19	0.29	8
DS05	CVAE s1	C	0.95	2.08	0.96	1.13	1.08	1.15	1.45	1.15	10
DS05	CVAE s1	Q	0.74	1.66	1.02	0.92	0.66	0.83	1.37	0.81	10
DS05	CVAE s2	P	0.27	7.88	0.25	7.62	6.88	4.97	17.38	3.93	9
DS05	CVAE s2	C	1.72	4.99	0.98	4.01	3.99	3.17	10.60	2.13	9
DS05	CVAE s2	Q	0.80	1.83	1.33	1.02	0.83	0.92	1.31	0.67	9
DS06	PCA	P	1.05	2.08	3.62	2.57	2.62	2.42	4.86	2.42	9
DS06	PCA	C	6.67	3.19	2.10	4.58	5.67	4.92	8.53	4.92	7
DS06	PCA	Q	18.46	7.90	4.26	14.20	17.46	15.01	27.22	14.69	7
DS06	IF s0	P	0.43	0.44	1.41	0.98	0.57	0.71	0.91	0.58	9
DS06	IF s0	C	0.93	0.45	1.00	0.55	0.55	0.74	1.26	0.58	8
DS06	IF s0	Q	2.82	2.21	2.20	0.63	1.82	2.13	2.93	1.53	9
DS06	IF s1	P	0.48	0.45	1.08	0.63	0.55	0.72	0.89	0.59	9
DS06	IF s1	C	1.01	0.43	0.95	0.58	0.57	0.76	1.24	0.62	8
DS06	IF s1	Q	4.22	1.95	2.08	2.27	3.22	3.12	4.62	2.87	9
DS06	IF s2	P	0.48	0.47	1.31	0.84	0.53	0.71	0.91	0.56	9
DS06	IF s2	C	0.95	0.48	0.99	0.52	0.52	0.71	1.24	0.55	8
DS06	IF s2	Q	3.09	1.57	1.79	1.52	2.09	2.22	3.57	1.83	9
DS06	LSTM s0	P	0.60	1.09	1.56	0.96	0.56	0.61	0.85	0.51	9
DS06	LSTM s0	C	1.01	1.42	1.15	0.41	0.42	0.63	0.91	0.31	7
DS06	LSTM s0	Q	3.95	1.87	1.46	2.49	2.95	2.98	4.92	2.38	7
DS06	LSTM s1	P	0.69	1.38	1.56	0.87	0.56	0.56	0.91	0.51	7
DS06	LSTM s1	C	1.05	1.60	1.21	0.55	0.60	0.71	0.99	0.44	7
DS06	LSTM s1	Q	6.56	2.88	1.64	4.92	5.56	5.34	10.81	4.42	7
DS06	LSTM s2	P	0.76	1.28	1.66	0.91	0.66	0.66	1.00	0.61	9
DS06	LSTM s2	C	1.20	1.69	1.27	0.49	0.69	0.92	1.10	0.51	9
DS06	LSTM s2	Q	4.08	2.35	1.49	2.58	3.08	3.28	5.21	2.45	7
DS06	CVAE s0	P	0.82	0.86	1.48	0.66	0.48	0.74	1.10	0.50	7
DS06	CVAE s0	C	0.96	1.51	0.96	0.56	0.51	0.87	1.21	0.69	8
DS06	CVAE s0	Q	0.85	1.49	1.00	0.65	0.49	0.61	0.74	0.59	7
DS06	CVAE s1	P	0.46	1.23	1.39	0.93	0.54	0.61	0.91	0.52	7
DS06	CVAE s1	C	0.92	1.86	0.86	0.99	0.86	0.78	1.09	0.75	9
DS06	CVAE s1	Q	0.92	1.65	1.00	0.73	0.65	0.62	0.80	0.61	9
DS06	CVAE s2	P	1.07	1.79	1.47	0.72	0.79	1.36	2.65	0.49	9
DS06	CVAE s2	C	1.06	3.86	0.86	3.00	2.86	2.13	6.79	1.58	9
DS06	CVAE s2	Q	1.05	1.95	1.21	0.91	0.95	1.12	2.16	0.50	9
DS07	PCA	P	0.41	0.92	1.35	0.93	0.59	0.62	0.88	0.50	9
DS07	PCA	C	0.84	1.11	0.94	0.27	0.16	0.67	1.35	0.09	8
DS07	PCA	Q	0.52	1.47	0.97	0.95	0.48	0.80	1.15	0.45	9
DS07	IF s0	P	0.54	0.43	0.84	0.41	0.57	0.78	0.88	0.60	9
DS07	IF s0	C	0.88	0.37	0.81	0.51	0.63	0.78	1.24	0.66	8
DS07	IF s0	Q	0.50	1.47	1.35	0.96	0.50	0.80	1.03	0.53	9
DS07	IF s1	P	0.54	0.44	0.84	0.40	0.56	0.78	0.88	0.60	9
DS07	IF s1	C	0.89	0.38	0.80	0.51	0.62	0.77	1.22	0.66	8
DS07	IF s1	Q	0.57	1.26	1.31	0.74	0.43	0.67	1.03	0.47	10
DS07	IF s2	P	0.58	0.44	0.78	0.33	0.56	0.78	0.87	0.61	8
DS07	IF s2	C	0.88	0.39	0.76	0.49	0.61	0.80	1.22	0.65	8
DS07	IF s2	Q	0.63	1.25	1.12	0.62	0.37	0.68	1.06	0.42	10
DS07	LSTM s0	P	0.57	1.10	1.25	0.68	0.43	0.57	0.75	0.32	9
DS07	LSTM s0	C	0.95	1.16	1.00	0.22	0.16	0.74	1.40	0.17	8
DS07	LSTM s0	Q	1.29	1.66	1.27	0.39	0.66	1.11	1.66	0.61	7
DS07	LSTM s1	P	0.70	1.03	1.21	0.51	0.30	0.70	1.00	0.18	8
DS07	LSTM s1	C	0.98	1.06	1.00	0.08	0.06	0.79	1.58	0.21	8
DS07	LSTM s1	Q	1.40	1.41	1.21	0.20	0.41	0.98	1.72	0.46	7
DS07	LSTM s2	P	0.69	1.09	1.27	0.58	0.31	0.67	0.83	0.24	8
DS07	LSTM s2	C	1.02	1.24	1.02	0.23	0.24	0.86	1.48	0.26	8
DS07	LSTM s2	Q	1.46	1.48	1.31	0.17	0.48	1.16	1.67	0.59	7
DS07	CVAE s0	P	0.25	1.05	1.77	1.51	0.77	0.97	1.73	0.74	7
DS07	CVAE s0	C	0.75	1.67	0.89	0.92	0.67	0.76	0.82	0.70	8
DS07	CVAE s0	Q	0.61	1.50	0.99	0.88	0.50	0.55	0.87	0.54	10
DS07	CVAE s1	P	1.27	1.37	1.41	0.14	0.41	0.82	1.62	0.45	8
DS07	CVAE s1	C	1.20	2.02	1.12	0.90	1.02	1.15	1.53	0.98	8
DS07	CVAE s1	Q	0.87	1.66	1.40	0.79	0.66	0.70	0.97	0.70	10
DS07	CVAE s2	P	0.68	1.27	1.33	0.64	0.33	0.74	0.98	0.34	9
DS07	CVAE s2	C	0.76	1.97	0.95	1.21	0.97	0.82	1.75	0.74	9
DS07	CVAE s2	Q	0.71	1.54	1.10	0.82	0.54	0.76	1.17	0.68	10
DS08a	PCA	P	0.57	1.50	2.05	1.47	1.05	1.61	4.19	1.00	14
DS08a	PCA	C	0.57	2.01	1.65	1.44	1.01	1.52	5.39	0.83	14
DS08a	PCA	Q	1.54	2.63	1.17	1.46	1.63	2.27	9.70	1.32	14
DS08a	IF s0	P	0.32	2.03	2.31	1.99	1.31	2.27	6.58	1.32	14
DS08a	IF s0	C	0.79	3.12	1.53	2.34	2.12	2.08	9.22	1.74	14

Subset	Run	Arm	Climb FPR (%)	Cruise FPR (%)	Descent FPR (%)	Pooled A1 (pp)	Pooled A2 (pp)	ME-A2 (pp)	WE-A2 (pp)	EW-A2 (pp)	Worst engine
DS08a	IF s0	Q	3.57	4.01	1.40	2.61	3.01	3.12	11.52	2.37	14
DS08a	IF s1	P	0.40	2.26	2.43	2.03	1.43	2.49	7.50	1.42	14
DS08a	IF s1	C	0.78	3.26	1.63	2.47	2.26	2.27	9.39	1.82	14
DS08a	IF s1	Q	3.01	3.63	1.38	2.25	2.63	2.94	11.19	2.06	14
DS08a	IF s2	P	0.34	2.18	2.40	2.06	1.40	2.46	7.10	1.41	14
DS08a	IF s2	C	0.63	3.21	1.68	2.58	2.21	2.26	9.38	1.79	14
DS08a	IF s2	Q	3.12	3.43	1.40	2.03	2.43	2.78	9.93	1.90	14
DS08a	LSTM s0	P	0.58	2.11	1.78	1.53	1.11	1.56	4.90	0.82	14
DS08a	LSTM s0	C	0.68	2.18	1.64	1.50	1.18	1.53	5.00	0.88	14
DS08a	LSTM s0	Q	0.80	3.98	1.25	3.18	2.98	2.93	11.80	2.40	14
DS08a	LSTM s1	P	0.58	2.00	1.83	1.42	1.00	1.54	4.75	0.75	14
DS08a	LSTM s1	C	0.70	2.11	1.63	1.40	1.11	1.49	4.91	0.80	14
DS08a	LSTM s1	Q	0.90	4.31	1.20	3.41	3.31	3.19	11.40	2.67	14
DS08a	LSTM s2	P	0.67	1.95	1.92	1.27	0.95	1.65	5.00	0.85	14
DS08a	LSTM s2	C	0.72	2.19	1.69	1.47	1.19	1.65	5.35	0.87	14
DS08a	LSTM s2	Q	0.88	4.00	1.26	3.12	3.00	2.97	11.66	2.41	14
DS08a	CVAE s0	P	3.04	2.32	0.29	2.75	2.04	2.43	6.72	1.98	14
DS08a	CVAE s0	C	1.00	3.73	1.16	2.74	2.73	2.57	9.23	2.16	14
DS08a	CVAE s0	Q	1.24	2.60	1.37	1.36	1.60	1.70	3.98	1.17	14
DS08a	CVAE s1	P	0.43	1.95	1.47	1.52	0.95	1.55	6.19	0.69	14
DS08a	CVAE s1	C	0.81	1.73	1.41	0.92	0.73	1.48	5.53	0.47	14
DS08a	CVAE s1	Q	0.69	1.15	1.50	0.81	0.50	0.70	2.23	0.47	14
DS08a	CVAE s2	P	1.66	5.28	1.56	3.72	4.28	3.77	9.76	3.32	13
DS08a	CVAE s2	C	2.26	4.65	1.43	3.22	3.65	3.29	8.36	2.82	13
DS08a	CVAE s2	Q	1.63	4.23	1.39	2.84	3.23	2.74	7.93	2.54	13
DS08c	PCA	P	0.31	0.30	0.79	0.50	0.70	0.81	0.91	0.69	8
DS08c	PCA	C	0.76	0.26	0.58	0.49	0.74	0.79	0.92	0.72	8
DS08c	PCA	Q	0.22	0.31	0.84	0.62	0.78	0.82	0.96	0.78	10
DS08c	IF s0	P	0.45	0.19	0.88	0.69	0.81	0.83	0.96	0.80	8
DS08c	IF s0	C	0.83	0.15	0.70	0.68	0.85	0.84	0.96	0.84	8
DS08c	IF s0	Q	0.18	2.81	1.19	2.63	1.81	1.91	5.01	1.60	8
DS08c	IF s1	P	0.43	0.19	0.77	0.58	0.81	0.84	0.95	0.79	8
DS08c	IF s1	C	0.67	0.15	0.65	0.52	0.85	0.84	0.96	0.84	8
DS08c	IF s1	Q	0.28	3.45	1.19	3.17	2.45	2.52	6.39	2.22	8
DS08c	IF s2	P	0.46	0.20	0.80	0.61	0.80	0.83	0.94	0.79	8
DS08c	IF s2	C	0.77	0.14	0.70	0.63	0.86	0.85	0.96	0.85	8
DS08c	IF s2	Q	0.28	1.72	1.17	1.43	0.72	1.26	2.39	0.72	8
DS08c	LSTM s0	P	0.46	0.27	0.54	0.26	0.73	0.78	0.94	0.71	8
DS08c	LSTM s0	C	0.69	0.15	0.54	0.54	0.85	0.84	0.96	0.84	8
DS08c	LSTM s0	Q	0.45	2.33	2.00	1.88	1.33	2.11	4.31	1.43	7
DS08c	LSTM s1	P	0.60	0.28	0.56	0.32	0.72	0.74	0.93	0.70	8
DS08c	LSTM s1	C	0.91	0.16	0.58	0.75	0.84	0.83	0.96	0.83	8
DS08c	LSTM s1	Q	0.61	2.40	2.86	2.25	1.86	2.65	4.39	1.90	7
DS08c	LSTM s2	P	0.51	0.27	0.51	0.24	0.73	0.77	0.93	0.71	8
DS08c	LSTM s2	C	0.82	0.14	0.54	0.68	0.86	0.85	0.96	0.85	8
DS08c	LSTM s2	Q	0.78	3.08	2.93	2.31	2.08	3.02	4.82	2.21	7
DS08c	CVAE s0	P	0.20	0.19	0.14	0.06	0.86	0.88	0.93	0.86	9
DS08c	CVAE s0	C	0.79	0.03	0.69	0.76	0.97	0.97	1.00	0.97	10
DS08c	CVAE s0	Q	0.63	1.10	1.16	0.53	0.37	0.48	0.73	0.37	7
DS08c	CVAE s1	P	0.53	0.19	0.23	0.35	0.81	0.85	0.94	0.80	9
DS08c	CVAE s1	C	0.89	0.04	0.86	0.84	0.96	0.95	0.99	0.95	9
DS08c	CVAE s1	Q	0.62	1.40	0.96	0.79	0.40	0.55	0.97	0.43	7
DS08c	CVAE s2	P	0.24	0.18	0.18	0.06	0.82	0.88	0.99	0.82	8
DS08c	CVAE s2	C	1.42	0.01	0.53	1.41	0.99	0.99	1.00	0.99	9
DS08c	CVAE s2	Q	0.75	1.15	1.21	0.46	0.25	0.44	0.89	0.25	7

Source: results/extension/(subset)/audit/transport_summary.csv. A1 = max minus min phase FPR; A2 = max over phases of $|FPR - \alpha|$; ME-A2 = mean over audit engines of per-engine A2; WE-A2 = worst engine; EW-A2 = A2 of the equal-engine average FPR.

Table S6: Paired comparisons of C and Q against P for every detector run at the 1% target (negative differences favour the conditioned arm).

Subset	Run	Arm	$\Delta A1$ (pp)	$\Delta A2$ (pp)	$\Delta ME-A2$ (pp)	$\Delta WE-A2$ (pp)	$\Delta EW-A2$ (pp)	Engines worse / better / all	All engines improved
DS02	PCA	C	-3.01	-2.06	-1.15	+0.34	-1.40	1 / 2 / 3	no
DS02	PCA	Q	-1.91	-1.33	-1.21	-1.77	-1.89	1 / 2 / 3	no
DS02	IF s0	C	-1.13	-0.49	+0.93	+4.91	+0.74	1 / 2 / 3	no
DS02	IF s0	Q	-0.30	-0.10	+1.03	+2.12	+0.10	3 / 0 / 3	no
DS02	IF s1	C	-1.85	-1.13	+0.37	+3.67	+0.11	1 / 2 / 3	no
DS02	IF s1	Q	-0.90	-0.73	-0.51	-0.71	-0.87	0 / 3 / 3	yes
DS02	IF s2	C	-1.40	-0.74	+0.85	+4.87	+0.67	1 / 2 / 3	no
DS02	IF s2	Q	-0.30	0.00	+1.00	+2.42	+0.17	3 / 0 / 3	no

Subset	Run	Arm – P	$\Delta A1$ (pp)	$\Delta A2$ (pp)	$\Delta ME-A2$ (pp)	$\Delta WE-A2$ (pp)	$\Delta EW-A2$ (pp)	Engines worse / better / all	All engines improved
DS02	LSTM s0	C	−1.32	−0.69	+0.63	+3.05	+0.40	1 / 2 / 3	no
DS02	LSTM s0	Q	−0.80	−0.53	−0.19	−0.06	−0.31	1 / 2 / 3	no
DS02	LSTM s1	C	−1.31	−0.82	+0.36	+2.22	+0.16	1 / 2 / 3	no
DS02	LSTM s1	Q	−0.90	−0.66	−0.52	−0.88	−0.65	0 / 3 / 3	yes
DS02	LSTM s2	C	−1.07	−0.60	+0.64	+3.28	+0.58	1 / 2 / 3	no
DS02	LSTM s2	Q	+0.80	+1.08	+2.15	+2.88	+1.01	3 / 0 / 3	no
DS02	CVAE s0	C	−0.87	−0.20	+0.71	+1.87	+0.52	2 / 1 / 3	no
DS02	CVAE s0	Q	−0.34	−0.22	+0.27	−0.13	−0.25	1 / 2 / 3	no
DS02	CVAE s1	C	−0.52	−0.19	+0.10	+0.74	+0.01	2 / 1 / 3	no
DS02	CVAE s1	Q	−0.24	−0.41	−0.47	−1.99	−0.76	2 / 1 / 3	no
DS02	CVAE s2	C	−0.46	−0.06	+0.37	+2.06	+0.33	1 / 2 / 3	no
DS02	CVAE s2	Q	−0.95	−0.87	−0.89	−2.86	−1.28	1 / 2 / 3	no
DS03	PCA	C	−0.48	−0.12	+0.01	+0.07	−0.12	3 / 3 / 6	no
DS03	PCA	Q	+0.15	+0.68	+0.82	+6.12	+0.41	3 / 3 / 6	no
DS03	IF s0	C	−1.36	−0.51	−0.30	+0.25	−0.50	2 / 4 / 6	no
DS03	IF s0	Q	−1.40	−0.21	+0.61	+4.20	−0.34	4 / 2 / 6	no
DS03	IF s1	C	−1.33	−0.46	−0.33	−0.10	−0.45	1 / 5 / 6	no
DS03	IF s1	Q	−1.36	−0.26	+0.55	+4.12	−0.44	2 / 4 / 6	no
DS03	IF s2	C	−1.25	−0.39	−0.25	+0.33	−0.38	2 / 4 / 6	no
DS03	IF s2	Q	−1.28	−0.23	+0.60	+4.20	−0.42	3 / 3 / 6	no
DS03	LSTM s0	C	−1.13	−0.40	0.00	−0.66	−0.28	3 / 3 / 6	no
DS03	LSTM s0	Q	−0.90	−0.54	−0.04	+0.98	−0.50	3 / 3 / 6	no
DS03	LSTM s1	C	−1.15	−0.38	−0.03	−0.73	−0.19	3 / 3 / 6	no
DS03	LSTM s1	Q	−0.97	−0.50	+0.15	+2.01	−0.46	4 / 2 / 6	no
DS03	LSTM s2	C	−1.15	−0.40	−0.06	−1.06	−0.21	3 / 3 / 6	no
DS03	LSTM s2	Q	−0.85	−0.56	−0.12	+0.51	−0.51	4 / 2 / 6	no
DS03	CVAE s0	C	−1.22	−0.39	−0.05	+0.90	−0.18	3 / 3 / 6	no
DS03	CVAE s0	Q	−1.04	+0.42	+0.23	+4.64	+0.41	2 / 4 / 6	no
DS03	CVAE s1	C	−0.02	+0.12	+0.20	+1.42	+0.37	3 / 3 / 6	no
DS03	CVAE s1	Q	−0.21	+0.02	−0.10	−0.02	+0.05	2 / 4 / 6	no
DS03	CVAE s2	C	−0.66	−0.08	+0.10	−0.21	−0.07	3 / 3 / 6	no
DS03	CVAE s2	Q	−0.70	−0.53	−0.31	−0.67	−0.47	2 / 4 / 6	no
DS01	PCA	C	−0.38	+0.08	+0.12	−0.19	+0.11	3 / 1 / 4	no
DS01	PCA	Q	−0.59	+0.30	+0.29	+1.08	0.00	2 / 2 / 4	no
DS01	IF s0	C	−1.30	−0.76	−0.50	−0.61	−0.65	0 / 4 / 4	yes
DS01	IF s0	Q	−1.19	−0.77	−0.35	−0.85	−0.70	1 / 3 / 4	no
DS01	IF s1	C	−1.16	−0.70	−0.48	−0.35	−0.60	0 / 4 / 4	yes
DS01	IF s1	Q	−1.14	−0.67	−0.28	−0.61	−0.60	0 / 4 / 4	yes
DS01	IF s2	C	−1.20	−0.73	−0.51	−0.51	−0.63	0 / 4 / 4	yes
DS01	IF s2	Q	−1.21	−0.75	−0.30	−0.79	−0.69	1 / 3 / 4	no
DS01	LSTM s0	C	−0.33	−0.08	−0.01	+0.07	−0.08	3 / 1 / 4	no
DS01	LSTM s0	Q	−0.47	−0.09	+0.20	+0.48	−0.04	3 / 1 / 4	no
DS01	LSTM s1	C	−0.39	−0.13	−0.06	+0.13	−0.11	2 / 2 / 4	no
DS01	LSTM s1	Q	−0.59	−0.12	+0.27	+0.73	−0.06	3 / 1 / 4	no
DS01	LSTM s2	C	−0.37	−0.13	−0.02	+0.21	−0.12	2 / 2 / 4	no
DS01	LSTM s2	Q	−0.46	−0.02	+0.20	+0.73	−0.07	3 / 1 / 4	no
DS01	CVAE s0	C	−0.52	−0.15	−0.21	−0.50	−0.17	2 / 2 / 4	no
DS01	CVAE s0	Q	−0.52	−0.07	+0.09	+0.09	−0.12	3 / 1 / 4	no
DS01	CVAE s1	C	−0.36	+0.02	−0.04	−0.44	+0.05	2 / 2 / 4	no
DS01	CVAE s1	Q	−0.87	−0.24	+0.02	+0.01	−0.24	1 / 3 / 4	no
DS01	CVAE s2	C	−0.56	−0.18	−0.10	−0.43	−0.11	2 / 2 / 4	no
DS01	CVAE s2	Q	−0.97	−0.44	−0.15	−0.59	−0.47	1 / 3 / 4	no
DS04	PCA	C	−0.10	−0.05	−0.05	−0.08	−0.05	1 / 3 / 4	no
DS04	PCA	Q	−0.22	−0.06	−0.07	−0.06	−0.06	1 / 2 / 4	no
DS04	IF s0	C	−0.90	−0.27	−0.34	−0.57	−0.27	0 / 4 / 4	yes
DS04	IF s0	Q	−0.96	−0.41	−0.47	−0.64	−0.41	0 / 4 / 4	yes
DS04	IF s1	C	−0.91	−0.27	−0.35	−0.60	−0.26	0 / 4 / 4	yes
DS04	IF s1	Q	−0.88	−0.38	−0.47	−0.65	−0.38	0 / 4 / 4	yes
DS04	IF s2	C	−0.86	−0.26	−0.33	−0.56	−0.26	0 / 4 / 4	yes
DS04	IF s2	Q	−0.87	−0.37	−0.42	−0.57	−0.37	0 / 4 / 4	yes
DS04	LSTM s0	C	−0.34	−0.17	−0.19	−0.22	−0.17	0 / 4 / 4	yes
DS04	LSTM s0	Q	−0.75	−0.51	−0.44	−0.31	−0.53	0 / 4 / 4	yes
DS04	LSTM s1	C	−0.26	−0.12	−0.12	−0.15	−0.12	0 / 4 / 4	yes
DS04	LSTM s1	Q	−0.62	−0.46	−0.38	−0.27	−0.46	0 / 4 / 4	yes
DS04	LSTM s2	C	−0.33	−0.15	−0.15	−0.18	−0.14	0 / 4 / 4	yes
DS04	LSTM s2	Q	−0.59	−0.48	−0.38	−0.31	−0.49	0 / 4 / 4	yes
DS04	CVAE s0	C	+1.28	+0.15	+0.26	+0.44	+0.15	4 / 0 / 4	no
DS04	CVAE s0	Q	−0.36	−0.75	−0.13	+0.26	−0.76	1 / 3 / 4	no
DS04	CVAE s1	C	+1.51	+0.64	+0.85	+2.91	+0.63	4 / 0 / 4	no
DS04	CVAE s1	Q	−0.27	+0.01	+0.29	+2.03	−0.01	1 / 3 / 4	no
DS04	CVAE s2	C	+1.37	+0.27	+0.42	+0.95	+0.26	4 / 0 / 4	no
DS04	CVAE s2	Q	−0.13	−0.59	−0.01	+0.33	−0.59	1 / 3 / 4	no
DS05	PCA	C	+0.90	+2.08	+1.82	+1.63	+1.82	4 / 0 / 4	no

Subset	Run	Arm – P	$\Delta A1$ (pp)	$\Delta A2$ (pp)	$\Delta ME-A2$ (pp)	$\Delta WE-A2$ (pp)	$\Delta EW-A2$ (pp)	Engines worse / better / all	All engines improved
DS05	PCA	Q	+10.73	+13.77	+11.60	+19.23	+11.30	3 / 1 / 4	no
DS05	IF s0	C	−0.15	+0.02	+0.02	+0.42	+0.03	2 / 2 / 4	no
DS05	IF s0	Q	+0.82	+1.69	+1.89	+3.01	+1.48	3 / 1 / 4	no
DS05	IF s1	C	−0.05	+0.03	+0.05	+0.45	+0.05	1 / 3 / 4	no
DS05	IF s1	Q	+0.43	+1.46	+2.08	+3.37	+1.35	3 / 1 / 4	no
DS05	IF s2	C	−0.36	−0.05	−0.07	+0.39	0.00	1 / 3 / 4	no
DS05	IF s2	Q	−0.21	+1.23	+1.33	+2.07	+1.00	3 / 1 / 4	no
DS05	LSTM s0	C	−0.66	−0.14	+0.17	+0.22	−0.24	2 / 2 / 4	no
DS05	LSTM s0	Q	+0.51	+1.45	+1.66	+2.99	+0.86	4 / 0 / 4	no
DS05	LSTM s1	C	−0.56	−0.09	+0.25	+0.49	−0.16	2 / 2 / 4	no
DS05	LSTM s1	Q	+0.36	+1.30	+1.57	+2.40	+0.78	4 / 0 / 4	no
DS05	LSTM s2	C	−0.19	+0.18	+0.48	+0.40	+0.02	4 / 0 / 4	no
DS05	LSTM s2	Q	+2.39	+3.15	+3.08	+5.76	+2.18	4 / 0 / 4	no
DS05	CVAE s0	C	−0.19	+0.23	−0.08	−0.36	+0.16	1 / 3 / 4	no
DS05	CVAE s0	Q	−0.71	−0.33	−0.43	−0.92	−0.30	1 / 3 / 4	no
DS05	CVAE s1	C	+0.96	+0.82	+0.33	+0.26	+0.86	3 / 1 / 4	no
DS05	CVAE s1	Q	+0.75	+0.40	+0.01	+0.18	+0.51	1 / 3 / 4	no
DS05	CVAE s2	C	−3.62	−2.89	−1.79	−6.78	−1.80	0 / 4 / 4	yes
DS05	CVAE s2	Q	−6.60	−6.05	−4.05	−16.07	−3.26	1 / 3 / 4	no
DS06	PCA	C	+2.00	+3.05	+2.50	+3.67	+2.50	4 / 0 / 4	no
DS06	PCA	Q	+11.62	+14.84	+12.60	+22.36	+12.28	3 / 1 / 4	no
DS06	IF s0	C	−0.43	−0.02	+0.03	+0.35	−0.01	1 / 3 / 4	no
DS06	IF s0	Q	−0.35	+1.25	+1.42	+2.02	+0.95	4 / 0 / 4	no
DS06	IF s1	C	−0.05	+0.02	+0.03	+0.35	+0.03	2 / 2 / 4	no
DS06	IF s1	Q	+1.64	+2.68	+2.40	+3.74	+2.28	3 / 1 / 4	no
DS06	IF s2	C	−0.32	−0.01	0.00	+0.33	−0.01	1 / 3 / 4	no
DS06	IF s2	Q	+0.69	+1.56	+1.51	+2.66	+1.27	4 / 0 / 4	no
DS06	LSTM s0	C	−0.55	−0.13	+0.02	+0.06	−0.20	2 / 2 / 4	no
DS06	LSTM s0	Q	+1.53	+2.39	+2.37	+4.07	+1.86	4 / 0 / 4	no
DS06	LSTM s1	C	−0.32	+0.05	+0.15	+0.08	−0.07	4 / 0 / 4	no
DS06	LSTM s1	Q	+4.05	+5.00	+4.78	+9.90	+3.91	4 / 0 / 4	no
DS06	LSTM s2	C	−0.42	+0.02	+0.26	+0.09	−0.11	4 / 0 / 4	no
DS06	LSTM s2	Q	+1.67	+2.41	+2.63	+4.21	+1.84	4 / 0 / 4	no
DS06	CVAE s0	C	−0.10	+0.03	+0.13	+0.11	+0.19	2 / 2 / 4	no
DS06	CVAE s0	Q	−0.01	+0.01	−0.13	−0.35	+0.09	1 / 3 / 4	no
DS06	CVAE s1	C	+0.06	+0.32	+0.17	+0.18	+0.23	3 / 1 / 4	no
DS06	CVAE s1	Q	−0.20	+0.11	0.00	−0.11	+0.09	3 / 1 / 4	no
DS06	CVAE s2	C	+2.28	+2.07	+0.77	+4.14	+1.09	1 / 3 / 4	no
DS06	CVAE s2	Q	+0.18	+0.16	−0.24	−0.49	+0.01	1 / 3 / 4	no
DS07	PCA	C	−0.66	−0.43	+0.05	+0.46	−0.42	1 / 3 / 4	no
DS07	PCA	Q	+0.02	−0.11	+0.19	+0.27	−0.06	3 / 1 / 4	no
DS07	IF s0	C	+0.10	+0.07	0.00	+0.36	+0.05	2 / 2 / 4	no
DS07	IF s0	Q	+0.56	−0.07	+0.03	+0.14	−0.07	2 / 2 / 4	no
DS07	IF s1	C	+0.11	+0.06	−0.01	+0.34	+0.05	2 / 2 / 4	no
DS07	IF s1	Q	+0.34	−0.13	−0.12	+0.14	−0.14	1 / 3 / 4	no
DS07	IF s2	C	+0.15	+0.05	+0.01	+0.35	+0.05	2 / 2 / 4	no
DS07	IF s2	Q	+0.29	−0.19	−0.10	+0.19	−0.19	1 / 3 / 4	no
DS07	LSTM s0	C	−0.46	−0.27	+0.17	+0.65	−0.16	2 / 2 / 4	no
DS07	LSTM s0	Q	−0.29	+0.23	+0.54	+0.92	+0.29	4 / 0 / 4	no
DS07	LSTM s1	C	−0.43	−0.24	+0.09	+0.59	+0.04	2 / 2 / 4	no
DS07	LSTM s1	Q	−0.31	+0.11	+0.28	+0.72	+0.28	2 / 2 / 4	no
DS07	LSTM s2	C	−0.36	−0.07	+0.19	+0.65	+0.02	3 / 1 / 4	no
DS07	LSTM s2	Q	−0.41	+0.17	+0.49	+0.84	+0.35	2 / 2 / 4	no
DS07	CVAE s0	C	−0.59	−0.10	−0.21	−0.90	−0.04	2 / 2 / 4	no
DS07	CVAE s0	Q	−0.63	−0.27	−0.42	−0.86	−0.20	1 / 3 / 4	no
DS07	CVAE s1	C	+0.76	+0.62	+0.34	−0.08	+0.53	3 / 1 / 4	no
DS07	CVAE s1	Q	+0.65	+0.25	−0.12	−0.65	+0.25	2 / 2 / 4	no
DS07	CVAE s2	C	+0.57	+0.65	+0.08	+0.77	+0.40	1 / 3 / 4	no
DS07	CVAE s2	Q	+0.18	+0.21	+0.02	+0.19	+0.33	2 / 2 / 4	no
DS08a	PCA	C	−0.03	−0.04	−0.09	+1.20	−0.17	1 / 5 / 6	no
DS08a	PCA	Q	−0.02	+0.58	+0.66	+5.51	+0.32	3 / 3 / 6	no
DS08a	IF s0	C	+0.35	+0.81	−0.19	+2.65	+0.43	1 / 5 / 6	no
DS08a	IF s0	Q	+0.62	+1.70	+0.85	+4.94	+1.05	3 / 3 / 6	no
DS08a	IF s1	C	+0.45	+0.83	−0.22	+1.89	+0.40	2 / 4 / 6	no
DS08a	IF s1	Q	+0.22	+1.20	+0.44	+3.69	+0.64	3 / 3 / 6	no
DS08a	IF s2	C	+0.52	+0.80	−0.20	+2.28	+0.38	1 / 5 / 6	no
DS08a	IF s2	Q	−0.03	+1.02	+0.32	+2.83	+0.49	2 / 4 / 6	no
DS08a	LSTM s0	C	−0.03	+0.07	−0.03	+0.10	+0.06	2 / 4 / 6	no
DS08a	LSTM s0	Q	+1.65	+1.87	+1.37	+6.90	+1.58	3 / 3 / 6	no
DS08a	LSTM s1	C	−0.02	+0.11	−0.05	+0.16	+0.05	1 / 5 / 6	no
DS08a	LSTM s1	Q	+1.99	+2.31	+1.65	+6.66	+1.92	4 / 2 / 6	no
DS08a	LSTM s2	C	+0.19	+0.24	−0.01	+0.35	+0.03	1 / 5 / 6	no
DS08a	LSTM s2	Q	+1.85	+2.05	+1.31	+6.66	+1.57	2 / 4 / 6	no

Subset	Run	Arm – P	$\Delta A1$ (pp)	$\Delta A2$ (pp)	$\Delta ME-A2$ (pp)	$\Delta WE-A2$ (pp)	$\Delta EW-A2$ (pp)	Engines worse / better / all	All engines improved
DS08a	CVAE s0	C	−0.01	+0.69	+0.14	+2.51	+0.18	2 / 4 / 6	no
DS08a	CVAE s0	Q	−1.39	−0.44	−0.73	−2.74	−0.81	3 / 3 / 6	no
DS08a	CVAE s1	C	−0.60	−0.23	−0.07	−0.65	−0.21	3 / 3 / 6	no
DS08a	CVAE s1	Q	−0.71	−0.45	−0.85	−3.96	−0.21	0 / 6 / 6	yes
DS08a	CVAE s2	C	−0.50	−0.63	−0.49	−1.40	−0.50	2 / 4 / 6	no
DS08a	CVAE s2	Q	−0.88	−1.05	−1.03	−1.83	−0.78	0 / 6 / 6	yes
DS08c	PCA	C	0.00	+0.03	−0.02	+0.01	+0.03	3 / 1 / 4	no
DS08c	PCA	Q	+0.12	+0.07	+0.01	+0.05	+0.09	3 / 1 / 4	no
DS08c	IF s0	C	−0.01	+0.04	0.00	+0.01	+0.04	3 / 1 / 4	no
DS08c	IF s0	Q	+1.94	+1.00	+1.08	+4.05	+0.80	4 / 0 / 4	no
DS08c	IF s1	C	−0.06	+0.04	0.00	+0.01	+0.05	3 / 1 / 4	no
DS08c	IF s1	Q	+2.59	+1.64	+1.68	+5.43	+1.42	3 / 1 / 4	no
DS08c	IF s2	C	+0.02	+0.06	+0.02	+0.02	+0.06	3 / 1 / 4	no
DS08c	IF s2	Q	+0.83	−0.08	+0.43	+1.44	−0.07	3 / 1 / 4	no
DS08c	LSTM s0	C	+0.28	+0.13	+0.07	+0.03	+0.14	4 / 0 / 4	no
DS08c	LSTM s0	Q	+1.61	+0.61	+1.33	+3.38	+0.72	3 / 1 / 4	no
DS08c	LSTM s1	C	+0.43	+0.12	+0.09	+0.03	+0.13	4 / 0 / 4	no
DS08c	LSTM s1	Q	+1.93	+1.14	+1.92	+3.46	+1.19	4 / 0 / 4	no
DS08c	LSTM s2	C	+0.44	+0.13	+0.08	+0.04	+0.14	4 / 0 / 4	no
DS08c	LSTM s2	Q	+2.06	+1.35	+2.26	+3.89	+1.50	4 / 0 / 4	no
DS08c	CVAE s0	C	+0.70	+0.11	+0.09	+0.07	+0.11	4 / 0 / 4	no
DS08c	CVAE s0	Q	+0.47	−0.50	−0.40	−0.20	−0.50	0 / 4 / 4	yes
DS08c	CVAE s1	C	+0.50	+0.14	+0.11	+0.05	+0.15	4 / 0 / 4	no
DS08c	CVAE s1	Q	+0.44	−0.41	−0.30	+0.03	−0.37	1 / 3 / 4	no
DS08c	CVAE s2	C	+1.35	+0.17	+0.10	+0.01	+0.17	4 / 0 / 4	no
DS08c	CVAE s2	Q	+0.40	−0.56	−0.44	−0.09	−0.56	1 / 3 / 4	no

Source: results/extension/⟨subset⟩/audit/paired_transport.csv.

Table S7: Per-engine nominal calibration error A2 at the 1% target for every audit engine: residual runs (PCA, IF and LSTM seeds; median and range over the seven runs) and CVAE seeds.

Subset	Engine (class)	Class in calibration	Residual A2 under P (pp) median [range]			under C (pp)		under Q (pp)	Residual runs worse: C / Q	CVAE A2 P / C / Q (pp, seed median)	CVAE seeds worse: C / Q
DS02	11 (3)	yes	1.36	[1.23, 1.78]		0.36	[0.25, 0.38]	1.45 [0.44, 4.25]	0/7 / 4/7	1.06 / 0.26 / 1.04	0/3 / 1/3
DS02	14 (1)	no	1.51	[0.99, 3.97]		4.65	[3.74, 6.59]	1.58 [0.54, 2.20]	7/7 / 4/7	6.29 / 7.37 / 3.43	3/3 / 0/3
DS02	15 (2)	no	1.40	[0.66, 2.67]		0.27	[0.14, 0.30]	1.08 [0.58, 4.18]	0/7 / 3/7	0.84 / 1.25 / 1.27	2/3 / 3/3
DS03	10 (3)	yes	2.23	[1.45, 3.08]		1.69	[1.52, 2.56]	6.27 [3.59, 7.57]	3/7 / 7/7	1.54 / 1.42 / 1.49	2/3 / 1/3
DS03	11 (3)	yes	1.32	[1.17, 1.67]		0.91	[0.49, 0.99]	0.84 [0.56, 0.89]	0/7 / 0/7	1.74 / 1.12 / 1.57	0/3 / 0/3
DS03	12 (1)	no	0.68	[0.39, 0.85]		1.07	[0.82, 2.17]	0.69 [0.40, 0.74]	7/7 / 4/7	0.86 / 1.53 / 0.42	3/3 / 1/3
DS03	13 (3)	yes	1.66	[1.25, 1.81]		1.22	[0.77, 1.47]	1.24 [0.58, 1.69]	0/7 / 2/7	1.34 / 1.01 / 0.56	0/3 / 0/3
DS03	14 (1)	no	0.43	[0.31, 0.74]		0.47	[0.37, 0.76]	0.78 [0.30, 0.97]	3/7 / 6/7	0.77 / 0.90 / 0.76	3/3 / 2/3
DS03	15 (2)	yes	0.62	[0.38, 1.15]		0.54	[0.25, 1.28]	0.61 [0.49, 0.66]	4/7 / 4/7	0.58 / 0.36 / 0.78	1/3 / 2/3
DS01	7 (1)	yes	0.90	[0.68, 1.57]		0.96	[0.75, 1.05]	0.62 [0.43, 0.92]	4/7 / 3/7	0.68 / 0.62 / 0.56	1/3 / 1/3
DS01	8 (2)	yes	0.69	[0.58, 1.50]		0.44	[0.11, 0.79]	0.52 [0.34, 2.57]	1/7 / 1/7	0.97 / 1.03 / 1.62	2/3 / 2/3
DS01	9 (1)	yes	0.55	[0.38, 0.97]		0.56	[0.46, 1.31]	0.64 [0.23, 0.73]	4/7 / 3/7	0.44 / 0.84 / 0.41	2/3 / 1/3
DS01	10 (3)	yes	0.64	[0.49, 0.99]		0.23	[0.11, 1.30]	1.16 [0.66, 1.88]	1/7 / 6/7	1.53 / 0.63 / 0.99	0/3 / 1/3
DS04	7 (2)	yes	0.74	[0.42, 0.91]		0.55	[0.39, 0.59]	0.28 [0.17, 0.66]	0/7 / 0/7	0.83 / 1.77 / 1.19	3/3 / 3/3
DS04	8 (2)	yes	0.74	[0.41, 0.78]		0.57	[0.33, 0.66]	0.39 [0.35, 0.46]	0/7 / 0/7	0.90 / 1.00 / 0.68	3/3 / 0/3
DS04	9 (2)	yes	0.67	[0.32, 0.88]		0.59	[0.37, 0.67]	0.35 [0.05, 0.57]	1/7 / 1/7	0.77 / 1.00 / 0.49	3/3 / 0/3
DS04	10 (3)	yes	0.78	[0.47, 1.28]		0.53	[0.33, 0.55]	0.42 [0.33, 0.45]	0/7 / 0/7	0.82 / 1.00 / 0.55	3/3 / 0/3
DS05	7 (2)	yes	0.97	[0.76, 2.27]		0.72	[0.67, 6.57]	2.96 [2.86, 24.17]	2/7 / 7/7	0.91 / 1.16 / 0.66	1/3 / 1/3
DS05	8 (1)	yes	0.65	[0.41, 1.07]		1.36	[1.20, 1.46]	0.56 [0.48, 0.63]	7/7 / 3/7	0.87 / 0.85 / 0.66	0/3 / 0/3
DS05	9 (3)	yes	0.90	[0.76, 4.95]		0.74	[0.35, 5.33]	3.38 [2.68, 21.82]	2/7 / 7/7	0.74 / 1.06 / 0.45	2/3 / 0/3
DS05	10 (1)	yes	0.50	[0.17, 1.49]		0.50	[0.36, 3.76]	2.33 [1.72, 9.59]	5/7 / 7/7	0.75 / 0.88 / 0.75	1/3 / 2/3
DS06	7 (2)	yes	0.83	[0.75, 2.44]		0.91	[0.69, 8.53]	4.92 [1.27, 27.22]	4/7 / 7/7	1.03 / 0.74 / 0.74	1/3 / 1/3
DS06	8 (1)	yes	0.62	[0.32, 0.91]		1.24	[0.52, 1.40]	0.64 [0.47, 0.75]	7/7 / 5/7	0.86 / 0.66 / 0.62	2/3 / 1/3
DS06	9 (3)	yes	0.91	[0.83, 4.86]		0.62	[0.40, 1.14]	4.38 [2.93, 22.86]	3/7 / 7/7	0.69 / 1.09 / 0.80	2/3 / 1/3
DS06	10 (1)	yes	0.53	[0.16, 1.46]		0.52	[0.27, 2.61]	2.89 [2.25, 9.33]	4/7 / 7/7	0.61 / 0.58 / 0.66	1/3 / 2/3
DS07	7 (2)	yes	0.66	[0.60, 0.85]		0.51	[0.32, 0.77]	0.69 [0.56, 1.72]	0/7 / 4/7	0.89 / 0.83 / 0.57	1/3 / 0/3
DS07	8 (1)	yes	0.83	[0.53, 1.00]		1.35	[1.22, 1.58]	0.61 [0.37, 0.68]	7/7 / 1/7	0.55 / 0.82 / 0.64	1/3 / 1/3
DS07	9 (3)	yes	0.86	[0.75, 0.88]		0.64	[0.49, 0.79]	0.71 [0.43, 1.15]	1/7 / 3/7	0.92 / 1.10 / 0.43	2/3 / 1/3
DS07	10 (1)	yes	0.46	[0.37, 0.59]		0.60	[0.28, 0.67]	1.06 [0.97, 1.57]	6/7 / 7/7	0.57 / 0.74 / 0.97	2/3 / 3/3
DS08a	10 (1)	yes	0.77	[0.58, 1.48]		0.56	[0.37, 0.83]	0.77 [0.45, 0.86]	0/7 / 3/7	0.48 / 0.58 / 0.33	2/3 / 1/3
DS08a	11 (2)	yes	2.01	[1.60, 2.65]		1.50	[1.17, 1.84]	0.75 [0.70, 1.02]	0/7 / 0/7	0.77 / 0.63 / 0.42	0/3 / 0/3
DS08a	12 (3)	yes	1.09	[0.93, 1.33]		0.85	[0.72, 1.23]	0.69 [0.49, 0.88]	1/7 / 0/7	0.86 / 0.79 / 1.16	0/3 / 1/3
DS08a	13 (3)	yes	0.73	[0.37, 1.30]		0.57	[0.37, 0.68]	3.36 [0.48, 4.76]	0/7 / 7/7	2.44 / 3.84 / 3.09	2/3 / 1/3
DS08a	14 (3)	yes	5.00	[4.19, 7.50]		5.39	[4.91, 9.39]	11.40 [9.70, 11.80]	7/7 / 7/7	6.72 / 7.56 / 3.98	1/3 / 0/3
DS08a	15 (1)	yes	0.70	[0.44, 2.06]		0.57	[0.29, 0.76]	0.93 [0.68, 1.49]	1/7 / 3/7	0.66 / 0.52 / 0.29	2/3 / 0/3
DS08c	7 (2)	no	0.68	[0.65, 0.81]		0.86	[0.70, 0.87]	2.08 [0.55, 4.82]	7/7 / 6/7	0.79 / 0.88 / 0.89	3/3 / 2/3
DS08c	8 (2)	no	0.94	[0.91, 0.96]		0.96	[0.92, 0.96]	2.98 [0.86, 6.39]	7/7 / 6/7	0.92 / 0.99 / 0.23	3/3 / 0/3
DS08c	9 (2)	no	0.82	[0.79, 0.86]		0.89	[0.82, 0.89]	0.95 [0.82, 3.29]	7/7 / 7/7	0.93 / 1.00 / 0.53	3/3 / 0/3
DS08c	10 (2)	no	0.71	[0.55, 0.90]		0.67	[0.64, 0.74]	0.70 [0.63, 1.00]	3/7 / 5/7	0.83 / 1.00 / 0.46	3/3 / 0/3

Source: results/extension/⟨subset⟩/audit/paired_transport_engines.csv.

A.3 Uncertainty and target sensitivity

Table S8: Per-subset hierarchical bootstrap U-EXT1 (engine then flight, 2,000 replicates, thresholds re-estimated in every replicate) for C minus P at the 1% target.

Subset	Run	$\Delta A1$ (C - P) pp [95% interval]	$\Delta A2$ (C - P) pp [95% interval]	$\Delta ME-A2$ (C - P) pp [95% interval]
DS02	PCA	-3.01 [-4.16, +0.26]	-2.06 [-3.17, +0.95]	-1.15 [-2.97, +2.77]
DS02	IF s0	-1.13 [-2.97, +3.66]	-0.49 [-2.19, +3.33]	+0.93 [-1.83, +4.99]
DS02	IF s1	-1.85 [-3.25, +2.79]	-1.13 [-2.42, +2.50]	+0.37 [-2.11, +4.15]
DS02	IF s2	-1.40 [-2.96, +4.03]	-0.74 [-2.12, +3.64]	+0.85 [-1.83, +5.38]
DS02	LSTM s0	-1.32 [-2.00, +3.30]	-0.69 [-1.42, +3.21]	+0.63 [-1.12, +5.18]
DS02	LSTM s1	-1.31 [-2.17, +2.80]	-0.82 [-1.56, +2.40]	+0.36 [-1.25, +2.97]
DS02	LSTM s2	-1.07 [-2.06, +3.32]	-0.60 [-1.54, +3.10]	+0.64 [-1.31, +5.24]
DS02	CVAE s0	-0.87 [-1.97, +2.97]	-0.20 [-1.08, +3.25]	+0.71 [-0.74, +4.54]
DS02	CVAE s1	-0.52 [-1.95, +3.22]	-0.19 [-1.16, +3.39]	+0.10 [-1.11, +3.85]
DS02	CVAE s2	-0.46 [-1.91, +4.39]	-0.06 [-1.03, +4.26]	+0.37 [-0.85, +5.46]
DS03	PCA	-0.48 [-1.20, +2.53]	-0.12 [-0.60, +2.72]	+0.01 [-0.40, +2.39]
DS03	IF s0	-1.36 [-2.01, +4.19]	-0.51 [-1.10, +4.62]	-0.30 [-0.90, +4.64]
DS03	IF s1	-1.33 [-2.01, +4.01]	-0.46 [-1.06, +4.51]	-0.33 [-0.88, +4.28]
DS03	IF s2	-1.25 [-1.94, +4.70]	-0.39 [-0.96, +5.08]	-0.25 [-0.79, +5.16]
DS03	LSTM s0	-1.13 [-1.82, +1.76]	-0.40 [-0.83, +1.98]	0.00 [-0.69, +2.19]
DS03	LSTM s1	-1.15 [-1.91, +1.79]	-0.38 [-0.86, +2.00]	-0.03 [-0.66, +2.16]
DS03	LSTM s2	-1.15 [-1.76, +2.54]	-0.40 [-0.83, +2.74]	-0.06 [-0.66, +2.83]
DS03	CVAE s0	-1.22 [-2.57, +9.45]	-0.39 [-1.05, +10.31]	-0.05 [-0.74, +8.55]
DS03	CVAE s1	-0.02 [-0.89, +2.32]	+0.12 [-0.42, +2.11]	+0.20 [-0.31, +1.88]
DS03	CVAE s2	-0.66 [-1.46, +1.19]	-0.08 [-0.81, +1.30]	+0.10 [-0.54, +1.34]
DS01	PCA	-0.38 [-1.40, +1.41]	+0.08 [-0.82, +1.64]	+0.12 [-0.59, +1.60]
DS01	IF s0	-1.30 [-1.71, +0.81]	-0.76 [-1.12, +0.99]	-0.50 [-0.92, +1.04]
DS01	IF s1	-1.16 [-1.65, +0.83]	-0.70 [-1.06, +1.03]	-0.48 [-0.88, +1.09]
DS01	IF s2	-1.20 [-1.59, +0.90]	-0.73 [-1.08, +1.04]	-0.51 [-0.88, +1.16]
DS01	LSTM s0	-0.33 [-0.89, +1.09]	-0.08 [-0.46, +0.98]	-0.01 [-0.35, +1.02]
DS01	LSTM s1	-0.39 [-0.90, +1.15]	-0.13 [-0.49, +1.05]	-0.06 [-0.38, +1.13]
DS01	LSTM s2	-0.37 [-0.85, +1.38]	-0.13 [-0.46, +1.26]	-0.02 [-0.36, +1.26]
DS01	CVAE s0	-0.52 [-1.18, +0.61]	-0.15 [-0.70, +0.70]	-0.21 [-0.60, +0.54]
DS01	CVAE s1	-0.36 [-1.24, +1.47]	+0.02 [-0.76, +1.35]	-0.04 [-0.55, +1.26]
DS01	CVAE s2	-0.56 [-1.36, +1.04]	-0.18 [-0.76, +0.91]	-0.10 [-0.64, +0.96]
DS04	PCA	-0.10 [-0.62, +0.62]	-0.05 [-0.32, +0.17]	-0.05 [-0.29, +0.16]
DS04	IF s0	-0.90 [-1.76, +0.32]	-0.27 [-1.02, -0.03]	-0.34 [-0.89, +0.03]
DS04	IF s1	-0.91 [-1.78, +0.29]	-0.27 [-1.01, -0.05]	-0.35 [-0.90, -0.03]
DS04	IF s2	-0.86 [-1.73, +0.38]	-0.26 [-0.99, -0.01]	-0.33 [-0.85, +0.04]
DS04	LSTM s0	-0.34 [-1.09, +0.50]	-0.17 [-0.61, 0.00]	-0.19 [-0.58, 0.00]
DS04	LSTM s1	-0.26 [-0.96, +0.51]	-0.12 [-0.52, +0.05]	-0.12 [-0.50, +0.04]
DS04	LSTM s2	-0.33 [-1.04, +0.45]	-0.15 [-0.55, +0.01]	-0.15 [-0.53, 0.00]
DS04	CVAE s0	+1.28 [-1.09, +2.77]	+0.15 [-0.60, +1.66]	+0.26 [-0.53, +1.54]
DS04	CVAE s1	+1.51 [-0.85, +4.61]	+0.64 [-0.44, +3.34]	+0.85 [-0.38, +3.38]
DS04	CVAE s2	+1.37 [-0.82, +3.04]	+0.27 [-0.43, +1.78]	+0.42 [-0.34, +1.83]
DS05	PCA	+0.90 [-3.10, +9.92]	+2.08 [-1.46, +10.39]	+1.82 [-1.19, +9.77]
DS05	IF s0	-0.15 [-1.53, +1.22]	+0.02 [-1.03, +1.15]	+0.02 [-0.72, +1.15]
DS05	IF s1	-0.05 [-1.57, +1.22]	+0.03 [-1.13, +1.11]	+0.05 [-0.76, +1.04]
DS05	IF s2	-0.36 [-1.52, +1.11]	-0.05 [-1.06, +1.17]	-0.07 [-0.75, +1.16]
DS05	LSTM s0	-0.66 [-1.01, +1.15]	-0.14 [-0.48, +1.09]	+0.17 [-0.33, +1.17]
DS05	LSTM s1	-0.56 [-0.92, +1.12]	-0.09 [-0.42, +1.06]	+0.25 [-0.26, +1.16]
DS05	LSTM s2	-0.19 [-0.79, +1.36]	+0.18 [-0.32, +1.25]	+0.48 [-0.19, +1.28]
DS05	CVAE s0	-0.19 [-1.39, +1.26]	+0.23 [-0.88, +1.37]	-0.08 [-0.72, +1.10]
DS05	CVAE s1	+0.96 [-0.31, +2.33]	+0.82 [-0.16, +1.93]	+0.33 [-0.23, +1.82]
DS05	CVAE s2	-3.62 [-7.72, +4.74]	-2.89 [-6.63, +4.09]	-1.79 [-5.35, +3.21]
DS06	PCA	+2.00 [-2.76, +12.95]	+3.05 [-1.03, +13.26]	+2.50 [-0.92, +12.11]
DS06	IF s0	-0.43 [-1.74, +0.98]	-0.02 [-1.33, +0.75]	+0.03 [-1.03, +0.83]
DS06	IF s1	-0.05 [-1.68, +1.31]	+0.02 [-1.26, +0.88]	+0.03 [-0.95, +0.90]
DS06	IF s2	-0.32 [-1.60, +1.02]	-0.01 [-1.25, +0.85]	0.00 [-0.94, +0.89]
DS06	LSTM s0	-0.55 [-0.95, +1.05]	-0.13 [-0.44, +0.92]	+0.02 [-0.30, +1.11]
DS06	LSTM s1	-0.32 [-0.79, +1.15]	+0.05 [-0.36, +1.23]	+0.15 [-0.21, +1.07]
DS06	LSTM s2	-0.42 [-0.83, +1.17]	+0.02 [-0.38, +1.08]	+0.26 [-0.25, +1.19]
DS06	CVAE s0	-0.10 [-0.80, +2.59]	+0.03 [-0.49, +2.27]	+0.13 [-0.32, +2.67]
DS06	CVAE s1	+0.06 [-0.82, +1.43]	+0.32 [-0.36, +1.32]	+0.17 [-0.29, +1.33]
DS06	CVAE s2	+2.28 [-1.14, +6.21]	+2.07 [-0.65, +6.00]	+0.77 [-0.56, +4.19]
DS07	PCA	-0.66 [-1.06, +3.12]	-0.43 [-0.52, +3.20]	+0.05 [-0.30, +2.42]
DS07	IF s0	+0.10 [-1.35, +1.29]	+0.07 [-0.97, +1.27]	0.00 [-0.67, +0.94]
DS07	IF s1	+0.11 [-1.24, +1.26]	+0.06 [-0.86, +1.25]	-0.01 [-0.54, +0.94]
DS07	IF s2	+0.15 [-1.10, +1.46]	+0.05 [-0.69, +1.44]	+0.01 [-0.38, +1.11]
DS07	LSTM s0	-0.46 [-0.67, +1.39]	-0.27 [-0.39, +1.19]	+0.17 [-0.16, +1.12]
DS07	LSTM s1	-0.43 [-0.60, +1.21]	-0.24 [-0.37, +1.05]	+0.09 [-0.18, +0.97]

Subset	Run	$\Delta A1$ (C - P) pp [95% interval]	$\Delta A2$ (C - P) pp [95% interval]	$\Delta ME-A2$ (C - P) pp [95% interval]
DS07	LSTM s2	-0.36 [-0.64, +1.53]	-0.07 [-0.38, +1.40]	+0.19 [-0.17, +1.13]
DS07	CVAE s0	-0.59 [-1.55, +1.29]	-0.10 [-0.99, +1.19]	-0.21 [-0.76, +1.33]
DS07	CVAE s1	+0.76 [-0.50, +1.91]	+0.62 [-0.17, +1.59]	+0.34 [-0.14, +1.38]
DS07	CVAE s2	+0.57 [-0.74, +1.86]	+0.65 [-0.41, +1.68]	+0.08 [-0.34, +1.27]
DS08a	PCA	-0.03 [-1.15, +1.40]	-0.04 [-0.74, +1.34]	-0.09 [-0.49, +0.85]
DS08a	IF s0	+0.35 [-1.88, +2.31]	+0.81 [-1.07, +2.59]	-0.19 [-0.97, +1.54]
DS08a	IF s1	+0.45 [-1.83, +2.27]	+0.83 [-1.09, +2.53]	-0.22 [-1.00, +1.24]
DS08a	IF s2	+0.52 [-1.81, +2.32]	+0.80 [-1.08, +2.53]	-0.20 [-0.94, +1.43]
DS08a	LSTM s0	-0.03 [-0.75, +1.35]	+0.07 [-0.32, +0.95]	-0.03 [-0.21, +0.63]
DS08a	LSTM s1	-0.02 [-0.83, +1.63]	+0.11 [-0.38, +1.30]	-0.05 [-0.22, +0.94]
DS08a	LSTM s2	+0.19 [-0.76, +1.83]	+0.24 [-0.40, +1.69]	-0.01 [-0.21, +1.19]
DS08a	CVAE s0	-0.01 [-2.23, +2.20]	+0.69 [-1.46, +2.83]	+0.14 [-1.52, +1.96]
DS08a	CVAE s1	-0.60 [-1.38, +0.85]	-0.23 [-0.71, +0.94]	-0.07 [-0.45, +0.63]
DS08a	CVAE s2	-0.50 [-2.44, +2.40]	-0.63 [-2.17, +2.22]	-0.49 [-1.71, +1.74]
DS08c	PCA	0.00 [-0.60, +2.70]	+0.03 [-0.30, +2.64]	-0.02 [-0.29, +2.48]
DS08c	IF s0	-0.01 [-1.55, +1.84]	+0.04 [-0.82, +1.40]	0.00 [-0.67, +1.41]
DS08c	IF s1	-0.06 [-1.38, +1.86]	+0.04 [-0.75, +1.43]	0.00 [-0.66, +1.41]
DS08c	IF s2	+0.02 [-1.49, +1.65]	+0.06 [-0.90, +1.15]	+0.02 [-0.79, +1.16]
DS08c	LSTM s0	+0.28 [-0.60, +1.12]	+0.13 [-0.25, +0.74]	+0.07 [-0.31, +0.68]
DS08c	LSTM s1	+0.43 [-0.76, +1.23]	+0.12 [-0.36, +0.91]	+0.09 [-0.32, +0.78]
DS08c	LSTM s2	+0.44 [-0.54, +1.25]	+0.13 [-0.23, +0.85]	+0.08 [-0.20, +0.76]
DS08c	CVAE s0	+0.70 [+0.09, +1.35]	+0.11 [-0.05, +0.39]	+0.09 [-0.07, +0.38]
DS08c	CVAE s1	+0.50 [-0.63, +1.23]	+0.14 [-0.56, +0.36]	+0.11 [-0.46, +0.38]
DS08c	CVAE s2	+1.35 [-0.03, +1.90]	+0.17 [-0.01, +0.79]	+0.10 [-0.05, +0.70]

Source: results/extension/(subset)/audit/bootstrap_uext1_ci.csv. Percentile intervals; Q has point estimates only (the quantile-regression fit is not repeated per replicate).

Table S9: Cross-dataset hierarchical bootstrap U-EXT2 (family, then subset, then a U-EXT1 replicate; 2,000 replicates; five top-level units, so intervals are crude) at the 1% target.

Detector family	Quantity	Median over families (pp)	95% interval (pp)	Subsets negative
PCA	$\Delta A1$ (C - P)	-0.03	[-0.55, +1.32]	5/7
PCA	$\Delta A2$ (C - P)	+0.03	[-0.29, +1.55]	3/7
PCA	$\Delta ME-A2$ (C - P)	-0.02	[-0.20, +1.41]	3/7
IF	$\Delta A1$ (C - P)	-0.11	[-1.15, +0.48]	5/7
IF	$\Delta A2$ (C - P)	+0.02	[-0.59, +0.47]	3/7
IF	$\Delta ME-A2$ (C - P)	-0.20	[-0.53, +0.22]	3/7
LSTM	$\Delta A1$ (C - P)	-0.31	[-0.52, +0.53]	5/7
LSTM	$\Delta A2$ (C - P)	-0.08	[-0.25, +0.32]	5/7
LSTM	$\Delta ME-A2$ (C - P)	-0.03	[-0.21, +0.34]	3/7
CVAE	$\Delta A1$ (C - P)	+0.01	[-0.56, +0.99]	3/7
CVAE	$\Delta A2$ (C - P)	+0.14	[-0.29, +0.56]	3/7
CVAE	$\Delta ME-A2$ (C - P)	-0.03	[-0.22, +0.46]	3/7
CVAE vs residual	CVAE - best residual ME-A2 (P)	+0.52	[+0.05, +1.31]	0/7

Source: results/extension/summary/uext2_cross_dataset.csv.

Table S10: Target sensitivity: medians over the seven residual runs of pooled A2 and ME-A2 at the 0.5%, 1% and 2% targets, with ME-A2 relative to α .

Subset	Arm	$\alpha = 0.5\%$ pooled A2 / ME-A2 (pp)	$\alpha = 0.5\%$ ME-A2 / α	$\alpha = 1\%$ pooled A2 / ME-A2 (pp)	$\alpha = 1\%$ ME-A2 / α	$\alpha = 2\%$ pooled A2 / ME-A2 (pp)	$\alpha = 2\%$ ME-A2 / α
DS02	P	0.57 / 0.54	1.08	1.34 / 1.38	1.38	4.82 / 5.17	2.59
DS02	C	0.30 / 1.15	2.30	0.46 / 1.77	1.77	0.97 / 3.35	1.67
DS02	Q	1.04 / 1.46	2.92	1.12 / 1.60	1.60	0.99 / 1.56	0.78
DS03	P	0.53 / 0.62	1.23	1.17 / 1.21	1.21	2.38 / 2.46	1.23
DS03	C	0.37 / 0.55	1.09	0.80 / 1.03	1.03	1.97 / 2.58	1.29
DS03	Q	0.50 / 0.89	1.78	0.87 / 1.74	1.74	1.77 / 2.48	1.24
DS01	P	0.23 / 0.46	0.91	0.89 / 0.91	0.91	1.70 / 2.02	1.01
DS01	C	0.28 / 0.39	0.79	0.35 / 0.52	0.52	0.57 / 1.13	0.57
DS01	Q	0.26 / 0.53	1.07	0.37 / 0.75	0.75	0.52 / 1.30	0.65
DS04	P	0.42 / 0.45	0.90	0.71 / 0.71	0.71	1.00 / 1.00	0.50
DS04	C	0.28 / 0.30	0.60	0.57 / 0.57	0.57	0.84 / 0.84	0.42
DS04	Q	0.17 / 0.20	0.41	0.33 / 0.34	0.34	0.49 / 0.52	0.26
DS05	P	0.24 / 0.43	0.86	0.57 / 0.72	0.72	1.36 / 1.39	0.70
DS05	C	0.18 / 0.53	1.07	0.54 / 0.79	0.79	1.15 / 1.29	0.65

Subset	Arm	$\alpha = 0.5\%$	$\alpha = 0.5\%$	$\alpha = 1\%$	$\alpha = 1\%$	$\alpha = 2\%$	$\alpha = 2\%$
		pooled A2 / ME-A2 (pp)	ME-A2 / α	pooled A2 / ME-A2 (pp)	ME-A2 / α	pooled A2 / ME-A2 (pp)	ME-A2 / α
DS05	Q	0.92 / 1.10	2.21	2.06 / 2.60	2.60	4.66 / 4.48	2.24
DS06	P	0.26 / 0.41	0.82	0.56 / 0.71	0.71	1.39 / 1.48	0.74
DS06	C	0.24 / 0.37	0.73	0.57 / 0.74	0.74	1.27 / 1.43	0.71
DS06	Q	1.46 / 1.75	3.49	3.08 / 3.12	3.12	5.55 / 5.19	2.59
DS07	P	0.20 / 0.43	0.87	0.56 / 0.70	0.70	1.13 / 1.34	0.67
DS07	C	0.16 / 0.40	0.79	0.24 / 0.78	0.78	0.50 / 1.23	0.62
DS07	Q	0.30 / 0.53	1.05	0.48 / 0.80	0.80	1.10 / 1.41	0.71
DS08a	P	0.77 / 1.26	2.53	1.11 / 1.65	1.65	1.95 / 2.71	1.36
DS08a	C	1.05 / 1.27	2.55	1.19 / 1.65	1.65	1.42 / 2.17	1.09
DS08a	Q	1.89 / 1.94	3.88	2.98 / 2.94	2.94	3.54 / 3.94	1.97
DS08c	P	0.40 / 0.43	0.86	0.73 / 0.81	0.81	1.58 / 1.62	0.81
DS08c	C	0.44 / 0.44	0.88	0.85 / 0.84	0.84	1.59 / 1.57	0.79
DS08c	Q	0.40 / 0.61	1.22	1.81 / 2.11	2.11	1.52 / 2.43	1.22

Source: results/extension/(subset)/audit/transport_summary.csv.

A.4 Calibration-fleet composition

Table S11: Calibration-fleet composition at fixed volume N, per detector run and arm (1% target; draw means over five nested subsamples). Δcov is the within-engine coverage effect CE1 (positive = calibrating on the engine's own class transports better); the balance check compares the class-balanced design with the mean single-engine design.

Subset	Run	Arm	Eligible engines	Median (pp)	Δcov	Supports coverage	ME-A2 balanced (pp)	Mean ME-A2 single (pp)	Balanced – best single (pp)	Balance check
DS01	CVAE s0	C	4	+0.22	yes		0.64	0.94	−0.16	yes
DS01	CVAE s0	P	4	+0.17	yes		0.86	0.88	+0.06	yes
DS01	CVAE s1	C	4	+0.25	yes		0.84	1.22	−0.12	yes
DS01	CVAE s1	P	4	+0.04	yes		0.93	0.95	+0.21	yes
DS01	CVAE s2	C	4	+0.55	yes		0.57	1.08	−0.34	yes
DS01	CVAE s2	P	4	+0.04	yes		0.80	0.88	+0.13	yes
DS01	IF s0	C	4	+1.01	yes		0.44	1.59	−0.27	yes
DS01	IF s0	P	4	+0.39	yes		0.93	1.07	+0.17	yes
DS01	IF s1	C	4	+1.14	yes		0.42	1.59	−0.20	yes
DS01	IF s1	P	4	+0.40	yes		0.84	0.97	+0.19	yes
DS01	IF s2	C	4	+1.31	yes		0.41	1.52	−0.25	yes
DS01	IF s2	P	4	+0.38	yes		0.89	1.02	+0.18	yes
DS01	LSTM s0	C	4	+1.04	yes		0.48	1.16	+0.14	yes
DS01	LSTM s0	P	4	−0.04	no		0.51	0.60	+0.05	yes
DS01	LSTM s1	C	4	+1.30	yes		0.46	1.28	+0.09	yes
DS01	LSTM s1	P	4	−0.09	no		0.55	0.61	+0.06	yes
DS01	LSTM s2	C	4	+1.41	yes		0.47	1.35	+0.13	yes
DS01	LSTM s2	P	4	−0.07	no		0.55	0.62	+0.08	yes
DS01	PCA	C	4	+0.51	yes		1.04	1.69	+0.34	yes
DS01	PCA	P	4	+0.08	yes		0.89	1.02	+0.11	yes
DS04	CVAE s0	C	4	+0.37	yes		1.01	0.94	+0.40	no
DS04	CVAE s0	P	4	+0.06	yes		0.94	0.99	+0.07	yes
DS04	CVAE s1	C	4	+0.49	yes		1.97	1.69	+0.62	no
DS04	CVAE s1	P	4	−0.03	no		0.96	1.37	−0.04	yes
DS04	CVAE s2	C	4	+0.14	yes		1.20	0.90	+0.54	no
DS04	CVAE s2	P	4	+0.01	yes		0.92	0.79	+0.25	no
DS04	IF s0	C	4	+0.29	yes		0.62	0.54	+0.25	no
DS04	IF s0	P	4	+0.04	yes		0.95	0.96	+0.06	yes
DS04	IF s1	C	4	+0.25	yes		0.63	0.57	+0.20	no
DS04	IF s1	P	4	+0.06	yes		0.98	0.99	+0.08	yes
DS04	IF s2	C	4	+0.29	yes		0.62	0.57	+0.20	no
DS04	IF s2	P	4	+0.04	yes		0.95	0.96	+0.07	yes
DS04	LSTM s0	C	4	+0.11	yes		0.59	0.46	+0.27	no
DS04	LSTM s0	P	4	−0.05	no		0.72	0.77	+0.01	yes
DS04	LSTM s1	C	4	+0.15	yes		0.60	0.44	+0.33	no
DS04	LSTM s1	P	4	+0.02	yes		0.67	0.70	+0.06	yes
DS04	LSTM s2	C	4	+0.14	yes		0.62	0.47	+0.31	no
DS04	LSTM s2	P	4	−0.01	no		0.73	0.76	+0.04	yes
DS04	PCA	C	4	+0.08	yes		0.43	0.40	+0.21	no
DS04	PCA	P	4	+0.03	yes		0.44	0.50	+0.05	yes
DS05	CVAE s0	C	4	+0.48	yes		0.83	1.03	+0.14	yes
DS05	CVAE s0	P	4	+0.07	yes		0.96	1.11	+0.12	yes
DS05	CVAE s1	C	4	+0.77	yes		1.01	1.44	+0.29	yes
DS05	CVAE s1	P	4	+0.19	yes		0.84	0.89	+0.17	yes

Subset	Run	Arm	Eligible engines	Median Δcov (pp)	Supports coverage	ME-A2 balanced (pp)	Mean ME-A2 single (pp)	Balanced – best single (pp)	Balance check
DS05	CVAE s2	C	4	+3.75	yes	3.46	11.32	+1.47	yes
DS05	CVAE s2	P	4	+2.31	yes	5.39	10.95	+1.98	yes
DS05	IF s0	C	4	+0.36	yes	0.72	1.03	−0.14	yes
DS05	IF s0	P	4	+0.15	yes	0.72	0.99	−0.05	yes
DS05	IF s1	C	4	+0.38	yes	0.76	1.06	−0.15	yes
DS05	IF s1	P	4	+0.31	yes	0.70	1.13	−0.09	yes
DS05	IF s2	C	4	+0.35	yes	0.65	1.08	−0.17	yes
DS05	IF s2	P	4	+0.19	yes	0.74	1.03	+0.01	yes
DS05	LSTM s0	C	4	+0.86	yes	0.74	0.96	+0.09	yes
DS05	LSTM s0	P	4	+0.09	yes	0.64	0.70	+0.04	yes
DS05	LSTM s1	C	4	+0.78	yes	0.79	1.00	+0.22	yes
DS05	LSTM s1	P	4	+0.13	yes	0.61	0.66	+0.02	yes
DS05	LSTM s2	C	4	+0.88	yes	1.11	1.26	+0.05	yes
DS05	LSTM s2	P	4	+0.05	yes	0.70	0.78	+0.13	yes
DS05	PCA	C	4	+1.19	yes	3.49	5.38	+1.03	yes
DS05	PCA	P	4	+0.41	yes	2.30	2.53	+0.25	yes
DS06	CVAE s0	C	4	+1.36	yes	0.82	1.77	+0.06	yes
DS06	CVAE s0	P	4	+0.33	yes	0.75	0.92	+0.09	yes
DS06	CVAE s1	C	4	+0.40	yes	0.84	1.06	+0.13	yes
DS06	CVAE s1	P	4	+0.16	yes	0.63	0.67	+0.02	yes
DS06	CVAE s2	C	4	+0.75	yes	2.24	2.68	−0.32	yes
DS06	CVAE s2	P	4	+0.11	yes	1.36	1.51	+0.48	yes
DS06	IF s0	C	4	+0.29	yes	0.74	0.86	+0.08	yes
DS06	IF s0	P	4	+0.33	yes	0.70	1.14	−0.04	yes
DS06	IF s1	C	4	+0.26	yes	0.74	0.90	0.00	yes
DS06	IF s1	P	4	+0.27	yes	0.71	1.14	−0.07	yes
DS06	IF s2	C	4	+0.32	yes	0.71	0.88	+0.02	yes
DS06	IF s2	P	4	+0.22	yes	0.72	1.02	−0.03	yes
DS06	LSTM s0	C	4	+0.65	yes	0.68	0.87	+0.08	yes
DS06	LSTM s0	P	4	+0.13	yes	0.62	0.69	−0.02	yes
DS06	LSTM s1	C	4	+0.62	yes	0.74	0.94	+0.08	yes
DS06	LSTM s1	P	4	+0.08	yes	0.58	0.66	+0.05	yes
DS06	LSTM s2	C	4	+0.77	yes	0.95	1.08	+0.03	yes
DS06	LSTM s2	P	4	+0.08	yes	0.72	0.78	+0.13	yes
DS06	PCA	C	4	+0.79	yes	5.05	5.97	+2.60	yes
DS06	PCA	P	4	+0.08	yes	2.45	2.46	+0.60	yes
DS07	CVAE s0	C	4	+1.07	yes	0.71	1.20	+0.02	yes
DS07	CVAE s0	P	4	+0.09	yes	0.98	1.04	+0.15	yes
DS07	CVAE s1	C	4	+0.11	yes	1.26	1.20	+0.24	no
DS07	CVAE s1	P	4	+0.07	yes	0.83	0.83	+0.15	no
DS07	CVAE s2	C	4	+0.32	yes	0.85	1.00	+0.07	yes
DS07	CVAE s2	P	4	+0.08	yes	0.77	0.78	+0.10	yes
DS07	IF s0	C	4	+0.42	yes	0.78	0.95	−0.08	yes
DS07	IF s0	P	4	+0.21	yes	0.77	1.01	+0.03	yes
DS07	IF s1	C	4	+0.41	yes	0.77	0.93	−0.08	yes
DS07	IF s1	P	4	+0.16	yes	0.78	0.95	+0.03	yes
DS07	IF s2	C	4	+0.42	yes	0.79	0.96	−0.03	yes
DS07	IF s2	P	4	+0.05	yes	0.77	0.85	+0.04	yes
DS07	LSTM s0	C	4	+0.97	yes	0.71	1.00	+0.07	yes
DS07	LSTM s0	P	4	+0.07	yes	0.57	0.67	0.00	yes
DS07	LSTM s1	C	4	+0.72	yes	0.79	0.96	+0.12	yes
DS07	LSTM s1	P	4	−0.07	no	0.70	0.73	+0.05	yes
DS07	LSTM s2	C	4	+0.96	yes	0.84	1.06	+0.15	yes
DS07	LSTM s2	P	4	+0.03	yes	0.68	0.77	−0.01	yes
DS07	PCA	C	4	+1.30	yes	0.59	2.00	−0.15	yes
DS07	PCA	P	4	+0.46	yes	0.64	1.06	+0.02	yes
DS08a	CVAE s0	C	6	+0.35	yes	2.17	3.31	+0.65	yes
DS08a	CVAE s0	P	6	−0.58	no	2.34	3.31	+0.93	yes
DS08a	CVAE s1	C	6	+0.36	yes	1.50	1.83	−0.06	yes
DS08a	CVAE s1	P	6	+0.17	yes	1.53	1.68	+0.08	yes
DS08a	CVAE s2	C	6	+2.46	yes	2.73	4.72	+1.02	yes
DS08a	CVAE s2	P	6	+1.19	yes	3.73	4.35	+1.01	yes
DS08a	IF s0	C	6	+0.51	yes	2.04	2.59	+0.43	yes
DS08a	IF s0	P	6	−0.14	no	2.06	2.33	+0.56	yes
DS08a	IF s1	C	6	+0.55	yes	2.21	2.85	+0.39	yes
DS08a	IF s1	P	6	−0.08	no	2.35	2.61	+0.50	yes
DS08a	IF s2	C	6	+0.46	yes	2.18	2.79	+0.40	yes
DS08a	IF s2	P	6	−0.15	no	2.25	2.50	+0.71	yes
DS08a	LSTM s0	C	6	+0.67	yes	1.48	1.92	+0.07	yes
DS08a	LSTM s0	P	6	+0.11	yes	1.51	1.78	+0.03	yes
DS08a	LSTM s1	C	6	+0.66	yes	1.46	2.09	+0.06	yes
DS08a	LSTM s1	P	6	+0.01	yes	1.51	1.75	+0.04	yes
DS08a	LSTM s2	C	6	+0.76	yes	1.58	2.20	0.00	yes

Subset	Run	Arm	Eligible engines	Median Δ cov (pp)	Supports coverage	ME-A2 balanced (pp)	Mean ME-A2 single (pp)	Balanced – best single (pp)	Balance check
DS08a	LSTM s2	P	6	+0.12	yes	1.59	1.77	+0.10	yes
DS08a	PCA	C	6	+0.73	yes	1.53	2.07	+0.19	yes
DS08a	PCA	P	6	−0.18	no	1.64	1.72	+0.37	yes

Source: results/extension/summary/composition_run_summary.csv. DS03 is a reference; DS02 and DS08c are not composition-eligible (single-class calibration pools).

Table S12: Composition bootstrap (family, then subset, then audit engine; 2,000 replicates) of the median over families of the family-median per-engine coverage effect (1% target).

Arm	Statistic	Point (pp)	95% interval (pp)	Engines	Families
P	median over families of family-median engine CE1 (median over runs)	+0.11	[−0.04, +0.18]	26	4
C	median over families of family-median engine CE1 (median over runs)	+0.55	[+0.18, +1.11]	26	4

Source: results/extension/summary/composition_bootstrap.csv.

Table S13: CE4, engine count versus class coverage: ME-A2 of a two-engine same-class design minus a two-engine mixed-class design sharing one engine (1% target; positive = the mixed design transports better).

Subset	Arm	Same-class design	Mixed design	Residual median (pp)	Residual runs same-class worse	CVAE median (pp)
DS04	C	same_class_3	mixed_23	+0.05	7/7	+0.10
DS04	P	same_class_3	mixed_23	+0.03	4/7	0.00
DS08a	C	same_class_1	mixed_12	+0.29	6/7	−0.09
DS08a	C	same_class_1	mixed_13	−0.06	3/7	+0.26
DS08a	C	same_class_2	balanced	+0.20	7/7	+0.59
DS08a	P	same_class_1	mixed_12	+0.31	6/7	−0.13
DS08a	P	same_class_1	mixed_13	−0.16	3/7	+0.07
DS08a	P	same_class_2	balanced	+0.07	7/7	+0.16

Source: results/extension/summary/composition_ce4.csv.

A.5 Alarm persistence, matched-burden delay and abnormal-state alarm rates

Table S14: Alarm persistence at the locked κ (1% target): pooled and worst-engine healthy-flight false-flag rates (nominal 5%), alarm events per healthy flight and the worst-engine persistence calibration error PA2 (relative to the calibration-set persistent-alarm rate).

Subset	Arm	Rule	Pooled FFR (%) residual median	WE-FFR (%) residual median [range]	Residual runs WE-FFR $\geq$ 10%	Events per healthy flight (median)	WE-PA2 (median)	CVAE WE-FFR (%) seed median
DS02	P	R0	11.8	20.0 [8.6, 25.7]	6/7	15.0	1.52	54.3
DS02	P	R1	13.2	20.0 [17.1, 22.9]	7/7	5.6	1.37	48.6
DS02	P	R2	11.8	17.4 [8.6, 22.9]	6/7	5.0	1.54	48.6
DS02	C	R0	13.2	22.9 [14.3, 37.1]	7/7	17.2	4.66	62.9
DS02	C	R1	18.4	34.3 [17.1, 40.0]	7/7	5.9	7.10	57.1
DS02	C	R2	13.2	22.9 [17.1, 31.4]	7/7	5.5	4.73	51.4
DS02	Q	R0	18.4	22.9 [11.4, 28.6]	7/7	28.4	2.12	28.6
DS02	Q	R1	22.4	28.6 [22.9, 51.4]	7/7	6.1	4.58	40.0
DS02	Q	R2	19.7	28.6 [22.9, 51.4]	7/7	6.8	3.43	34.3
DS03	P	R0	8.8	11.8 [8.3, 41.2]	6/7	41.9	2.23	11.8
DS03	P	R1	7.4	17.6 [11.1, 35.3]	7/7	8.3	2.11	5.9
DS03	P	R2	9.5	17.6 [11.1, 41.2]	7/7	8.9	2.29	11.8
DS03	C	R0	8.1	13.9 [11.1, 47.1]	7/7	37.4	2.06	16.7
DS03	C	R1	7.4	13.9 [5.9, 35.3]	4/7	8.3	2.17	11.1
DS03	C	R2	7.4	13.9 [11.1, 47.1]	7/7	8.5	2.16	11.8
DS03	Q	R0	6.1	23.5 [17.6, 29.4]	7/7	47.2	6.26	29.4
DS03	Q	R1	6.1	29.4 [23.5, 35.3]	7/7	9.1	9.43	23.5
DS03	Q	R2	6.8	23.5 [17.6, 29.4]	7/7	10.1	7.12	23.5
DS01	P	R0	5.4	14.3 [8.6, 16.0]	4/7	26.2	1.31	12.0
DS01	P	R1	3.6	11.4 [8.6, 17.6]	6/7	6.2	1.59	12.0
DS01	P	R2	5.4	14.3 [8.6, 17.6]	4/7	7.2	1.18	12.0
DS01	C	R0	5.4	11.4 [5.9, 16.0]	4/7	25.7	0.96	16.0
DS01	C	R1	3.6	5.9 [2.9, 16.0]	3/7	5.9	1.51	16.0
DS01	C	R2	3.6	11.4 [5.9, 16.0]	4/7	6.5	1.14	16.0
DS01	Q	R0	4.5	17.6 [8.0, 36.0]	5/7	32.4	1.19	20.0

Subset	Arm	Rule	Pooled FFR (%) residual median	WE-FFR (%) residual median [range]	Residual runs WE-FFR $\geq$ 10%	Events per healthy flight (median)	WE-PA2 (median)	CVAE WE-FFR (%) seed median
DS01	Q	R1	4.5	16.0 [8.0, 36.0]	6/7	7.1	3.06	24.0
DS01	Q	R2	3.6	16.0 [8.0, 36.0]	5/7	7.6	1.76	24.0
DS04	P	R0	3.8	9.5 [0.0, 19.0]	1/7	43.4	0.78	9.5
DS04	P	R1	8.9	14.3 [6.2, 23.8]	4/7	5.5	1.31	4.8
DS04	P	R2	8.9	19.0 [6.2, 19.0]	4/7	6.1	1.02	9.5
DS04	C	R0	6.3	9.5 [6.2, 23.8]	2/7	51.0	0.59	4.8
DS04	C	R1	8.9	14.3 [4.8, 23.8]	4/7	4.9	0.89	4.8
DS04	C	R2	7.6	12.5 [6.2, 19.0]	4/7	5.4	0.77	14.3
DS04	Q	R0	10.1	19.0 [4.8, 33.3]	4/7	87.4	0.44	9.5
DS04	Q	R1	5.1	9.5 [4.8, 19.0]	2/7	3.1	1.07	9.5
DS04	Q	R2	11.4	19.0 [4.8, 33.3]	4/7	5.7	1.04	9.5
DS05	P	R0	9.8	16.7 [11.1, 43.5]	7/7	29.4	0.97	17.1
DS05	P	R1	10.7	14.3 [8.7, 34.8]	5/7	6.5	1.15	13.0
DS05	P	R2	9.8	14.3 [11.4, 38.9]	7/7	7.1	0.92	11.4
DS05	C	R0	11.6	16.7 [11.1, 50.0]	7/7	29.2	1.36	17.1
DS05	C	R1	9.8	17.1 [13.0, 55.6]	7/7	6.7	2.19	13.0
DS05	C	R2	11.6	17.1 [4.3, 55.6]	6/7	7.0	1.52	13.0
DS05	Q	R0	18.8	33.3 [17.4, 94.4]	7/7	37.0	4.01	11.1
DS05	Q	R1	17.9	33.3 [26.1, 88.9]	7/7	10.2	7.14	8.7
DS05	Q	R2	19.6	33.3 [17.4, 88.9]	7/7	10.0	5.26	8.7
DS06	P	R0	10.0	14.3 [9.7, 56.2]	6/7	27.2	0.91	16.1
DS06	P	R1	8.0	14.3 [9.5, 42.9]	5/7	6.4	1.04	19.0
DS06	P	R2	9.0	14.3 [9.7, 50.0]	6/7	6.8	0.90	16.1
DS06	C	R0	11.0	14.3 [12.9, 57.1]	7/7	28.9	1.24	16.1
DS06	C	R1	9.0	16.1 [14.3, 56.2]	7/7	6.2	2.15	16.1
DS06	C	R2	10.0	14.3 [6.2, 57.1]	6/7	6.7	1.37	16.1
DS06	Q	R0	24.0	43.8 [19.0, 93.8]	7/7	39.8	4.66	16.1
DS06	Q	R1	23.0	43.8 [28.6, 93.8]	7/7	10.5	8.91	14.3
DS06	Q	R2	22.0	42.9 [19.0, 93.8]	7/7	10.4	6.23	16.1
DS07	P	R0	8.0	11.1 [5.6, 16.7]	5/7	22.4	0.88	13.0
DS07	P	R1	4.4	13.0 [5.6, 16.7]	6/7	4.0	1.34	13.0
DS07	P	R2	8.0	13.0 [4.3, 16.7]	5/7	5.5	0.97	11.1
DS07	C	R0	6.2	11.1 [2.8, 27.8]	4/7	20.6	1.35	11.1
DS07	C	R1	7.1	16.7 [0.0, 16.7]	4/7	4.1	2.02	13.0
DS07	C	R2	7.1	13.0 [2.8, 22.2]	4/7	5.1	1.77	13.0
DS07	Q	R0	8.0	16.7 [8.3, 21.7]	6/7	30.1	1.18	8.3
DS07	Q	R1	8.0	16.7 [11.1, 21.7]	7/7	6.0	1.91	16.7
DS07	Q	R2	7.1	13.9 [8.3, 21.7]	6/7	6.7	1.55	13.0
DS08a	P	R0	11.5	18.5 [10.0, 33.3]	7/7	27.4	5.00	26.7
DS08a	P	R1	9.0	20.0 [10.0, 25.9]	7/7	7.5	8.12	20.0
DS08a	P	R2	9.8	20.0 [10.0, 25.9]	7/7	7.6	6.25	20.0
DS08a	C	R0	7.4	13.3 [10.0, 18.5]	7/7	25.7	5.39	33.3
DS08a	C	R1	8.2	17.6 [10.0, 20.0]	7/7	7.3	8.51	33.3
DS08a	C	R2	9.0	17.6 [10.0, 20.0]	7/7	7.1	6.52	33.3
DS08a	Q	R0	9.8	28.6 [20.0, 33.3]	7/7	32.8	11.34	46.7
DS08a	Q	R1	9.8	28.6 [20.0, 33.3]	7/7	7.8	20.87	28.6
DS08a	Q	R2	9.8	28.6 [20.0, 33.3]	7/7	7.9	13.47	40.0
DS08c	P	R0	1.2	4.8 [0.0, 13.6]	1/7	9.8	0.94	0.0
DS08c	P	R1	1.2	4.8 [0.0, 13.6]	1/7	3.6	0.96	0.0
DS08c	P	R2	1.2	4.8 [0.0, 13.6]	1/7	3.4	0.94	0.0
DS08c	C	R0	1.2	4.5 [0.0, 9.1]	0/7	10.5	0.96	9.1
DS08c	C	R1	1.2	4.5 [0.0, 9.1]	0/7	3.7	0.98	9.1
DS08c	C	R2	0.0	0.0 [0.0, 4.5]	0/7	3.4	0.97	9.1
DS08c	Q	R0	9.3	18.2 [0.0, 22.7]	6/7	24.6	4.36	9.1
DS08c	Q	R1	10.5	18.2 [0.0, 28.6]	6/7	6.5	7.28	18.2
DS08c	Q	R2	9.3	18.2 [0.0, 22.7]	6/7	6.7	5.38	13.6

Source: results/extension/(subset)/audit/flight_false_flags.csv and persistence_calibration.csv. R0 single row; R1 three consecutive rows; R2 three of five rows.

Table S15: Matched-burden detection delay (frozen adversarial Part B per rule; 1% target): pre-specified labels of C and Q against P over the residual runs, the number of runs with a lower curve-level mean delay, and the CVAE seeds' labels.

Subset	Arm vs P	Rule	Evaluable				Lower curve		CVAE labels
			residual runs	Earlier	Equal	Later	Mixed	mean	
DS02	C	R0	7/7	3	0	0	4	6	earlier, mixed, equal
DS02	C	R1	6/7	1	0	0	5	5	earlier, mixed, equal
DS02	C	R2	7/7	4	0	0	3	7	earlier, later, equal
DS02	Q	R0	6/7	2	0	0	4	6	earlier, n/e, n/e
DS02	Q	R1	7/7	3	0	0	4	6	earlier, mixed, earlier

Subset	Arm vs P	Rule	Evaluable				Lower curve mean	CVAE labels
			residual runs	Earlier	Equal	Later	Mixed	
DS02	Q	R2	7/7	3	0	0	4	6 n/e, mixed, earlier
DS03	C	R0	7/7	0	2	0	5	6 mixed, mixed, mixed
DS03	C	R1	6/7	1	1	0	4	3 mixed, mixed, equal
DS03	C	R2	7/7	0	0	0	7	6 mixed, mixed, equal
DS03	Q	R0	7/7	3	0	0	4	5 mixed, earlier, mixed
DS03	Q	R1	7/7	3	0	0	4	4 earlier, earlier, mixed
DS03	Q	R2	7/7	4	0	0	3	5 earlier, earlier, mixed
DS01	C	R0	7/7	0	3	0	4	3 equal, mixed, mixed
DS01	C	R1	6/7	2	1	0	3	5 equal, mixed, mixed
DS01	C	R2	7/7	1	2	0	4	4 mixed, equal, mixed
DS01	Q	R0	7/7	3	0	0	4	6 mixed, mixed, mixed
DS01	Q	R1	7/7	4	0	0	3	6 mixed, mixed, mixed
DS01	Q	R2	7/7	3	0	0	4	6 mixed, mixed, mixed
DS04	C	R0	7/7	3	0	0	4	5 earlier, mixed, mixed
DS04	C	R1	7/7	0	1	0	6	4 mixed, mixed, mixed
DS04	C	R2	7/7	1	0	0	6	1 mixed, mixed, mixed
DS04	Q	R0	7/7	3	0	0	4	3 mixed, mixed, mixed
DS04	Q	R1	7/7	7	0	0	0	6 mixed, mixed, mixed
DS04	Q	R2	7/7	7	0	0	0	5 mixed, later, earlier
DS05	C	R0	7/7	0	5	0	2	0 mixed, mixed, earlier
DS05	C	R1	7/7	0	5	0	2	0 earlier, equal, mixed
DS05	C	R2	6/7	0	3	0	3	0 earlier, equal, earlier
DS05	Q	R0	7/7	4	0	1	2	1 mixed, n/e, earlier
DS05	Q	R1	7/7	4	0	0	3	2 earlier, mixed, earlier
DS05	Q	R2	7/7	4	0	0	3	2 earlier, mixed, earlier
DS06	C	R0	7/7	0	0	0	7	6 equal, equal, earlier
DS06	C	R1	7/7	2	0	0	5	6 equal, equal, earlier
DS06	C	R2	7/7	1	0	0	6	6 equal, equal, earlier
DS06	Q	R0	7/7	2	0	1	4	4 mixed, equal, earlier
DS06	Q	R1	7/7	3	0	2	2	4 mixed, equal, earlier
DS06	Q	R2	7/7	2	0	2	3	4 mixed, equal, earlier
DS07	C	R0	7/7	0	6	0	1	2 equal, equal, mixed
DS07	C	R1	7/7	0	3	0	4	2 mixed, n/e, mixed
DS07	C	R2	7/7	0	4	0	3	2 n/e, equal, mixed
DS07	Q	R0	7/7	4	0	0	3	3 equal, mixed, mixed
DS07	Q	R1	7/7	6	0	0	1	4 equal, earlier, mixed
DS07	Q	R2	7/7	4	0	0	3	2 mixed, earlier, mixed
DS08a	C	R0	7/7	0	2	0	5	5 mixed, mixed, equal
DS08a	C	R1	7/7	0	3	0	4	5 mixed, equal, equal
DS08a	C	R2	7/7	0	4	0	3	5 mixed, equal, equal
DS08a	Q	R0	7/7	3	0	0	4	6 equal, mixed, earlier
DS08a	Q	R1	7/7	4	0	0	3	7 mixed, earlier, mixed
DS08a	Q	R2	7/7	4	0	0	3	6 equal, mixed, mixed
DS08c	C	R0	7/7	4	2	0	1	6 earlier, mixed, mixed
DS08c	C	R1	7/7	3	1	0	3	7 earlier, earlier, mixed
DS08c	C	R2	7/7	3	1	0	3	4 earlier, mixed, mixed
DS08c	Q	R0	7/7	4	0	0	3	5 earlier, mixed, mixed
DS08c	Q	R1	7/7	4	0	0	3	7 mixed, earlier, mixed
DS08c	Q	R2	7/7	4	0	0	3	5 earlier, earlier, mixed

Source: results/extension/(subset)/audit/matched_labels.csv. n/e = not evaluable (fewer than 3 of 5 anchors supported for both arms).

Table S16: Abnormal-state ($hs = 0$) row alarm rates at the locked thresholds (1% target, R0): residual median [range] over seven runs, and the CVAE seed median. These rates are relative to the simulated onset label, not to a discrete fault event.

Subset	Arm	Abnormal-state row alarm rate (%)	
		all post-onset flights	first 10 post-onset flights
DS02	P	62.0 [50.8, 63.7]; CVAE 51.9	
DS02	C	64.6 [59.0, 68.4]; CVAE 52.7	
DS02	Q	64.3 [58.5, 73.0]; CVAE 53.0	
DS03	P	58.0 [39.5, 59.3]; CVAE 62.3	
DS03	C	57.9 [42.0, 59.0]; CVAE 62.2	
DS03	Q	55.6 [42.3, 59.9]; CVAE 64.7	
DS01	P	54.9 [45.2, 58.5]; CVAE 48.0	
DS01	C	56.4 [46.6, 58.9]; CVAE 48.8	
DS01	Q	58.3 [51.0, 60.1]; CVAE 50.5	
DS04	P	34.9 [24.9, 43.3]; CVAE 20.8	

Subset	Arm	Abnormal-state row alarm rate (%)		Abnormal-state row alarm rate (%)	
		all post-onset flights		first 10 post-onset flights	
DS04	C	35.0	[25.0, 46.0]; CVAE 18.1	0.9	[0.7, 0.9]; CVAE 0.5
DS04	Q	35.5	[25.3, 47.6]; CVAE 23.7	1.1	[0.9, 1.1]; CVAE 0.8
DS05	P	61.3	[50.6, 70.1]; CVAE 44.4	2.1	[1.1, 4.7]; CVAE 2.3
DS05	C	61.7	[52.8, 71.1]; CVAE 45.9	2.1	[1.1, 7.3]; CVAE 2.6
DS05	Q	60.2	[58.8, 72.3]; CVAE 47.6	5.9	[3.6, 20.0]; CVAE 2.2
DS06	P	58.7	[47.2, 65.2]; CVAE 30.4	1.9	[1.0, 2.9]; CVAE 2.5
DS06	C	59.0	[48.8, 66.1]; CVAE 31.7	1.9	[1.0, 4.5]; CVAE 2.6
DS06	Q	58.5	[56.4, 67.3]; CVAE 33.1	5.2	[3.4, 16.1]; CVAE 2.3
DS07	P	49.3	[37.1, 51.6]; CVAE 40.6	1.6	[0.8, 1.8]; CVAE 2.3
DS07	C	49.6	[39.8, 51.7]; CVAE 41.1	1.6	[0.8, 1.7]; CVAE 2.6
DS07	Q	49.3	[47.9, 51.5]; CVAE 42.1	3.3	[2.5, 5.0]; CVAE 2.3
DS08a	P	54.4	[45.8, 55.9]; CVAE 31.7	1.6	[1.6, 1.8]; CVAE 1.4
DS08a	C	54.3	[48.6, 56.9]; CVAE 31.9	1.7	[1.5, 2.0]; CVAE 1.4
DS08a	Q	54.0	[52.1, 57.7]; CVAE 33.7	2.6	[2.0, 4.5]; CVAE 1.5
DS08c	P	44.4	[43.3, 48.9]; CVAE 16.6	0.6	[0.5, 0.7]; CVAE 0.2
DS08c	C	43.8	[42.4, 47.8]; CVAE 18.2	0.8	[0.5, 0.9]; CVAE 0.7
DS08c	Q	49.7	[47.4, 51.2]; CVAE 24.2	2.6	[1.1, 6.7]; CVAE 1.1

Source: results/extension/(subset)/audit/abnormal_alarm_rates.csv.

A.6 Reference reproduction

Table S17: Reference-run reproduction gates (DS02 and DS03 rescored with the extension code before any new-subset audit; Amendment 3).

Subset	Reproduction gate	Result
DS02	adversarial part d q healthy counts	equal (336 cells)
DS02	stage c healthy audit fingerprints	equal (7 residual runs)
DS02	stage e trajectory alarm counts	equal (42 run x target x arm checks)
DS03	adversarial part d q healthy counts	equal (588 cells)
DS03	stage c healthy audit fingerprints	equal (7 residual runs)
DS03	stage e trajectory alarm counts	equal (42 run x target x arm checks)

Source: results/extension/DS02/audit/audit_record.json and DS03/audit/audit_record.json.

A.7 Post hoc checks (article Section 4.9; not pre-specified)

Table S18: Post hoc: worst-engine healthy-flight false-flag rate (WE-FFR) under P against a perfect-calibration reference (each healthy flight flagged independently with probability p, p drawn from the exceedance distribution of the empirical 0.95 quantile of the calibration flights; residual runs, 1% target).

Subset	Engines	Healthy flights	Calibration	Null WE-FFR (%)	Null P(WE-FFR	Observed median (%)	Runs above null
		per engine	flights	median / 95th pct.	≥ 10%)	R0 / R1 / R2	95th pct. R0 / R1 / R2
DS01	4	17–35	79	8.6 / 17.6	0.41	14.3 / 11.4 / 14.3	0/7 / 0/7 / 0/7
DS04	4	16–21	53	9.5 / 23.8	0.42	9.5 / 14.3 / 19.0	0/7 / 0/7 / 0/7
DS05	4	18–36	77	8.7 / 19.4	0.44	16.7 / 14.3 / 14.3	2/7 / 2/7 / 2/7
DS06	4	16–32	71	9.5 / 19.4	0.42	14.3 / 14.3 / 14.3	2/7 / 1/7 / 1/7
DS07	4	18–36	78	8.7 / 19.4	0.43	11.1 / 13.0 / 13.0	0/7 / 0/7 / 0/7
DS08a	6	14–29	131	11.8 / 21.4	0.69	18.5 / 20.0 / 20.0	2/7 / 3/7 / 2/7
DS08c	4	21–22	47	9.5 / 23.8	0.43	4.8 / 4.8 / 4.8	0/7 / 0/7 / 0/7

Source: paper/mssp_extended/evidence/post_hoc_checks.json (post_hoc_checks.py).

Table S19: Post hoc: approximate sampling reference for per-engine A2 (each phase FPR equal to α with a binomial standard error inflated by the engine's flight-clustered design effect at the locked threshold; engine-run pairs of the residual runs, 1% target, R0). The design effect includes true between-flight variation, so the reference is conservative.

Subset	Arm	Median design effect	Median SE of engine FPR (pp)	Null P(A2 ≥ 0.5 pp)	Observed share A2 ≥ 0.5 pp	Observed share above null 95th pct.	Material errors under-alarming
DS01	C	37	0.16	0.22	0.57	0.43	0.00
DS01	P	45	0.15	0.22	0.86	0.61	0.00
DS01	Q	51	0.16	0.32	0.82	0.39	0.48
DS04	C	23	0.10	0.09	0.82	0.54	1.00

Subset	Arm	Median design effect	Median SE of engine FPR (pp)	Null $P(A2 \geq 0.5 \text{ pp})$	Observed share $A2 \geq 0.5 \text{ pp}$	Observed share above null 95th pct.	Material errors under-alarming
DS04	P	27	0.10	0.10	0.86	1.00	0.79
DS04	Q	14	0.08	0.02	0.18	0.61	1.00
DS05	C	50	0.16	0.29	0.79	0.54	0.23
DS05	P	56	0.17	0.29	0.75	0.54	0.43
DS05	Q	238	0.52	0.64	0.96	0.75	0.19
DS06	C	53	0.16	0.34	0.79	0.46	0.27
DS06	P	61	0.17	0.35	0.79	0.32	0.50
DS06	Q	200	0.51	0.73	0.96	0.75	0.15
DS07	C	48	0.13	0.23	0.75	0.46	0.57
DS07	P	49	0.14	0.24	0.86	0.50	0.75
DS07	Q	64	0.19	0.41	0.86	0.43	0.42
DS08a	C	141	0.32	0.62	0.86	0.19	0.17
DS08a	P	136	0.29	0.63	0.95	0.45	0.30
DS08a	Q	151	0.27	0.61	0.93	0.52	0.49
DS08c	C	38	0.10	0.14	1.00	0.93	1.00
DS08c	P	45	0.10	0.18	1.00	0.86	1.00
DS08c	Q	363	0.45	0.67	1.00	0.64	0.43

Source: paper/mssp_extended/evidence/post_hoc_engine_noise.csv.

Table S20: Post hoc: two-consecutive-flight confirmation at the locked κ (not pre-specified; R0 rows, residual runs, 1% target): worst-engine healthy confirmed-alert rate and median detection delay.

Arm	WE-FFR single flight (median %)	Runs $\geq 10\%$ single	WE two-flight confirmed (median %)	Runs at 0 / $\geq 10\%$ (two-flight)	Median delay (flights) single $\rightarrow$ two-flight
P	13.0	31/49	0.0	36 / 2	9.5 $\rightarrow$ 17
C	12.9	31/49	0.0	35 / 2	7.5 $\rightarrow$ 16.5
Q	20.0	42/49	7.1	17 / 20	4.5 $\rightarrow$ 13.5

Source: paper/mssp_extended/evidence/post_hoc_checks.json; delays from detection_delay.csv (two_flight_persistence_delay).

A.8 Final focused validation (article Sections 3.10 and 4.8; post hoc)

Table S21: Final focused validation, component 1: fleet-calibrated per-engine A2 against the engine’s own cross-fitted self-calibration reference (medians over the seven residual runs; 5-fold cross-fit, 200 repetitions; 1% target). The reference describes what calibrating on the engine’s own flights achieves and understates flight-to-flight sampling noise (article Section 3.10).

Subset	Engine (class)	Class in calibration	Healthy flights	Arm	Observed A2 (pp)	NF50 (pp)	NF95 (pp)	Residual runs above NF95	Clearly exceeds	CVAE runs above NF95
DS01	7 (1)	yes	35	P	0.90	0.09	0.20	7/7	yes	3/3
DS01	7 (1)	yes	35	C	0.96	0.09	0.20	7/7	yes	3/3
DS01	7 (1)	yes	35	Q	0.62	0.09	0.20	7/7	yes	3/3
DS01	8 (2)	yes	25	P	0.69	0.12	0.44	7/7	yes	3/3
DS01	8 (2)	yes	25	C	0.44	0.12	0.44	3/7	no	3/3
DS01	8 (2)	yes	25	Q	0.52	0.12	0.44	4/7	yes	3/3
DS01	9 (1)	yes	34	P	0.55	0.09	0.22	7/7	yes	3/3
DS01	9 (1)	yes	34	C	0.56	0.09	0.22	7/7	yes	3/3
DS01	9 (1)	yes	34	Q	0.64	0.09	0.22	7/7	yes	3/3
DS01	10 (3)	yes	17	P	0.64	0.06	0.21	7/7	yes	3/3
DS01	10 (3)	yes	17	C	0.23	0.06	0.21	4/7	yes	3/3
DS01	10 (3)	yes	17	Q	1.16	0.06	0.21	7/7	yes	3/3
DS04	7 (2)	yes	21	P	0.74	0.03	0.06	7/7	yes	1/3
DS04	7 (2)	yes	21	C	0.55	0.03	0.06	7/7	yes	2/3
DS04	7 (2)	yes	21	Q	0.28	0.03	0.06	7/7	yes	2/3
DS04	8 (2)	yes	21	P	0.74	0.03	0.08	7/7	yes	3/3
DS04	8 (2)	yes	21	C	0.57	0.03	0.08	7/7	yes	3/3
DS04	8 (2)	yes	21	Q	0.39	0.03	0.08	7/7	yes	3/3
DS04	9 (2)	yes	21	P	0.67	0.02	0.04	7/7	yes	3/3
DS04	9 (2)	yes	21	C	0.59	0.02	0.04	7/7	yes	3/3
DS04	9 (2)	yes	21	Q	0.35	0.02	0.04	7/7	yes	0/3
DS04	10 (3)	yes	16	P	0.78	0.02	0.05	7/7	yes	2/3
DS04	10 (3)	yes	16	C	0.53	0.02	0.05	7/7	yes	3/3
DS04	10 (3)	yes	16	Q	0.42	0.02	0.05	7/7	yes	2/3
DS05	7 (2)	yes	23	P	0.97	0.14	0.39	7/7	yes	3/3
DS05	7 (2)	yes	23	C	0.72	0.14	0.39	7/7	yes	2/3
DS05	7 (2)	yes	23	Q	2.96	0.14	0.39	7/7	yes	2/3

Subset	Engine (class)	Class in calibration	Healthy flights	Arm	Observed A2 (pp)	NF50 (pp)	NF95 (pp)	Residual runs above NF95	Clearly exceeds	CVAE runs above NF95
DS05	8 (1)	yes	35	P	0.65	0.15	0.34	7/7	yes	2/3
DS05	8 (1)	yes	35	C	1.36	0.15	0.34	7/7	yes	2/3
DS05	8 (1)	yes	35	Q	0.56	0.15	0.34	7/7	yes	2/3
DS05	9 (3)	yes	18	P	0.90	0.16	0.48	7/7	yes	3/3
DS05	9 (3)	yes	18	C	0.74	0.16	0.48	4/7	yes	3/3
DS05	9 (3)	yes	18	Q	3.38	0.16	0.48	7/7	yes	2/3
DS05	10 (1)	yes	36	P	0.50	0.09	0.21	6/7	yes	1/3
DS05	10 (1)	yes	36	C	0.50	0.09	0.21	7/7	yes	2/3
DS05	10 (1)	yes	36	Q	2.33	0.09	0.21	7/7	yes	2/3
DS06	7 (2)	yes	21	P	0.83	0.23	0.48	7/7	yes	3/3
DS06	7 (2)	yes	21	C	0.91	0.23	0.48	7/7	yes	3/3
DS06	7 (2)	yes	21	Q	4.92	0.23	0.48	7/7	yes	3/3
DS06	8 (1)	yes	31	P	0.62	0.17	0.43	7/7	yes	2/3
DS06	8 (1)	yes	31	C	1.24	0.17	0.43	7/7	yes	3/3
DS06	8 (1)	yes	31	Q	0.64	0.17	0.43	7/7	yes	2/3
DS06	9 (3)	yes	16	P	0.91	0.26	0.71	7/7	yes	3/3
DS06	9 (3)	yes	16	C	0.62	0.26	0.71	4/7	yes	3/3
DS06	9 (3)	yes	16	Q	4.38	0.26	0.71	7/7	yes	2/3
DS06	10 (1)	yes	32	P	0.53	0.09	0.38	4/7	yes	2/3
DS06	10 (1)	yes	32	C	0.52	0.09	0.38	5/7	yes	3/3
DS06	10 (1)	yes	32	Q	2.89	0.09	0.38	7/7	yes	3/3
DS07	7 (2)	yes	23	P	0.66	0.23	0.49	7/7	yes	3/3
DS07	7 (2)	yes	23	C	0.51	0.23	0.49	5/7	yes	2/3
DS07	7 (2)	yes	23	Q	0.69	0.23	0.49	7/7	yes	2/3
DS07	8 (1)	yes	36	P	0.83	0.17	0.40	7/7	yes	3/3
DS07	8 (1)	yes	36	C	1.35	0.17	0.40	7/7	yes	2/3
DS07	8 (1)	yes	36	Q	0.61	0.17	0.40	6/7	yes	2/3
DS07	9 (3)	yes	18	P	0.86	0.28	0.74	6/7	yes	3/3
DS07	9 (3)	yes	18	C	0.64	0.28	0.74	3/7	no	3/3
DS07	9 (3)	yes	18	Q	0.71	0.28	0.74	5/7	yes	2/3
DS07	10 (1)	yes	36	P	0.46	0.12	0.25	6/7	yes	1/3
DS07	10 (1)	yes	36	C	0.60	0.12	0.25	6/7	yes	1/3
DS07	10 (1)	yes	36	Q	1.06	0.12	0.25	7/7	yes	2/3
DS08a	10 (1)	yes	29	P	0.77	0.15	0.38	7/7	yes	3/3
DS08a	10 (1)	yes	29	C	0.56	0.15	0.38	6/7	yes	3/3
DS08a	10 (1)	yes	29	Q	0.77	0.15	0.38	7/7	yes	3/3
DS08a	11 (2)	yes	20	P	2.01	0.41	1.33	7/7	yes	3/3
DS08a	11 (2)	yes	20	C	1.50	0.41	1.33	5/7	yes	3/3
DS08a	11 (2)	yes	20	Q	0.75	0.41	1.33	0/7	no	2/3
DS08a	12 (3)	yes	17	P	1.09	0.37	1.19	3/7	no	2/3
DS08a	12 (3)	yes	17	C	0.85	0.37	1.19	2/7	no	2/3
DS08a	12 (3)	yes	17	Q	0.69	0.37	1.19	1/7	no	2/3
DS08a	13 (3)	yes	14	P	0.73	0.24	0.45	6/7	yes	3/3
DS08a	13 (3)	yes	14	C	0.57	0.24	0.45	5/7	yes	3/3
DS08a	13 (3)	yes	14	Q	3.36	0.24	0.45	7/7	yes	3/3
DS08a	14 (3)	yes	15	P	5.00	3.35	4.13	5/7	yes	2/3
DS08a	14 (3)	yes	15	C	5.39	3.35	4.13	7/7	yes	2/3
DS08a	14 (3)	yes	15	Q	11.40	3.35	4.13	7/7	yes	2/3
DS08a	15 (1)	yes	27	P	0.70	0.10	0.24	7/7	yes	3/3
DS08a	15 (1)	yes	27	C	0.57	0.10	0.24	7/7	yes	2/3
DS08a	15 (1)	yes	27	Q	0.93	0.10	0.24	7/7	yes	2/3
DS08c	7 (2)	no	22	P	0.68	0.11	0.35	6/7	yes	3/3
DS08c	7 (2)	no	22	C	0.86	0.11	0.35	6/7	yes	3/3
DS08c	7 (2)	no	22	Q	2.08	0.11	0.35	6/7	yes	3/3
DS08c	8 (2)	no	22	P	0.94	0.14	0.39	7/7	yes	3/3
DS08c	8 (2)	no	22	C	0.96	0.14	0.39	7/7	yes	3/3
DS08c	8 (2)	no	22	Q	2.98	0.14	0.39	7/7	yes	0/3
DS08c	9 (2)	no	21	P	0.82	0.13	0.35	5/7	yes	3/3
DS08c	9 (2)	no	21	C	0.89	0.13	0.35	5/7	yes	3/3
DS08c	9 (2)	no	21	Q	0.95	0.13	0.35	5/7	yes	2/3
DS08c	10 (2)	no	21	P	0.71	0.09	0.30	7/7	yes	3/3
DS08c	10 (2)	no	21	C	0.67	0.09	0.30	7/7	yes	3/3
DS08c	10 (2)	no	21	Q	0.70	0.09	0.30	7/7	yes	2/3

Source: results/extension/focused_validation/summary/engine_summary.csv (plan docs/extension/FOCUSED_VALIDATION_PLAN.md).

Table S22: Final focused validation, component 3: local recalibration on each audit engine's first five healthy flights, evaluated on its remaining healthy flights (medians over the seven residual runs; 1% target). Change = fleet minus local A2 (positive = local better); NF95K = 95th percentile of the matched reference from 200 random five-flight subsets.

Subset	Engine (class)	Arm	Evaluation flights	Fleet A2 (pp)	Local A2 (pp)	Change (pp)	Runs improved	NF95K (pp)	Local within NF95K (runs)	Fleet within NF95K (runs)	Local within NF95 (runs)
DS01	7 (1)	C	30	0.49	0.65	-0.36	1/7	2.35	7/7	7/7	1/7
DS01	7 (1)	P	30	0.90	0.78	+0.11	4/7	2.35	7/7	7/7	0/7
DS01	8 (2)	C	20	0.25	0.36	-0.16	1/7	3.06	7/7	7/7	3/7
DS01	8 (2)	P	20	0.82	0.82	+0.19	4/7	3.06	7/7	7/7	3/7
DS01	9 (1)	C	29	0.60	0.52	+0.15	5/7	2.98	7/7	7/7	0/7
DS01	9 (1)	P	29	0.51	0.51	-0.04	2/7	2.98	7/7	7/7	0/7
DS01	10 (3)	C	12	0.20	3.04	-1.72	0/7	2.91	5/7	7/7	0/7
DS01	10 (3)	P	12	0.73	0.91	-0.16	0/7	2.91	7/7	7/7	0/7
DS04	7 (2)	C	16	0.53	0.34	+0.05	4/7	0.86	7/7	7/7	0/7
DS04	7 (2)	P	16	0.73	0.83	-0.17	0/7	0.86	6/7	7/7	0/7
DS04	8 (2)	C	16	0.59	0.42	+0.23	6/7	1.07	7/7	7/7	0/7
DS04	8 (2)	P	16	0.69	0.71	-0.14	1/7	1.07	4/7	7/7	0/7
DS04	9 (2)	C	16	0.53	0.31	+0.14	7/7	0.72	7/7	7/7	0/7
DS04	9 (2)	P	16	0.67	0.52	+0.09	6/7	0.72	7/7	7/7	0/7
DS04	10 (3)	C	11	0.47	0.36	+0.11	6/7	0.98	7/7	7/7	0/7
DS04	10 (3)	P	11	0.77	0.70	0.00	3/7	0.98	4/7	6/7	0/7
DS05	7 (2)	C	18	0.82	0.58	+0.32	6/7	3.48	7/7	6/7	0/7
DS05	7 (2)	P	18	0.80	0.84	-0.02	1/7	3.48	7/7	7/7	0/7
DS05	8 (1)	C	30	1.58	1.91	-0.37	1/7	3.42	7/7	7/7	0/7
DS05	8 (1)	P	30	0.73	0.99	-0.18	1/7	3.42	7/7	7/7	0/7
DS05	9 (3)	C	13	0.68	0.74	-0.03	3/7	3.03	7/7	5/7	1/7
DS05	9 (3)	P	13	1.03	0.86	+0.17	6/7	3.03	7/7	6/7	0/7
DS05	10 (1)	C	31	0.50	2.89	-2.43	0/7	2.41	2/7	7/7	0/7
DS05	10 (1)	P	31	0.43	1.51	-1.08	0/7	2.41	6/7	7/7	0/7
DS06	7 (2)	C	16	0.85	0.61	+0.38	6/7	3.54	7/7	6/7	0/7
DS06	7 (2)	P	16	0.81	0.83	-0.05	1/7	3.54	7/7	7/7	0/7
DS06	8 (1)	C	26	1.51	1.96	-0.44	1/7	3.11	7/7	7/7	0/7
DS06	8 (1)	P	26	0.57	0.82	-0.06	2/7	3.11	7/7	7/7	0/7
DS06	9 (3)	C	11	0.70	0.69	+0.08	4/7	3.15	7/7	6/7	3/7
DS06	9 (3)	P	11	1.04	0.87	+0.26	5/7	3.15	7/7	6/7	0/7
DS06	10 (1)	C	27	0.51	2.91	-2.30	0/7	2.31	4/7	7/7	0/7
DS06	10 (1)	P	27	0.46	1.70	-1.24	0/7	2.31	6/7	7/7	0/7
DS07	7 (2)	C	18	0.67	0.60	-0.06	3/7	3.96	7/7	7/7	1/7
DS07	7 (2)	P	18	0.83	0.84	0.00	3/7	3.96	7/7	7/7	0/7
DS07	8 (1)	C	31	1.50	1.84	-0.31	1/7	3.81	7/7	7/7	0/7
DS07	8 (1)	P	31	1.06	1.20	-0.14	0/7	3.81	7/7	7/7	0/7
DS07	9 (3)	C	13	0.73	0.74	+0.01	4/7	5.53	7/7	7/7	2/7
DS07	9 (3)	P	13	0.84	0.81	0.00	4/7	5.53	7/7	7/7	2/7
DS07	10 (1)	C	31	0.56	1.16	-0.56	0/7	2.25	6/7	7/7	0/7
DS07	10 (1)	P	31	0.41	1.06	-0.52	0/7	2.25	7/7	7/7	0/7
DS08a	10 (1)	C	24	0.71	1.03	-0.44	1/7	2.59	7/7	7/7	1/7
DS08a	10 (1)	P	24	0.75	0.80	-0.17	2/7	2.59	7/7	7/7	0/7
DS08a	11 (2)	C	15	2.30	4.22	-1.87	0/7	6.28	7/7	7/7	0/7
DS08a	11 (2)	P	15	2.90	4.63	-1.72	0/7	6.28	7/7	7/7	0/7
DS08a	12 (3)	C	12	1.27	2.24	-0.51	0/7	4.93	7/7	7/7	1/7
DS08a	12 (3)	P	12	0.86	0.94	-0.05	3/7	4.93	7/7	7/7	5/7
DS08a	13 (3)	C	9	0.54	0.88	-0.37	0/7	3.48	7/7	7/7	0/7
DS08a	13 (3)	P	9	0.72	0.94	+0.01	4/7	3.48	7/7	7/7	1/7
DS08a	14 (3)	C	10	0.82	0.92	+0.03	6/7	9.30	7/7	7/7	7/7
DS08a	14 (3)	P	10	0.85	0.96	-0.11	0/7	9.30	7/7	7/7	7/7
DS08a	15 (1)	C	22	0.61	0.87	-0.25	0/7	3.39	7/7	7/7	0/7
DS08a	15 (1)	P	22	0.73	0.78	-0.02	3/7	3.39	7/7	7/7	0/7
DS08c	7 (2)	C	17	0.87	0.69	+0.19	7/7	3.39	7/7	7/7	1/7
DS08c	7 (2)	P	17	0.78	0.75	+0.02	4/7	3.39	7/7	7/7	1/7
DS08c	8 (2)	C	17	0.95	1.52	-0.57	0/7	2.88	7/7	7/7	0/7
DS08c	8 (2)	P	17	0.93	1.23	-0.34	2/7	2.88	7/7	7/7	0/7
DS08c	9 (2)	C	16	0.89	0.61	+0.28	7/7	4.74	7/7	7/7	2/7
DS08c	9 (2)	P	16	0.83	0.79	+0.02	4/7	4.74	7/7	7/7	1/7
DS08c	10 (2)	C	16	0.66	0.81	-0.17	2/7	2.85	7/7	7/7	0/7
DS08c	10 (2)	P	16	0.68	0.47	+0.27	7/7	2.85	7/7	7/7	3/7

Source: results/extension/focused_validation/summary/local_engine_summary.csv (plan docs/extension/FOCUSED_VALIDATION_PLAN.md).

Table S23: Final focused validation, component 2: per-subset bootstrap intervals with calibration engines resampled within flight class against the original U-EXT1 design (identical audit plans; medians over the seven residual runs; 1% target).

Subset	Metric	Width ratio (class-preserving / original)	Share of width from class omission	Runs including zero original / class-preserving
DS01	diff_ME_A2	0.571	43%	7/7 / 7/7
DS01	diff_WE_A2	0.623	38%	7/7 / 7/7
DS01	diff_pooled_A1	0.669	33%	7/7 / 4/7
DS01	diff_pooled_A2	0.622	38%	7/7 / 6/7
DS01	phase_conditioned.ME_A2	0.527	47%	0/7 / 0/7
DS01	phase_conditioned.WE_A2	0.547	45%	0/7 / 0/7
DS01	pooled.ME_A2	0.895	10%	0/7 / 0/7

Subset	Metric	Width ratio (class-preserving / original)	Share of width from class omission	Runs including zero original / class-preserving
DS01	pooled.WE_A2	0.914	9%	0/7 / 0/7
DS04	diff_ME_A2	0.989	1%	4/7 / 4/7
DS04	diff_WE_A2	1.015	-1%	7/7 / 7/7
DS04	diff_pooled_A1	0.962	4%	7/7 / 7/7
DS04	diff_pooled_A2	1.008	-1%	4/7 / 4/7
DS04	phase_conditioned.ME_A2	0.975	3%	0/7 / 0/7
DS04	phase_conditioned.WE_A2	0.972	3%	0/7 / 0/7
DS04	pooled.ME_A2	0.959	4%	0/7 / 0/7
DS04	pooled.WE_A2	0.975	2%	0/7 / 0/7
DS05	diff_ME_A2	0.907	9%	7/7 / 7/7
DS05	diff_WE_A2	0.900	10%	7/7 / 7/7
DS05	diff_pooled_A1	0.914	9%	7/7 / 7/7
DS05	diff_pooled_A2	0.907	9%	7/7 / 7/7
DS05	phase_conditioned.ME_A2	0.931	7%	0/7 / 0/7
DS05	phase_conditioned.WE_A2	0.958	4%	0/7 / 0/7
DS05	pooled.ME_A2	0.909	9%	0/7 / 0/7
DS05	pooled.WE_A2	0.912	9%	0/7 / 0/7
DS06	diff_ME_A2	0.860	14%	7/7 / 7/7
DS06	diff_WE_A2	0.836	16%	7/7 / 7/7
DS06	diff_pooled_A1	0.848	15%	7/7 / 7/7
DS06	diff_pooled_A2	0.852	15%	7/7 / 7/7
DS06	phase_conditioned.ME_A2	0.912	9%	0/7 / 0/7
DS06	phase_conditioned.WE_A2	0.926	7%	0/7 / 0/7
DS06	pooled.ME_A2	0.924	8%	0/7 / 0/7
DS06	pooled.WE_A2	0.932	7%	0/7 / 0/7
DS07	diff_ME_A2	0.857	14%	7/7 / 7/7
DS07	diff_WE_A2	0.796	20%	7/7 / 7/7
DS07	diff_pooled_A1	0.830	17%	7/7 / 7/7
DS07	diff_pooled_A2	0.790	21%	7/7 / 7/7
DS07	phase_conditioned.ME_A2	0.872	13%	0/7 / 0/7
DS07	phase_conditioned.WE_A2	0.867	13%	0/7 / 0/7
DS07	pooled.ME_A2	0.797	20%	0/7 / 0/7
DS07	pooled.WE_A2	0.880	12%	0/7 / 0/7
DS08a	diff_ME_A2	0.867	13%	7/7 / 7/7
DS08a	diff_WE_A2	0.853	15%	7/7 / 7/7
DS08a	diff_pooled_A1	0.881	12%	7/7 / 7/7
DS08a	diff_pooled_A2	0.893	11%	7/7 / 7/7
DS08a	phase_conditioned.ME_A2	0.970	3%	0/7 / 0/7
DS08a	phase_conditioned.WE_A2	0.993	1%	0/7 / 0/7
DS08a	pooled.ME_A2	0.990	1%	0/7 / 0/7
DS08a	pooled.WE_A2	0.997	0%	0/7 / 0/7
DS08c	diff_ME_A2	1.014	-1%	7/7 / 7/7
DS08c	diff_WE_A2	1.029	-3%	7/7 / 7/7
DS08c	diff_pooled_A1	1.016	-2%	7/7 / 7/7
DS08c	diff_pooled_A2	1.005	-0%	7/7 / 7/7
DS08c	phase_conditioned.ME_A2	1.057	-6%	0/7 / 0/7
DS08c	phase_conditioned.WE_A2	1.051	-5%	0/7 / 0/7
DS08c	pooled.ME_A2	1.008	-1%	0/7 / 0/7
DS08c	pooled.WE_A2	1.023	-2%	0/7 / 0/7

Source: results/extension/focused_validation/summary/bootstrap_width_comparison.csv (plan docs/extension/FOCUSED_VALIDATION_PLAN.md).

Table S24: Final focused validation, component 2: cross-dataset (U-EXT2) intervals recomputed from the class-preserving replicates with the same routine and seed.

Detector family	Quantity	Median over families (pp)	Original interval (pp)	Class-preserving interval (pp)	Width ratio
PCA	diff_pooled_A1	-0.03	[-0.55, +1.32]	[-0.54, +1.16]	0.91
PCA	diff_pooled_A2	+0.03	[-0.29, +1.55]	[-0.27, +1.36]	0.89
PCA	diff_ME_A2	-0.02	[-0.20, +1.41]	[-0.18, +1.26]	0.89
IF	diff_pooled_A1	-0.11	[-1.15, +0.48]	[-1.20, +0.50]	1.04
IF	diff_pooled_A2	+0.02	[-0.59, +0.47]	[-0.63, +0.28]	0.87
IF	diff_ME_A2	-0.20	[-0.53, +0.22]	[-0.54, +0.18]	0.97
LSTM	diff_pooled_A1	-0.31	[-0.52, +0.53]	[-0.57, +0.37]	0.90
LSTM	diff_pooled_A2	-0.08	[-0.25, +0.32]	[-0.27, +0.20]	0.82
LSTM	diff_ME_A2	-0.03	[-0.21, +0.34]	[-0.20, +0.24]	0.81
CVAE	diff_pooled_A1	+0.01	[-0.56, +0.99]	[-0.60, +0.93]	0.99
CVAE	diff_pooled_A2	+0.14	[-0.29, +0.56]	[-0.33, +0.47]	0.94
CVAE	diff_ME_A2	-0.03	[-0.22, +0.46]	[-0.26, +0.35]	0.90
CVAE vs residual	cvae_minus_best_residual_ME_A2_P	+0.52	[+0.05, +1.31]	[+0.06, +1.18]	0.90

Source: results/extension/focused_validation/summary/uext2_original_vs_class_preserving.csv (plan docs/extension/FOCUSED_VALIDATION_PLAN.m

Part B. Supplement of the pre-extension study (included unchanged)

The following pages reproduce, unchanged, the committed pre-extension supplement (SHA-256 `f1d38a515ca6e6ce...`); its table and figure numbers are its own. Source file:
`paper/mssp/supplement/supplement.pdf`.

Supplementary material for “Phase-Dependent False-Alarm Calibration and Its Transport Across Engines in Full-Flight Aero-Engine Anomaly Detection”

Jinghang Mei

This supplement accompanies the article. Every table is a formatted view of committed outputs of the frozen and post-confirmation analyses; values are rounded for display and nothing is recomputed. Healthy false-positive rates (FPR) are percentages of healthy rows; differences are in percentage points (pp). Runs are one PCA run (PCA) and three seeds each of Isolation Forest (IF s0–s2) and the past-only LSTM (LSTM s0–s2). Arms: P, pooled; C, retrospective phase-conditioned; C', past-only regime-conditioned; Q, continuous contextual baseline (healthy side only). The primary nominal target is 1%. Machine-readable versions of all tables are in the public repository (<https://github.com/surnamemei/ncmapss-phase-entanglement>; [results/](#) and [paper/mssp/evidence/](#)).

Contents

1	Provenance, reproduction and deviations	1
2	Frozen discovery and confirmation	2
3	Post-confirmation protocol v1.0	5
4	Adversarial validation	8

1 Provenance, reproduction and deviations

The study was staged: DS02 exploratory discovery, a protocol freeze, a one-shot frozen DS03 confirmation, and two post-confirmation stages (protocol v1.0 and the adversarial-validation plan), each committed before execution. Before every post-confirmation run, reproduction gates required exact agreement with the frozen evidence: rescored healthy score vectors equalled the frozen fingerprints bit for bit; recomputed healthy count tables, calibration-only κ locks and bootstrap summaries equalled the committed outputs; and the committed per-flight alarm fractions reproduced every committed detection delay. A hash manifest of 186 frozen files was verified before and after every run. Abnormal-state ($hs = 0$) rows were read once per dataset, after the calibration-only lock, and never again. Table S1 lists the governing documents and records.

Deviations and amendments. (i) Amendment A-1 corrected a repository-metadata statement about the published v1.0.0 release tag; it was recorded before any abnormal-state row was opened and changed no analysis. (ii) Dev-1: the error decomposition (Section 4) was added after the plan and is descriptive. (iii) Dev-2: operational definitions of the target-robustness claims were fixed in code before execution. (iv) Dev-3: after all runs, an output guard and a robustness fix for phases without rows were added to the analysis module; no output was regenerated, and every output names the module version that produced it. (v) Story-lock Amendment 1 corrected the count of audit engines from flight classes absent from calibration (four, three of them class 1) without changing any analysis.

Verification. The repository’s test suite re-derives the post-confirmation metrics and operating points from committed inputs, and a checker re-derives every number cited in the article and verifies that it appears verbatim.

Table S1: Provenance of the analyses reported in the article. Each run record also names the analysis-module version that produced its outputs.

Artefact	Path	SHA-256 (first 16 hex)
Frozen baseline manifest (186 files)	docs/mssp/frozen_baseline.sha256	37c0cef26826bef2
Post-confirmation protocol v1.0	docs/mssp/post_confirmation_mitigation_protocol.md	b4a669cdd00d4075
Protocol amendments	docs/mssp/AMENDMENTS.md	60522a5b4eb5be60
Adversarial validation plan	docs/mssp/adversarial_validation_plan.md	d993dd3f9e5ab286
Final story lock	docs/mssp/FINAL_STORY_LOCK.md	f6c5b2cf2d067ed0
Post-confirmation module	src/mssp_mitigation.py	2764f27113a0c7af
Adversarial-validation module	src/mssp_adversarial.py	0b2db2b4ad60e509
DS02 endpoint record (abnormal rows read once)	results/mssp_mitigation/ds02/stage_e/stage_e_record.json	a2ecfaa0eea41074
DS03 endpoint record (abnormal rows read once)	results/mssp_mitigation/ds03/stage_e/stage_e_record.json	e95e80c76d154ec7
DS02 adversarial run record (A–C)	results/mssp_adversarial/ds02/run_record_ABC.json	f676c2189422e2b8
DS03 adversarial run record (A–C)	results/mssp_adversarial/ds03/run_record_ABC.json	5fdd9e77f976964c
DS02 baseline run record (D)	results/mssp_adversarial/ds02/run_record_D.json	df0d765dd5ed67dd
DS03 baseline run record (D)	results/mssp_adversarial/ds03/run_record_D.json	1c48684e825e8c9c
Adversarial summary record	results/mssp_adversarial/summary/summary_record.json	1a59f7170292eead

2 Frozen discovery and confirmation

The values in this section are the frozen DS02 (exploratory) and DS03 (frozen confirmation) results, shown unchanged.

Table S2: Frozen DS02 PCA correction sensitivity (exploratory): pooled healthy FPR (%) by phase at the 1% target, primary phase rule, and worst off-diagonal cross-phase transfer FPR (%).

Correction	Overall	Climb	Cruise	Descent	Worst transfer
static	1.277	0.146	0.583	3.001	14.306
static-derivative	1.239	0.061	0.457	3.106	15.865
history	1.363	0.125	0.428	3.459	15.555

Table S3: Frozen DS02 alternative retrospective phase rule (exploratory; 85% altitude range and centred 60-s altitude rate at most 2 ft/s): pooled healthy FPR (%) at the 1% target.

Run	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
PCA	0.131	0.270	3.563	+3.432	+3.293

(continued)

Run	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
IF s0	0.276	0.379	2.515	+2.238	+2.135
IF s1	0.247	0.387	2.787	+2.540	+2.400
IF s2	0.259	0.396	2.629	+2.370	+2.233
LSTM s0	0.451	0.516	2.264	+1.813	+1.747
LSTM s1	0.459	0.524	2.316	+1.857	+1.792
LSTM s2	0.530	0.540	2.270	+1.741	+1.730

Table S4: Frozen individual-run pooled healthy FPR (%) at the 1% target with hierarchical (engine, then flight) bootstrap 95% intervals for the contrasts (2000 replicates, thresholds re-estimated). Intervals belong to individual runs; no seed-mean interval exists.

Dataset	Run	Climb	Cruise	Descent	Descent – climb, 95% CI (pp)	Descent – cruise, 95% CI (pp)
DS02	PCA	0.125	0.428	3.459	+3.335 [+2.070, +5.629]	+3.031 [+1.808, +5.261]
	IF s0	0.285	0.566	2.335	+2.050 [+0.850, +4.074]	+1.769 [+0.678, +3.749]
	IF s1	0.253	0.528	2.669	+2.416 [+1.313, +4.166]	+2.141 [+1.012, +3.943]
	IF s2	0.266	0.580	2.455	+2.190 [+1.063, +3.947]	+1.875 [+0.742, +3.709]
	LSTM s0	0.452	0.707	2.078	+1.626 [+0.317, +2.838]	+1.371 [+0.528, +2.490]
	LSTM s1	0.474	0.731	2.102	+1.628 [−0.022, +2.817]	+1.372 [+0.399, +2.454]
	LSTM s2	0.539	0.750	2.060	+1.521 [−0.347, +2.802]	+1.309 [+0.330, +2.511]
DS03	PCA	0.877	0.912	1.977	+1.100 [+0.434, +1.785]	+1.066 [+0.560, +1.573]
	IF s0	0.417	1.095	2.166	+1.749 [+0.901, +2.847]	+1.071 [+0.404, +1.994]
	IF s1	0.399	1.060	2.127	+1.728 [+0.891, +2.833]	+1.067 [+0.426, +1.976]
	IF s2	0.412	1.157	2.134	+1.722 [+0.871, +2.772]	+0.976 [+0.347, +1.862]
	LSTM s0	0.759	1.795	2.231	+1.472 [+0.536, +2.870]	+0.436 [−0.222, +1.201]
	LSTM s1	0.719	1.890	2.220	+1.501 [+0.512, +3.022]	+0.330 [−0.270, +1.158]
	LSTM s2	0.774	1.982	2.193	+1.419 [+0.379, +3.059]	+0.210 [−0.582, +1.025]

Table S5: Frozen per-engine pooled-threshold healthy FPR (%) at the 1% target (DS02 exploratory; DS03 frozen confirmation). Directional exceptions remain visible.

Dataset	Run	Engine	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
DS02	PCA	11	0.097	0.406	2.780	+2.683	+2.374
		14	0.294	0.491	4.972	+4.678	+4.481
		15	0.113	0.425	3.666	+3.553	+3.241
	IF s0	11	0.091	0.371	2.234	+2.143	+1.863
		14	1.487	1.351	2.510	+1.024	+1.159
		15	0.197	0.393	2.396	+2.199	+2.003
	IF s1	11	0.082	0.304	2.348	+2.266	+2.044
		14	1.291	1.393	3.287	+1.996	+1.894
		15	0.180	0.352	2.824	+2.644	+2.473
	IF s2	11	0.078	0.373	2.298	+2.220	+1.925
		14	1.389	1.432	2.724	+1.335	+1.292
		15	0.191	0.383	2.551	+2.360	+2.167
	LSTM s0	11	0.227	0.558	2.364	+2.137	+1.806
		14	1.887	1.282	2.071	+0.184	+0.789
		15	0.340	0.589	1.621	+1.281	+1.032
	LSTM s1	11	0.246	0.615	2.523	+2.277	+1.908
		14	1.993	1.264	1.835	−0.158	+0.571
		15	0.344	0.585	1.581	+1.237	+0.996
	LSTM s2	11	0.248	0.592	2.378	+2.130	+1.786
		14	2.296	1.384	1.775	−0.521	+0.391
		15	0.418	0.612	1.714	+1.296	+1.102
DS03	PCA	10	1.258	1.247	2.448	+1.190	+1.201
		11	0.383	0.878	2.173	+1.790	+1.294

(continued)

Dataset	Run	Engine	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
IF s0		12	0.322	1.039	1.061	+0.739	+0.022
		13	1.120	0.744	2.249	+1.129	+1.505
		14	0.744	0.587	1.431	+0.686	+0.843
		15	1.112	0.846	2.153	+1.041	+1.307
		10	0.374	1.197	3.113	+2.739	+1.917
		11	0.288	1.158	2.281	+1.993	+1.123
		12	0.847	0.617	1.845	+0.998	+1.229
		13	0.453	1.274	2.665	+2.212	+1.390
		14	0.288	1.095	1.282	+0.994	+0.187
		15	0.378	0.971	1.495	+1.117	+0.524
	IF s1	10	0.335	1.171	3.155	+2.820	+1.984
		11	0.307	1.117	2.318	+2.011	+1.201
		12	0.839	0.550	1.696	+0.857	+1.146
		13	0.419	1.378	2.604	+2.185	+1.226
		14	0.276	0.910	1.105	+0.829	+0.196
IF s2		15	0.355	0.901	1.516	+1.161	+0.615
		10	0.389	1.314	3.232	+2.844	+1.918
		11	0.296	1.213	2.290	+1.994	+1.077
		12	0.808	0.573	1.689	+0.881	+1.116
		13	0.450	1.428	2.619	+2.169	+1.190
		14	0.258	1.089	1.169	+0.911	+0.080
		15	0.372	0.981	1.446	+1.074	+0.466
	LSTM s0	10	0.703	2.519	3.696	+2.992	+1.176
		11	0.545	1.956	2.671	+2.126	+0.714
		12	1.141	0.585	1.237	+0.097	+0.652
		13	0.913	2.397	2.690	+1.777	+0.292
		14	0.693	0.904	0.992	+0.299	+0.088
LSTM s1		15	0.693	1.323	1.556	+0.863	+0.233
		10	0.705	2.793	3.906	+3.201	+1.113
		11	0.504	2.041	2.619	+2.115	+0.579
		12	1.093	0.614	1.210	+0.117	+0.595
		13	0.842	2.412	2.667	+1.825	+0.254
		14	0.681	0.849	0.918	+0.237	+0.070
		15	0.634	1.445	1.449	+0.815	+0.004
	LSTM s2	10	0.713	2.934	4.083	+3.370	+1.149
		11	0.550	2.141	2.652	+2.102	+0.510
		12	1.036	0.487	0.950	−0.086	+0.463
		13	0.951	2.810	2.705	+1.754	−0.105
		14	0.651	0.746	0.784	+0.133	+0.039
		15	0.792	1.378	1.344	+0.552	−0.034

Table S6: Frozen pooled cross-phase threshold-transfer matrices at the 1% target: healthy FPR (%) when the threshold calibrated within one phase is applied to each phase. Diagonal cells are the within-phase (phase-conditioned) FPRs.

Dataset	Run	Calibrated in	Applied to climb	Applied to cruise	Applied to descent
DS02	PCA	climb	1.403	3.430	15.555
		cruise	0.581	1.327	9.120
		descent	0.041	0.238	1.081
	IF s0	climb	1.655	3.357	29.982
		cruise	1.079	1.845	17.042
		descent	0.108	0.205	0.921
	IF s1	climb	1.544	3.000	25.461
		cruise	0.993	1.528	14.017
		descent	0.103	0.237	0.978
	IF s2	climb	1.686	2.914	26.400
		cruise	1.141	1.715	16.783
		descent	0.110	0.204	0.927
	LSTM s0	climb	1.385	2.371	5.505
		cruise	0.698	1.115	3.039
		descent	0.237	0.280	1.080
	LSTM s1	climb	1.243	2.437	6.774

(continued)

Dataset	Run	Calibrated in	Applied to climb	Applied to cruise	Applied to descent
DS03	LSTM s2	cruise	0.729	1.279	3.370
		descent	0.235	0.224	0.961
		climb	1.456	2.391	5.582
		cruise	0.766	1.073	2.953
		descent	0.251	0.269	1.001
	PCA	climb	1.856	3.281	6.716
		cruise	1.259	1.789	3.659
		descent	0.584	0.578	1.231
	IF s0	climb	1.656	3.294	9.194
		cruise	0.531	1.394	2.930
		descent	0.322	0.664	1.271
	IF s1	climb	1.663	3.389	9.438
		cruise	0.485	1.366	2.881
		descent	0.306	0.634	1.267
	IF s2	climb	1.745	3.666	9.525
		cruise	0.503	1.424	2.689
		descent	0.313	0.672	1.268
	LSTM s0	climb	1.777	3.978	5.112
		cruise	0.773	1.830	2.270
		descent	0.511	1.086	1.488
	LSTM s1	climb	1.845	4.313	5.617
		cruise	0.669	1.756	2.059
		descent	0.484	1.205	1.495
	LSTM s2	climb	1.796	4.193	5.195
		cruise	0.671	1.720	1.887
		descent	0.545	1.338	1.529

Table S7: Frozen flight-level burden under pooled calibration at the 1% target: percentage of healthy audit flights with any alarm in each phase, and alarm events per healthy flight. This endpoint is distinct from the κ -based healthy-flight false-flag rate of the post-confirmation analyses.

Dataset	Run	Any alarm, climb (%)	Any alarm, cruise (%)	Any alarm, descent (%)	Events per flight, climb	Events per flight, cruise	Events per flight, descent
DS02	PCA	13.2	50.0	98.7	0.32	4.43	16.03
	IF s0	68.4	44.7	86.8	1.03	3.07	9.08
	IF s1	60.5	44.7	85.5	0.88	2.76	8.36
	IF s2	67.1	46.1	86.8	0.97	3.07	9.61
	LSTM s0	97.4	77.6	100.0	1.76	4.66	8.62
	LSTM s1	92.1	64.5	100.0	1.45	4.12	9.68
	LSTM s2	98.7	68.4	100.0	1.75	4.49	8.86
DS03	PCA	67.6	73.6	99.3	4.57	10.40	30.66
	IF s0	81.1	67.6	99.3	2.93	6.76	22.95
	IF s1	81.1	65.5	99.3	2.59	6.44	20.99
	IF s2	77.7	66.2	99.3	2.81	7.07	21.39
	LSTM s0	100.0	95.3	100.0	5.96	13.47	26.52
	LSTM s1	100.0	92.6	100.0	5.68	13.28	23.97
	LSTM s2	100.0	94.6	99.3	5.39	13.12	23.39

LSTM training and convergence

Figures S1 and S2 reproduce the recorded training and epoch-selection validation losses without smoothing. The DS02 seed-0 selection reached the pre-specified 600-epoch ceiling while validation loss was still improving; it is reported as a convergence limitation and was not retuned. These diagnostics do not inspect test outcomes.

3 Post-confirmation protocol v1.0

These analyses are exploratory. They reuse the frozen detectors and engine partitions and are not part of the DS03 confirmation.

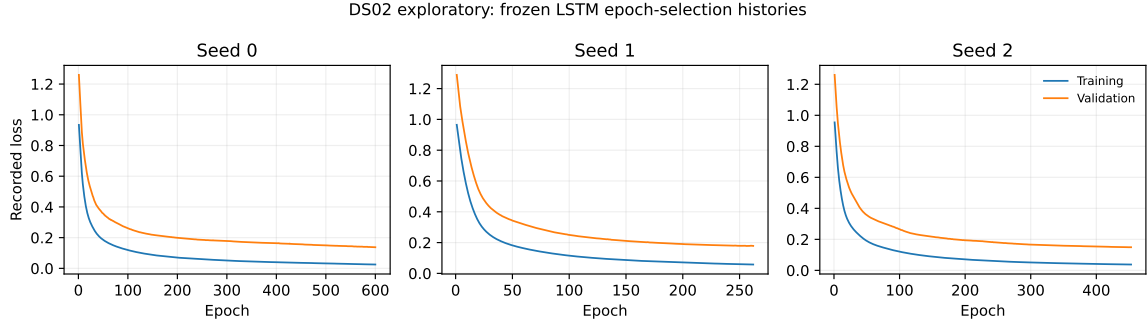


Figure S1: Frozen DS02 LSTM training and epoch-selection validation losses (seeds 0–2).

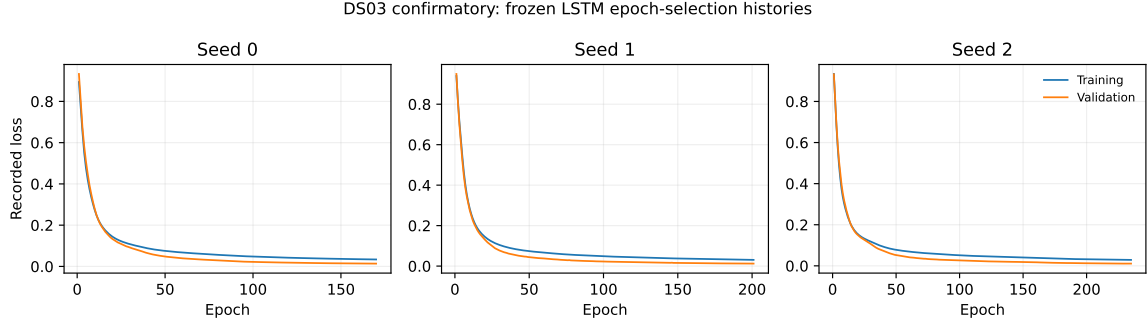


Figure S2: Frozen DS03 LSTM training and epoch-selection validation losses (seeds 0–2).

Table S8: Post-confirmation pooled audit healthy FPR (%) by phase at the 1% target under P, C and C' (protocol v1.0).

Dataset	Run	Arm	Climb	Cruise	Descent	Overall
DS02	PCA	P	0.125	0.428	3.459	1.363
		C	1.403	1.327	1.081	1.265
		C'	1.462	1.075	1.360	1.273
	IF s0	P	0.285	0.566	2.335	1.084
		C	1.655	1.845	0.921	1.485
		C'	1.525	1.119	1.669	1.410
	IF s1	P	0.253	0.528	2.669	1.172
		C	1.544	1.528	0.978	1.348
		C'	1.434	1.099	1.699	1.389
	IF s2	P	0.266	0.580	2.455	1.125
		C	1.686	1.715	0.927	1.443
		C'	1.560	1.125	1.680	1.426
	LSTM s0	P	0.452	0.707	2.078	1.099
		C	1.385	1.115	1.080	1.175
		C'	1.377	1.541	1.778	1.577
	LSTM s1	P	0.474	0.731	2.102	1.122
		C	1.243	1.279	0.961	1.163
		C'	1.200	1.390	1.725	1.452
	LSTM s2	P	0.539	0.750	2.060	1.133
		C	1.456	1.073	1.001	1.150
		C'	1.419	1.503	1.748	1.563
DS03	PCA	P	0.877	0.912	1.977	1.341
		C	1.856	1.789	1.231	1.577
		C'	1.977	1.108	1.685	1.564
	IF s0	P	0.417	1.095	2.166	1.364
		C	1.656	1.394	1.271	1.410
		C'	1.053	0.827	1.694	1.240
	IF s1	P	0.399	1.060	2.127	1.332
		C	1.663	1.366	1.267	1.400
		C'	1.037	0.791	1.672	1.215
	IF s2	P	0.412	1.157	2.134	1.370

(continued)

Dataset	Run	Arm	Climb	Cruise	Descent	Overall
	LSTM s0	C	1.745	1.424	1.268	1.441
		C'	1.083	0.882	1.719	1.276
		P	0.759	1.795	2.231	1.713
		C	1.777	1.830	1.488	1.676
	LSTM s1	C'	1.841	1.833	1.723	1.790
		P	0.719	1.890	2.220	1.730
		C	1.845	1.756	1.495	1.671
		C'	1.879	1.931	1.625	1.792
	LSTM s2	P	0.774	1.982	2.193	1.764
		C	1.796	1.720	1.529	1.661
		C'	1.877	2.004	1.746	1.866

Table S9: Paired arm-minus-P bootstrap 95% intervals (percentage points) at the 1% target for the row-pooled calibration metrics (frozen U1 resampling plans; engines then flights; thresholds re-estimated; 2000 replicates). A1, A2, A3 and A5 are row-pooled; the last column weights audit engines equally. Every interval includes zero.

Dataset	Run	Arm - P	A1 (pp)	A2 (pp)	A3 (pp)	A5 (pp)	A2, equal-engine (pp)
DS02	PCA	C	[-4.28, +0.63]	[-3.23, +1.21]	[-1.82, +0.85]	[-0.52, +1.21]	[-3.24, +2.90]
		C'	[-4.31, +1.49]	[-3.39, +2.83]	[-1.94, +1.77]	[-0.59, +1.74]	[-3.30, +5.81]
	IF s0	C	[-3.02, +4.92]	[-2.26, +4.34]	[-1.23, +3.03]	[-0.27, +1.86]	[-2.11, +5.41]
		C'	[-2.93, +3.06]	[-2.25, +3.92]	[-1.25, +2.80]	[-0.37, +1.99]	[-2.09, +5.04]
	IF s1	C	[-3.37, +3.92]	[-2.54, +3.23]	[-1.40, +2.36]	[-0.31, +1.26]	[-2.35, +4.52]
		C'	[-3.22, +2.06]	[-2.52, +3.22]	[-1.43, +2.35]	[-0.34, +1.69]	[-2.22, +4.24]
	IF s2	C	[-3.02, +5.30]	[-2.21, +4.51]	[-1.26, +3.25]	[-0.26, +1.77]	[-2.17, +5.94]
		C'	[-2.96, +3.31]	[-2.18, +4.28]	[-1.25, +2.89]	[-0.32, +1.97]	[-2.01, +5.69]
	LSTM s0	C	[-2.01, +3.89]	[-1.40, +3.65]	[-0.82, +2.13]	[-0.21, +1.25]	[-1.28, +5.65]
		C'	[-1.88, +2.31]	[-1.13, +3.38]	[-0.66, +2.47]	[-0.23, +2.20]	[-1.16, +5.32]
	LSTM s1	C	[-2.11, +3.18]	[-1.51, +2.66]	[-0.89, +1.66]	[-0.24, +1.03]	[-1.39, +3.37]
		C'	[-1.74, +1.86]	[-1.06, +2.49]	[-0.63, +2.09]	[-0.22, +1.86]	[-1.04, +3.11]
	LSTM s2	C	[-2.06, +4.03]	[-1.51, +3.67]	[-0.87, +2.16]	[-0.24, +1.01]	[-1.42, +5.95]
		C'	[-1.89, +2.62]	[-1.29, +3.35]	[-0.72, +2.56]	[-0.25, +2.17]	[-1.27, +5.51]
DS03	PCA	C	[-1.13, +2.56]	[-0.56, +2.61]	[-0.27, +1.51]	[-0.05, +0.92]	[-0.55, +2.43]
		C'	[-0.97, +2.91]	[-0.50, +3.67]	[-0.26, +2.21]	[-0.15, +1.48]	[-0.49, +3.61]
	IF s0	C	[-1.99, +4.43]	[-1.08, +4.85]	[-0.56, +2.68]	[-0.25, +1.34]	[-1.08, +5.00]
		C'	[-1.49, +1.82]	[-0.91, +2.49]	[-0.52, +1.39]	[-0.39, +0.99]	[-0.90, +2.51]
	IF s1	C	[-1.94, +4.25]	[-1.09, +4.70]	[-0.56, +2.63]	[-0.24, +1.35]	[-1.09, +4.69]
		C'	[-1.51, +1.71]	[-0.89, +2.46]	[-0.52, +1.40]	[-0.40, +1.01]	[-0.87, +2.42]
	IF s2	C	[-1.90, +4.88]	[-0.99, +5.27]	[-0.51, +2.96]	[-0.20, +1.44]	[-0.99, +5.47]
		C'	[-1.52, +2.12]	[-0.84, +2.85]	[-0.50, +1.65]	[-0.36, +1.11]	[-0.84, +2.84]
	LSTM s0	C	[-1.73, +1.82]	[-0.82, +1.97]	[-0.37, +1.08]	[-0.23, +0.45]	[-0.88, +2.45]
		C'	[-1.90, +1.08]	[-1.09, +1.65]	[-0.64, +0.97]	[-0.48, +0.77]	[-1.08, +2.04]
	LSTM s1	C	[-1.83, +1.84]	[-0.83, +1.89]	[-0.40, +1.01]	[-0.24, +0.43]	[-0.89, +2.35]
		C'	[-2.03, +1.11]	[-1.08, +1.52]	[-0.64, +0.91]	[-0.48, +0.68]	[-1.10, +1.92]
	LSTM s2	C	[-1.69, +2.52]	[-0.81, +2.48]	[-0.43, +1.38]	[-0.29, +0.59]	[-0.85, +3.05]
		C'	[-1.89, +1.57]	[-1.08, +2.07]	[-0.70, +1.18]	[-0.56, +0.92]	[-1.07, +2.49]

Table S10: Target robustness of the post-confirmation claims: number of runs (of 7) in which each claim holds at the nominal targets 0.5%, 1% and 2%, with the pre-specified classification.

Dataset	Claim	0.5%	1%	2%	Class
DS02	R1 descent is the highest pooled healthy-FPR phase (P)	7/7	7/7	7/7	invariant
	R2 C reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C reduces max-abs nominal calibration error (A2)	7/7	7/7	7/7	invariant
	R4 C reduces RMS nominal calibration error (A3)	7/7	7/7	7/7	invariant
	R5 C mainly redistributes: overall FPR change < 0.5 × A1 change	6/7	7/7	7/7	mostly invariant
	R6 C abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C is not earlier than P at matched false-flag burden	4/7	4/7	3/7	reversed
	R2 C' reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C' reduces max-abs nominal calibration error (A2)	4/7	7/7	7/7	mostly invariant
	R4 C' reduces RMS nominal calibration error (A3)	4/7	7/7	7/7	mostly invariant
	R5 C' mainly redistributes: overall FPR change < 0.5 × A1 change	2/7	7/7	7/7	reversed
	R6 C' abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant

(continued)

Dataset	Claim	0.5%	1%	2%	Class
	R8 C' is not earlier than P at matched false-flag burden	7/7	3/7	2/7	not supported at 1%
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (P)	7/7	7/7	7/7	invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C)	7/7	7/7	7/7	invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C')	7/7	7/7	7/7	invariant
DS03	R1 descent is the highest pooled healthy-FPR phase (P)	7/7	7/7	7/7	invariant
	R2 C reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C reduces max-abs nominal calibration error (A2)	7/7	7/7	6/7	mostly invariant
	R4 C reduces RMS nominal calibration error (A3)	7/7	6/7	6/7	mostly invariant
	R5 C mainly redistributes: $ \text{overall FPR change} < 0.5 \times \text{A1 change} $	7/7	7/7	7/7	invariant
	R6 C abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C is not earlier than P at matched false-flag burden	6/7	7/7	6/7	mostly invariant
	R2 C' reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C' reduces max-abs nominal calibration error (A2)	6/7	7/7	6/7	mostly invariant
	R4 C' reduces RMS nominal calibration error (A3)	4/7	6/7	5/7	target-sensitive
	R5 C' mainly redistributes: $ \text{overall FPR change} < 0.5 \times \text{A1 change} $	6/7	6/7	6/7	mostly invariant
	R6 C' abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C' is not earlier than P at matched false-flag burden	6/7	5/7	7/7	mostly invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (P)	7/7	6/7	7/7	mostly invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C)	4/7	4/7	7/7	target-sensitive
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C')	7/7	7/7	7/7	invariant

4 Adversarial validation

Table S11: Per-engine maximum nominal calibration error A2 (percentage points) at the 1% target for every detector run under P, C, C' and Q.

Dataset	Engine (class)	Run	P	C	C'	Q
DS02	11 (3)	PCA	1.78	0.38	0.63	2.00
		IF s0	1.23	0.25	0.97	1.45
		IF s1	1.35	0.35	0.99	1.28
		IF s2	1.30	0.25	0.97	1.48
		LSTM s0	1.36	0.36	1.11	0.61
		LSTM s1	1.52	0.38	1.15	0.44
		LSTM s2	1.38	0.36	1.07	4.25
	14 (1)	PCA	3.97	4.31	5.31	2.20
		IF s0	1.51	6.42	6.12	2.16
		IF s1	2.29	5.96	5.72	1.58
		IF s2	1.72	6.59	6.27	1.95
		LSTM s0	1.07	4.42	4.39	1.30
		LSTM s1	0.99	3.74	3.67	0.54
		LSTM s2	1.30	4.65	4.50	1.39
	15 (2)	PCA	2.67	0.28	0.40	0.58
		IF s0	1.40	0.27	0.57	3.63
		IF s1	1.82	0.26	0.56	1.08
		IF s2	1.55	0.28	0.55	4.15
		LSTM s0	0.66	0.19	0.23	0.62
		LSTM s1	0.66	0.14	0.29	0.64
		LSTM s2	0.71	0.30	0.27	4.18
DS03	10 (3)	PCA	1.45	1.52	1.77	7.57
		IF s0	2.11	2.37	1.30	6.31
		IF s1	2.16	2.06	1.36	6.27
		IF s2	2.23	2.56	1.33	6.44
		LSTM s0	2.70	1.56	1.93	3.67
		LSTM s1	2.91	1.57	2.21	4.91
		LSTM s2	3.08	1.69	2.39	3.59
	11 (3)	PCA	1.17	0.97	1.14	0.56
		IF s0	1.28	0.52	0.93	0.89
		IF s1	1.32	0.49	0.93	0.87
		IF s2	1.29	0.50	0.96	0.89
		LSTM s0	1.67	0.99	1.16	0.84
		LSTM s1	1.62	0.91	1.02	0.81
		LSTM s2	1.65	0.97	1.16	0.76
	12 (1)	PCA	0.68	0.82	0.73	0.69
		IF s0	0.85	1.07	0.75	0.40
		IF s1	0.70	0.92	0.70	0.49
		IF s2	0.69	0.96	0.77	0.50

(continued)

Dataset	Engine (class)	Run	P	C	C'	Q
	13 (3)	LSTM s0	0.41	2.04	2.24	0.72
		LSTM s1	0.39	2.17	2.47	0.70
		LSTM s2	0.51	2.02	2.39	0.74
		PCA	1.25	1.22	1.42	1.41
		IF s0	1.66	0.77	0.77	1.69
		IF s1	1.60	0.86	0.75	1.24
	14 (1)	IF s2	1.62	0.81	0.80	1.33
		LSTM s0	1.69	1.47	1.10	0.64
		LSTM s1	1.67	1.22	1.20	0.70
		LSTM s2	1.81	1.39	1.21	0.58
		PCA	0.43	0.37	0.43	0.30
		IF s0	0.71	0.44	0.50	0.94
	15 (2)	IF s1	0.72	0.44	0.44	0.95
		IF s2	0.74	0.47	0.45	0.97
		LSTM s0	0.31	0.64	0.74	0.73
		LSTM s1	0.32	0.76	0.84	0.66
		LSTM s2	0.35	0.63	0.79	0.78
		PCA	1.15	1.28	1.20	0.54
		IF s0	0.62	0.25	0.34	0.65
		IF s1	0.64	0.40	0.35	0.62
		IF s2	0.63	0.38	0.43	0.66
		LSTM s0	0.56	0.65	0.69	0.50
		LSTM s1	0.45	0.54	0.54	0.49
		LSTM s2	0.38	0.74	0.83	0.61

Table S12: Denominator audit: sign agreement between row-pooled and alternative weightings or endpoints, over runs, targets and arms. The frozen burden endpoint (healthy flights with any alarm) and the κ -based healthy-flight false-flag rate are distinct quantities.

Dataset	Check	Agreeing
DS02	row-pooled vs equal-engine sign of diff_spread	25 of 42
	row-pooled vs equal-engine sign of diff_max_abs_error	16 of 42
	row-pooled vs equal-engine sign of diff_rms_error	15 of 42
	equal-flight vs row-pooled sign of overall healthy FPR change (phase_conditioned)	19 of 21
	equal-flight vs row-pooled sign of overall healthy FPR change (causal_regime)	19 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (phase_conditioned)	8 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (causal_regime)	8 of 21
	row abnormal alarm rate higher vs flight delay earlier (phase_conditioned)	5 of 21
DS03	row abnormal alarm rate higher vs flight delay earlier (causal_regime)	13 of 21
	row-pooled vs equal-engine sign of diff_spread	42 of 42
	row-pooled vs equal-engine sign of diff_max_abs_error	39 of 42
	row-pooled vs equal-engine sign of diff_rms_error	36 of 42
	equal-flight vs row-pooled sign of overall healthy FPR change (phase_conditioned)	14 of 21
	equal-flight vs row-pooled sign of overall healthy FPR change (causal_regime)	19 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (phase_conditioned)	0 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (causal_regime)	4 of 21
	row abnormal alarm rate higher vs flight delay earlier (phase_conditioned)	10 of 21
	row abnormal alarm rate higher vs flight delay earlier (causal_regime)	13 of 21

Matched false-flag delay

Figure S3 shows the full operating characteristics: the lower-median detection delay against the realized pooled healthy-flight false-flag rate as κ is swept over the calibration-only grid. Tables S13 and S14 give the matched comparisons at the pre-specified anchors, and Table S15 gives per-engine delays at the locked κ together with their false-flag rates.

Table S13: Matched healthy-flight false-flag comparisons at the 1% row target (nearest-anchor rule): realized pooled false-flag rates, lower-median delays (flights from the labelled abnormal-state onset), per-engine earlier/same/late counts, and the paired lower-median difference (arm – P). Unsupported anchors are shown but were not used.

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/late	Paired median diff.
DS02	PCA	C	2.5	yes	2.6 / 2.6	19 / 2	3/0/0	–13
		C	5	yes	5.3 / 5.3	19 / 2	3/0/0	–13

(continued)

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/later	Paired median diff.
	IF s0	C	10	no	10.5 / 11.8	4 / 2	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 2	1/1/1	0
		C	20	yes	19.7 / 19.7	2 / 2	1/1/1	0
		C'	2.5	yes	2.6 / 1.3	19 / 2	3/0/0	-13
		C'	5	yes	5.3 / 5.3	19 / 2	3/0/0	-13
		C'	10	yes	10.5 / 10.5	4 / 2	1/1/1	0
		C'	15	yes	14.5 / 14.5	2 / 1	2/0/1	-2
		C'	20	no	19.7 / 18.4	2 / 0	2/0/1	-2
		C	2.5	yes	2.6 / 1.3	12 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	11 / 5	3/0/0	-6
		C	10	yes	10.5 / 10.5	4 / 3	2/1/0	-3
		C	15	yes	15.8 / 14.5	2 / 1	1/2/0	0
		C	20	no	18.4 / 19.7	2 / 1	1/0/2	1
		C'	2.5	yes	2.6 / 1.3	12 / 5	3/0/0	-7
		C'	5	yes	5.3 / 5.3	11 / 5	3/0/0	-6
		C'	10	yes	10.5 / 10.5	4 / 3	2/1/0	-3
		C'	15	yes	15.8 / 14.5	2 / 0	2/1/0	-1
		C'	20	no	18.4 / 19.7	2 / 0	1/1/1	0
	IF s1	C	2.5	yes	1.3 / 1.3	15 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C	10	yes	10.5 / 10.5	6 / 3	2/1/0	-5
		C	15	yes	14.5 / 14.5	3 / 1	2/1/0	-2
		C	20	yes	19.7 / 19.7	3 / 1	2/1/0	-2
		C'	2.5	yes	1.3 / 1.3	15 / 5	3/0/0	-8
		C'	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C'	10	yes	10.5 / 10.5	6 / 3	2/1/0	-5
		C'	15	yes	14.5 / 15.8	3 / 0	2/1/0	-2
		C'	20	yes	19.7 / 19.7	3 / 0	2/1/0	-2
	IF s2	C	2.5	yes	2.6 / 1.3	13 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	12 / 4	3/0/0	-7
		C	10	yes	10.5 / 10.5	3 / 3	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 1	2/0/1	-1
		C	20	yes	19.7 / 19.7	0 / 1	1/1/1	0
		C'	2.5	yes	2.6 / 1.3	13 / 5	3/0/0	-7
		C'	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C'	10	yes	10.5 / 10.5	3 / 1	2/1/0	-2
		C'	15	yes	14.5 / 14.5	2 / 0	2/0/1	-2
		C'	20	yes	19.7 / 19.7	0 / 0	0/2/1	0
	LSTM s0	C	2.5	no	3.9 / 1.3	14 / 10	3/0/0	-4
		C	5	yes	5.3 / 5.3	13 / 5	3/0/0	-8
		C	10	no	11.8 / 10.5	11 / 2	2/0/1	-4
		C	15	yes	14.5 / 14.5	2 / 2	0/2/1	0
		C	20	yes	19.7 / 21.1	2 / 2	0/2/1	0
		C'	2.5	no	3.9 / 1.3	14 / 11	3/0/0	-3
		C'	5	yes	5.3 / 5.3	13 / 7	3/0/0	-6
		C'	10	no	11.8 / 10.5	11 / 5	2/0/1	-3
		C'	15	yes	14.5 / 14.5	2 / 5	0/1/2	3
		C'	20	yes	19.7 / 19.7	2 / 5	0/1/2	3
	LSTM s1	C	2.5	yes	1.3 / 2.6	13 / 10	3/0/0	-6
		C	5	yes	5.3 / 5.3	13 / 7	3/0/0	-5
		C	10	yes	10.5 / 10.5	2 / 5	1/0/2	1
		C	15	yes	14.5 / 14.5	2 / 2	0/1/2	1
		C	20	yes	19.7 / 21.1	2 / 2	0/2/1	0
		C'	2.5	yes	1.3 / 1.3	13 / 10	3/0/0	-4
		C'	5	yes	5.3 / 5.3	13 / 7	3/0/0	-6
		C'	10	yes	10.5 / 10.5	2 / 7	1/0/2	1
		C'	15	yes	14.5 / 14.5	2 / 7	0/1/2	5
		C'	20	yes	19.7 / 19.7	2 / 7	0/1/2	5
	LSTM s2	C	2.5	no	3.9 / 2.6	13 / 10	2/1/0	-4
		C	5	yes	5.3 / 5.3	11 / 2	3/0/0	-8
		C	10	no	7.9 / 9.2	2 / 2	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 2	0/2/1	0
		C	20	yes	21.1 / 21.1	2 / 2	0/2/1	0
		C'	2.5	no	3.9 / 2.6	13 / 12	3/0/0	-2
		C'	5	yes	5.3 / 3.9	11 / 12	1/1/1	0
		C'	10	no	7.9 / 10.5	2 / 5	1/0/2	1
		C'	15	yes	14.5 / 14.5	2 / 5	0/1/2	3
		C'	20	yes	21.1 / 19.7	2 / 2	0/2/1	0
DS03	PCA	C	2.5	yes	2.7 / 2.7	14 / 12	3/0/3	-4
		C	5	no	4.1 / 4.7	2 / 10	3/0/3	-5
		C	10	yes	10.1 / 10.1	2 / 4	1/2/3	0
		C	15	no	14.2 / 15.5	2 / 2	1/4/1	0

(continued)

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/later	Paired median diff.
	IF s0	C	20	yes	19.6 / 20.3	2 / 2	1/3/2	0
		C'	2.5	yes	2.7 / 2.7	14 / 10	3/0/3	-4
		C'	5	no	4.1 / 4.7	2 / 10	3/0/3	-4
		C'	10	yes	10.1 / 10.1	2 / 5	1/0/5	1
		C'	15	no	14.2 / 15.5	2 / 5	2/0/4	1
		C'	20	yes	19.6 / 19.6	2 / 1	2/1/3	0
		C	2.5	yes	2.7 / 2.0	6 / 7	0/2/4	1
		C	5	yes	4.7 / 4.7	5 / 7	0/2/4	1
		C	10	yes	10.1 / 10.1	3 / 2	2/2/2	0
		C	15	yes	15.5 / 15.5	2 / 2	1/3/2	0
		C	20	yes	20.3 / 20.3	2 / 2	1/4/1	0
		C'	2.5	yes	2.7 / 2.7	6 / 6	2/4/0	0
		C'	5	yes	4.7 / 5.4	5 / 6	2/3/1	0
		C'	10	yes	10.1 / 10.1	3 / 5	3/1/2	-3
		C'	15	no	15.5 / 16.2	2 / 1	1/3/2	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
	IF s1	C	2.5	yes	2.7 / 2.0	6 / 7	1/3/2	0
		C	5	yes	4.7 / 4.7	6 / 7	0/4/2	0
		C	10	no	9.5 / 8.8	6 / 2	3/3/0	-1
		C	15	yes	14.9 / 14.9	2 / 2	1/4/1	0
		C	20	yes	20.3 / 20.3	2 / 2	1/4/1	0
		C'	2.5	yes	2.7 / 2.0	6 / 6	1/5/0	0
		C'	5	yes	4.7 / 5.4	6 / 6	1/5/0	0
		C'	10	yes	9.5 / 10.1	6 / 6	2/3/1	0
		C'	15	yes	14.9 / 14.9	2 / 2	1/3/2	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
		C	2.5	yes	2.0 / 2.7	7 / 7	2/2/2	0
		C	5	yes	5.4 / 4.7	6 / 7	1/3/2	0
	IF s2	C	10	yes	9.5 / 9.5	5 / 5	2/1/3	0
		C	15	no	15.5 / 14.2	2 / 2	1/4/1	0
		C	20	no	20.3 / 18.9	2 / 2	1/4/1	0
		C'	2.5	yes	2.0 / 2.0	7 / 6	5/1/0	-3
		C'	5	yes	5.4 / 4.7	6 / 6	2/4/0	0
		C'	10	yes	9.5 / 10.1	5 / 6	3/1/2	-1
		C'	15	yes	15.5 / 15.5	2 / 2	1/2/3	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
	LSTM s0	C	2.5	yes	2.7 / 2.7	10 / 5	4/2/0	-8
		C	5	yes	4.7 / 4.7	7 / 5	3/2/1	-2
		C	10	yes	10.1 / 10.1	4 / 2	2/3/1	0
		C	15	yes	14.9 / 14.9	4 / 2	2/2/2	0
		C	20	yes	20.3 / 19.6	2 / 1	1/3/2	0
		C'	2.5	yes	2.7 / 2.0	10 / 5	4/1/1	-3
		C'	5	yes	4.7 / 4.7	7 / 3	5/0/1	-8
		C'	10	yes	10.1 / 9.5	4 / 3	4/1/1	-1
		C'	15	yes	14.9 / 14.9	4 / 1	3/3/0	-1
		C'	20	yes	20.3 / 20.3	2 / 1	3/2/1	-1
	LSTM s1	C	2.5	yes	2.7 / 2.7	8 / 5	4/2/0	-4
		C	5	yes	4.7 / 5.4	7 / 5	4/2/0	-3
		C	10	yes	10.1 / 9.5	1 / 2	1/2/3	0
		C	15	yes	14.9 / 15.5	1 / 1	1/2/3	0
		C	20	yes	20.3 / 19.6	1 / 1	1/3/2	0
		C'	2.5	yes	2.7 / 2.7	8 / 5	5/0/1	-4
		C'	5	yes	4.7 / 5.4	7 / 5	4/0/2	-4
		C'	10	yes	10.1 / 9.5	1 / 3	1/4/1	0
		C'	15	yes	14.9 / 15.5	1 / 1	1/3/2	0
		C'	20	yes	20.3 / 20.3	1 / 1	1/3/2	0
	LSTM s2	C	2.5	yes	2.7 / 2.0	9 / 5	4/1/1	-4
		C	5	yes	4.7 / 4.7	5 / 5	3/2/1	-4
		C	10	yes	10.1 / 9.5	5 / 2	2/2/2	0
		C	15	yes	14.9 / 14.9	5 / 1	2/3/1	0
		C	20	yes	20.3 / 20.3	2 / 1	2/2/2	0
		C'	2.5	yes	2.7 / 2.7	9 / 5	5/0/1	-4
		C'	5	yes	4.7 / 4.7	5 / 3	4/1/1	-4
		C'	10	yes	10.1 / 9.5	5 / 3	2/3/1	0
		C'	15	no	14.9 / 14.2	5 / 1	2/4/0	0
		C'	20	no	20.3 / 18.9	2 / 1	2/2/2	0

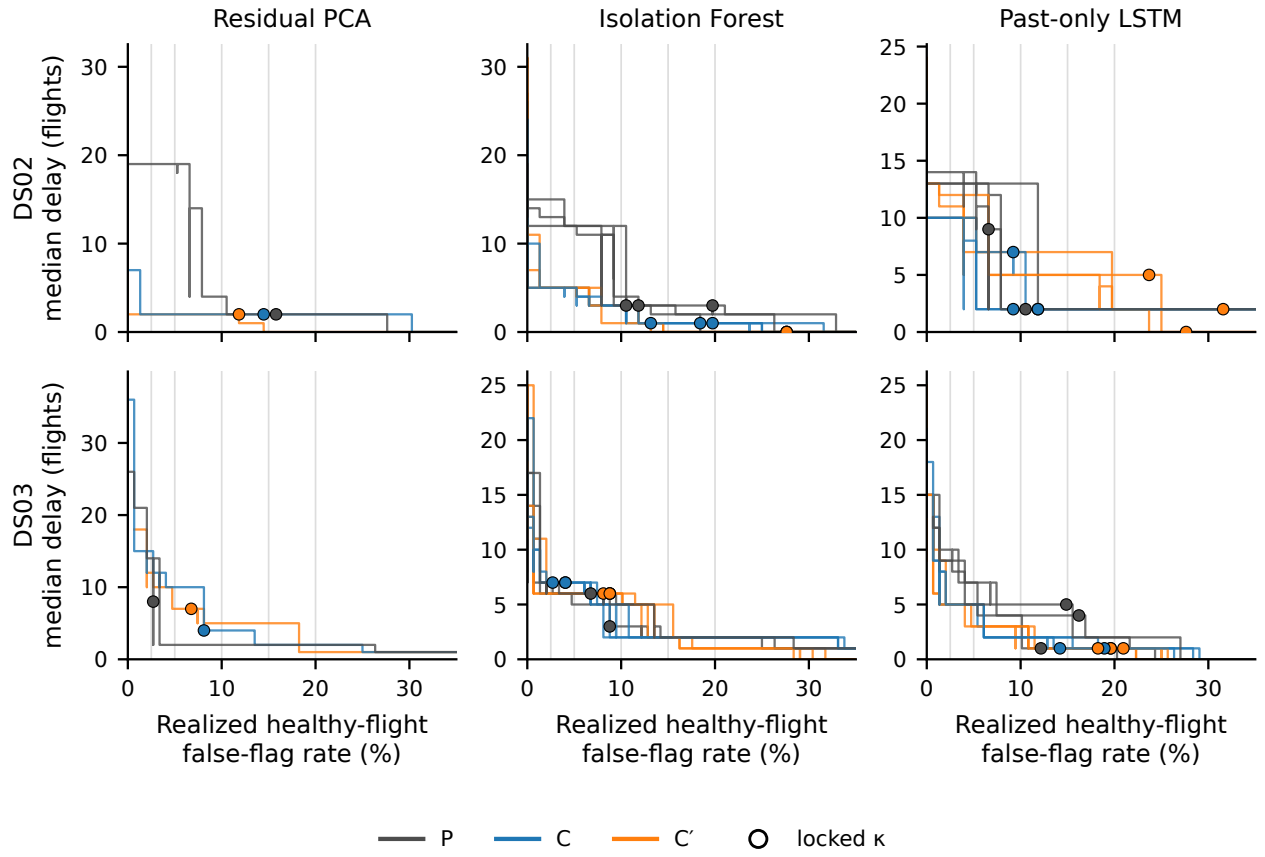


Figure S3: Median detection delay against realized pooled healthy-flight false-flag rate at the 1% row target, one step curve per seed and arm over the calibration-only κ grid. Grey vertical lines mark the matching anchors; filled markers are the locked- κ operating points (nominal 5%). Vertical segments mark κ intervals that share a realized rate.

Table S14: Pre-specified run labels at the 1% target under the nearest-anchor rule and the conservative (not-exceeding) sensitivity rule, curve-level mean delays over realized false-flag rates 2.5–20% (flights), and the Pareto class of the locked operating point against P. Labels: E, earlier at matched burden; L, later; M, mixed; =, equal; n/e, not evaluable.

Dataset	Run	Arm	Supported anchors	Label (nearest)	Label (conservative)	Curve mean P / arm	Locked point
DS02	PCA	C	4	M	M	6.93 / 2.00	arm dominates
	IF s0	C	4	E	E	6.09 / 2.38	trade-off
	IF s1	C	5	E	E	7.17 / 2.38	arm dominates
	IF s2	C	5	E	E	5.75 / 2.38	trade-off
	LSTM s0	C	3	M	M	6.74 / 3.23	identical
	LSTM s1	C	5	M	M	4.87 / 4.91	trade-off
	LSTM s2	C	3	M	M	4.85 / 3.01	trade-off
	PCA	C'	4	E	E	6.93 / 1.52	arm dominates
	IF s0	C'	4	E	E	6.09 / 2.33	trade-off
	IF s1	C'	5	E	E	7.17 / 2.27	trade-off
	IF s2	C'	5	E	E	5.75 / 1.58	trade-off
	LSTM s0	C'	3	M	M	6.74 / 5.97	arm dominated
	LSTM s1	C'	5	M	M	4.87 / 7.41	arm dominated
	LSTM s2	C'	3	M	M	4.85 / 6.09	trade-off
DS03	PCA	C	3	M	E	2.61 / 5.70	trade-off
	IF s0	C	5	M	M	3.54 / 3.28	trade-off
	IF s1	C	4	=	=	4.09 / 3.96	trade-off
	IF s2	C	3	=	=	4.53 / 3.84	trade-off
	LSTM s0	C	5	M	M	4.83 / 2.26	trade-off
	LSTM s1	C	5	M	=	3.00 / 2.31	arm dominated
	LSTM s2	C	5	M	E	4.72 / 2.42	arm dominates
	PCA	C'	3	M	M	2.61 / 5.59	trade-off
	IF s0	C'	4	M	M	3.54 / 3.94	arm dominated
	IF s1	C'	5	=	M	4.09 / 3.96	arm dominates
	IF s2	C'	5	M	E	4.53 / 4.23	arm dominated
	LSTM s0	C'	5	E	E	4.83 / 2.25	trade-off
	LSTM s1	C'	5	M	=	3.00 / 2.24	arm dominated
	LSTM s2	C'	3	E	E	4.72 / 2.12	trade-off

Table S15: Per-engine detection delay (flights from the labelled abnormal-state onset; cens. = no flagged post-onset flight) and healthy-flight false-flag rate at the locked κ (1% row target).

Dataset	Run	Engine	P: delay / FFR (%)	C: delay / FFR (%)	C': delay / FFR (%)
DS02	PCA	11	2 / 5.6	10 / 0.0	10 / 0.0
		14	4 / 17.1	1 / 28.6	1 / 25.7
		15	2 / 21.7	2 / 4.3	2 / 0.0
	IF s0	11	3 / 5.6	3 / 5.6	3 / 5.6
		14	4 / 17.1	1 / 34.3	0 / 54.3
		15	2 / 4.3	0 / 4.3	0 / 4.3
	IF s1	11	3 / 5.6	3 / 5.6	3 / 5.6
		14	4 / 25.7	1 / 22.9	0 / 54.3
		15	2 / 21.7	0 / 4.3	0 / 4.3
	IF s2	11	3 / 5.6	1 / 5.6	3 / 5.6
		14	4 / 17.1	1 / 37.1	0 / 54.3
		15	2 / 8.7	0 / 4.3	0 / 4.3
	LSTM s0	11	11 / 5.6	7 / 0.0	5 / 5.6
		14	0 / 22.9	1 / 22.9	0 / 57.1
		15	2 / 0.0	2 / 4.3	2 / 13.0
	LSTM s1	11	13 / 5.6	10 / 0.0	5 / 5.6
		14	0 / 20.0	1 / 20.0	0 / 45.7
		15	2 / 0.0	7 / 0.0	5 / 4.3
	LSTM s2	11	11 / 5.6	5 / 0.0	5 / 5.6
		14	9 / 8.6	1 / 14.3	0 / 51.4
		15	2 / 4.3	2 / 8.7	0 / 8.7
DS03	PCA	10	23 / 5.9	17 / 11.8	19 / 11.8
		11	8 / 0.0	2 / 0.0	5 / 5.6
		12	15 / 0.0	10 / 13.9	7 / 0.0
		13	2 / 5.6	2 / 11.1	1 / 16.7
		14	14 / 0.0	4 / 0.0	7 / 2.9
	IF s0	15	1 / 8.3	12 / 12.5	11 / 12.5
		10	18 / 5.9	19 / 5.9	18 / 11.8
		11	3 / 11.1	7 / 0.0	6 / 5.6
		12	0 / 8.3	1 / 2.8	0 / 8.3
		13	18 / 11.1	18 / 11.1	18 / 0.0
		14	5 / 11.4	6 / 5.7	2 / 11.4

(continued)

Dataset	Run	Engine	P: delay / FFR (%)	C: delay / FFR (%)	C': delay / FFR (%)
	IF s1	15	1 / 4.2	8 / 0.0	6 / 12.5
		10	19 / 11.8	19 / 5.9	18 / 11.8
		11	6 / 11.1	7 / 0.0	6 / 5.6
		12	0 / 5.6	0 / 2.8	0 / 8.3
		13	18 / 11.1	18 / 11.1	18 / 0.0
		14	6 / 11.4	6 / 5.7	3 / 11.4
	IF s2	15	1 / 4.2	8 / 0.0	6 / 8.3
		10	19 / 5.9	19 / 5.9	18 / 11.8
		11	6 / 5.6	7 / 0.0	6 / 5.6
		12	3 / 5.6	4 / 2.8	0 / 8.3
		13	18 / 11.1	18 / 11.1	18 / 0.0
		14	6 / 8.6	6 / 0.0	5 / 11.4
	LSTM s0	15	6 / 4.2	8 / 0.0	6 / 12.5
		10	6 / 35.3	6 / 41.2	5 / 29.4
		11	5 / 11.1	2 / 22.2	5 / 16.7
		12	4 / 5.6	1 / 11.1	0 / 19.4
		13	1 / 38.9	1 / 33.3	0 / 27.8
		14	5 / 8.6	5 / 8.6	3 / 11.4
	LSTM s1	15	1 / 16.7	1 / 20.8	1 / 20.8
		10	5 / 41.2	6 / 47.1	5 / 35.3
		11	2 / 11.1	2 / 16.7	5 / 16.7
		12	0 / 0.0	0 / 11.1	0 / 22.2
		13	1 / 27.8	1 / 27.8	1 / 27.8
		14	6 / 0.0	5 / 8.6	3 / 11.4
	LSTM s2	15	0 / 16.7	1 / 20.8	1 / 20.8
		10	5 / 41.2	8 / 35.3	5 / 29.4
		11	5 / 22.2	5 / 11.1	5 / 22.2
		12	9 / 2.8	1 / 8.3	0 / 16.7
		13	1 / 33.3	1 / 27.8	1 / 22.2
		14	7 / 0.0	5 / 0.0	5 / 11.4
		15	1 / 16.7	1 / 20.8	1 / 16.7

Table S16: Abnormal-state ($hs = 0$) row alarm rates (%) at the 1% target and the S4 separability of calibration p-values. S4 pAUC is the normalized partial area under the row-level ROC curve of abnormal versus healthy audit rows over row FPR in $[0, 2\%]$, computed on calibration p-values; it is threshold-free and is not a flight-level operating characteristic. Family guardrail ratios (arm/P, all post-onset rows) ranged 0.98–1.16.

Dataset	Run	Arm	All post-onset (%)	First 10 post-onset flights (%)	S4 pAUC, all	S4 pAUC, first 10
DS02	PCA	P	50.78	1.91	0.445	0.015
		C	59.00	4.03	0.517	0.032
		C'	59.00	5.03	0.529	0.039
	IF s0	P	61.00	1.89	0.537	0.017
		C	67.12	13.23	0.564	0.067
		C'	67.48	13.48	0.573	0.074
	IF s1	P	59.96	1.64	0.519	0.014
		C	66.44	11.44	0.565	0.067
		C'	67.42	12.75	0.569	0.071
	IF s2	P	62.05	2.06	0.545	0.018
		C	68.43	15.50	0.582	0.081
		C'	69.05	16.12	0.585	0.085
	LSTM s0	P	62.41	1.52	0.569	0.016
		C	63.62	4.18	0.583	0.034
		C'	65.98	5.17	0.572	0.028
	LSTM s1	P	63.66	1.43	0.575	0.015
		C	64.62	3.07	0.589	0.030
		C'	66.63	3.76	0.578	0.025
	LSTM s2	P	63.43	1.99	0.574	0.020
		C	63.79	4.19	0.581	0.034
		C'	65.25	5.08	0.565	0.028
DS03	PCA	P	39.51	1.37	0.331	0.010
		C	41.98	1.54	0.337	0.010
		C'	42.53	1.59	0.351	0.010
	IF s0	P	58.00	2.15	0.494	0.015
		C	58.57	2.21	0.502	0.015
		C'	56.87	2.37	0.505	0.018
	IF s1	P	56.88	1.86	0.482	0.014
		C	57.47	1.94	0.492	0.014
		C'	55.79	1.93	0.495	0.016
	IF s2	P	56.87	1.92	0.480	0.014
		C	57.50	2.10	0.490	0.014

(continued)

Dataset	Run	Arm	All post-onset (%)	First 10 post-onset flights (%)	S4 pAUC, all	S4 pAUC, first 10
	LSTM s0	C'	56.23	2.19	0.493	0.017
		P	58.37	2.11	0.497	0.012
		C	58.15	2.62	0.499	0.015
	LSTM s1	C'	57.64	3.76	0.500	0.018
		P	58.13	2.33	0.489	0.013
		C	57.87	2.71	0.492	0.015
	LSTM s2	C'	57.26	3.95	0.496	0.019
		P	59.26	2.29	0.507	0.012
		C	59.05	2.65	0.508	0.015
C'		58.85	4.70	0.512	0.021	

Table S17: Healthy flights with any alarm (the frozen burden endpoint) under P, C and C' at the 1% target, shown beside the κ -based healthy-flight false-flag rate. The two quantities are distinct and are never merged.

Dataset	Run	Arm	Any alarm, overall (%)	Any alarm, climb (%)	Any alarm, cruise (%)	Any alarm, descent (%)	κ -based FFR (%)
DS02	PCA	P	98.7	13.2	50.0	98.7	15.8
		C	100.0	56.6	81.6	86.8	14.5
		C'	96.1	44.7	81.6	68.4	11.8
	IF s0	P	100.0	68.4	44.7	86.8	10.5
		C	100.0	97.4	81.6	59.2	18.4
		C'	100.0	96.1	73.7	65.8	27.6
	IF s1	P	98.7	60.5	44.7	85.5	19.7
		C	100.0	97.4	77.6	64.5	13.2
		C'	100.0	96.1	72.4	68.4	27.6
	IF s2	P	100.0	67.1	46.1	86.8	11.8
		C	100.0	97.4	82.9	56.6	19.7
		C'	100.0	97.4	76.3	67.1	27.6
	LSTM s0	P	100.0	97.4	77.6	100.0	11.8
		C	100.0	100.0	90.8	88.2	11.8
		C'	100.0	100.0	90.8	71.1	31.6
	LSTM s1	P	100.0	92.1	64.5	100.0	10.5
		C	100.0	100.0	85.5	69.7	9.2
		C'	100.0	100.0	90.8	67.1	23.7
	LSTM s2	P	100.0	98.7	68.4	100.0	6.6
		C	100.0	100.0	78.9	76.3	9.2
		C'	100.0	100.0	90.8	65.8	27.6
DS03	PCA	P	100.0	67.6	73.6	99.3	2.7
		C	100.0	91.9	84.5	99.3	8.1
		C'	100.0	92.6	87.2	99.3	6.8
	IF s0	P	100.0	81.1	67.6	99.3	8.8
		C	100.0	99.3	69.6	99.3	4.1
		C'	100.0	93.2	64.9	99.3	8.8
	IF s1	P	100.0	81.1	65.5	99.3	8.8
		C	100.0	99.3	68.2	98.6	4.1
		C'	100.0	93.9	63.5	99.3	8.1
	IF s2	P	100.0	77.7	66.2	99.3	6.8
		C	100.0	98.0	68.2	99.3	2.7
		C'	100.0	93.9	63.5	98.0	8.8
	LSTM s0	P	100.0	100.0	95.3	100.0	16.2
		C	100.0	100.0	95.3	99.3	19.6
		C'	100.0	100.0	90.5	89.2	19.6
	LSTM s1	P	100.0	100.0	92.6	100.0	12.2
		C	100.0	100.0	91.9	99.3	18.9
		C'	100.0	100.0	92.6	87.8	20.9
	LSTM s2	P	100.0	100.0	94.6	99.3	14.9
		C	100.0	100.0	92.6	98.0	14.2
		C'	100.0	100.0	89.9	87.2	18.2

Continuous contextual baseline

Table S18: Continuous contextual baseline Q: linear quantile regression of the detector score at level $1 - \alpha$ on a quadratic surface of the four standardized operating descriptors (4 linear, 4 squared and 6 pairwise terms plus an intercept), solved exactly by linear programming (HiGHS, no penalty) on every fifth healthy calibration row, per run and target, with no tuning and no phase input. Fitted coefficients are in the machine-readable evidence files.

Dataset	Target	Run	Fit rows	In-sample alarm rate (%)	κ	Audit FFR at κ (%)
DS02	0.5%	PCA	77,329	0.52	0.0195	21.1
		IF s0	77,329	0.53	0.0116	31.6
		IF s1	77,329	0.52	0.0116	31.6
		IF s2	77,329	0.52	0.0106	32.9
		LSTM s0	77,329	0.52	0.0137	19.7
		LSTM s1	77,329	0.52	0.0164	18.4
		LSTM s2	77,329	0.52	0.0201	17.1
	1%	PCA	77,329	1.03	0.0434	9.2
		IF s0	77,329	1.00	0.0307	18.4
		IF s1	77,329	1.01	0.0235	17.1
		IF s2	77,329	1.00	0.0318	18.4
		LSTM s0	77,329	0.98	0.0247	18.4
		LSTM s1	77,329	1.03	0.0304	7.9
		LSTM s2	77,329	1.01	0.0250	22.4
	2%	PCA	77,329	1.99	0.0618	11.8
		IF s0	77,329	1.98	0.0704	17.1
		IF s1	77,329	2.00	0.0651	13.2
		IF s2	77,329	2.01	0.0864	9.2
		LSTM s0	77,329	1.99	0.0571	6.6
		LSTM s1	77,329	2.04	0.0647	6.6
		LSTM s2	77,329	2.03	0.0535	9.2
DS03	0.5%	PCA	91,621	0.49	0.0169	7.4
		IF s0	91,621	0.50	0.0149	11.5
		IF s1	91,621	0.50	0.0143	11.5
		IF s2	91,621	0.48	0.0149	10.1
		LSTM s0	91,621	0.50	0.0158	5.4
		LSTM s1	91,621	0.51	0.0173	3.4
		LSTM s2	91,621	0.48	0.0167	3.4
	1%	PCA	91,621	1.00	0.0394	6.1
		IF s0	91,621	1.00	0.0359	10.8
		IF s1	91,621	1.00	0.0448	5.4
		IF s2	91,621	1.00	0.0449	5.4
		LSTM s0	91,621	1.00	0.0305	7.4
		LSTM s1	91,621	0.97	0.0292	7.4
		LSTM s2	91,621	0.99	0.0340	4.1
	2%	PCA	91,621	2.05	0.1129	2.0
		IF s0	91,621	2.02	0.1079	5.4
		IF s1	91,621	2.02	0.1170	4.1
		IF s2	91,621	2.02	0.1249	3.4
		LSTM s0	91,621	2.01	0.0547	10.8
		LSTM s1	91,621	2.02	0.0613	9.5
		LSTM s2	91,621	2.03	0.0634	7.4

Table S19: Continuous contextual baseline Q: pooled audit healthy FPR (%) by phase and calibration metrics at the 1% target (healthy side only).

Dataset	Run	Climb	Cruise	Descent	Overall	A1 (pp)	A2 (pp)	A3 (pp)
DS02	PCA	0.702	2.125	1.271	1.462	1.42	1.12	0.69
	IF s0	0.490	2.239	2.138	1.742	1.75	1.24	1.01
	IF s1	0.421	1.339	1.937	1.296	1.52	0.94	0.67
	IF s2	0.563	2.455	2.113	1.840	1.89	1.46	1.09
	LSTM s0	0.717	1.546	1.294	1.242	0.83	0.55	0.39
	LSTM s1	0.562	1.289	0.967	0.989	0.73	0.44	0.30
	LSTM s2	0.819	2.711	3.140	2.353	2.32	2.14	1.59
DS03	PCA	2.657	1.798	1.405	1.853	1.25	1.66	1.09
	IF s0	1.959	1.607	1.704	1.735	0.35	0.96	0.77
	IF s1	1.869	1.502	1.509	1.597	0.37	0.87	0.65
	IF s2	1.903	1.461	1.589	1.625	0.44	0.90	0.68
	LSTM s0	1.513	1.121	1.690	1.454	0.57	0.69	0.50
	LSTM s1	1.724	1.192	1.714	1.541	0.53	0.72	0.60
	LSTM s2	1.493	1.064	1.630	1.405	0.57	0.63	0.46

Operating support and the phase effect within calibration engines

Table S20: Operating-support distances of healthy audit rows to the nearest healthy calibration row of the same retrospective phase (standardized altitude, Mach number, throttle-resolver angle and fan-inlet temperature), with the any-phase and 8-D (adding 120-s trailing slopes) variants, the fraction beyond the cross-calibration-engine 95th-percentile reference, and the fraction outside the calibration range.

Dataset	Engine (class)	Phase	Rows	Median d	95th pct d	Median d , any phase	Median d , 8-D	Beyond ref. (%)	Outside range (%)
DS02	11 (3)	climb	53,678	0.024	0.137	0.022	0.131	1.2	0.2
		cruise	86,535	0.024	0.348	0.013	0.152	0.0	0.0
		descent	73,093	0.050	0.256	0.043	0.292	2.6	0.0
	14 (1)	climb	12,241	0.045	0.280	0.042	0.240	4.6	2.3
		cruise	33,377	1.130	1.443	0.089	1.324	38.6	95.3
		descent	26,650	0.060	0.216	0.060	0.255	0.3	22.4
	15 (2)	climb	48,789	0.037	0.126	0.035	0.168	1.8	0.6
		cruise	53,466	0.068	0.319	0.023	0.241	0.0	1.4
		descent	45,284	0.049	0.193	0.048	0.306	0.2	0.1
DS03	10 (3)	climb	63,828	0.039	0.248	0.032	0.153	1.8	3.6
		cruise	95,181	0.069	0.428	0.048	0.178	2.4	7.3
		descent	90,839	0.043	0.238	0.037	0.251	0.5	0.6
	11 (3)	climb	63,479	0.026	0.138	0.023	0.132	0.0	0.0
		cruise	98,352	0.025	0.329	0.022	0.151	0.0	2.5
		descent	99,376	0.043	0.259	0.037	0.232	1.3	0.1
	12 (1)	climb	35,417	0.022	0.119	0.022	0.175	0.0	0.0
		cruise	47,845	0.736	1.096	0.060	1.036	14.6	84.3
		descent	72,178	0.034	0.149	0.034	0.269	0.0	0.0
	13 (3)	climb	66,056	0.026	0.188	0.025	0.129	0.1	0.5
		cruise	81,213	0.020	0.196	0.018	0.132	0.1	3.0
		descent	104,404	0.038	0.331	0.037	0.224	0.3	4.2
	14 (1)	climb	33,331	0.022	0.102	0.022	0.179	0.0	0.0
		cruise	52,432	0.878	1.108	0.061	1.125	18.0	82.1
		descent	73,818	0.029	0.100	0.029	0.244	0.0	0.0
	15 (2)	climb	64,502	0.020	0.125	0.019	0.117	0.0	0.2
		cruise	61,379	0.025	0.205	0.015	0.125	0.0	1.1
		descent	91,049	0.025	0.142	0.024	0.200	0.0	0.3

Table S21: Spearman correlations across engine $\times$ phase cells between median same-phase support distance and calibration error, per run at the 1% target, with leave-one-engine-out ranges. Pre-specified verdict: indeterminate due to small engine count.

Dataset	Run	Cells	$\rho(d, \text{err} \text{ C})$	$\rho(d, \text{err} \text{ C}')$	$\rho(d_{\text{all}}, \text{err} \text{ P})$	$\rho(d, \text{change } \text{P} \rightarrow \text{C})$	Leave-one-engine-out range
DS02	PCA	9	-0.23	0.33	0.25	-0.05	[-0.60, -0.03]
DS03	PCA	18	0.02	0.03	0.16	-0.11	[-0.25, 0.17]
DS02	IF s0	9	0.37	0.43	0.15	0.10	[0.03, 0.43]
DS03	IF s0	18	0.14	0.08	0.27	-0.11	[-0.10, 0.47]
DS02	IF s1	9	0.03	0.43	0.22	-0.08	[-0.49, 0.20]
DS03	IF s1	18	0.14	0.13	0.19	-0.15	[-0.07, 0.39]
DS02	IF s2	9	0.25	0.43	0.22	0.08	[-0.20, 0.43]
DS03	IF s2	18	0.08	0.08	0.17	-0.17	[-0.11, 0.36]
DS02	LSTM s0	9	-0.22	0.17	0.10	-0.05	[-0.60, -0.03]
DS03	LSTM s0	18	-0.06	-0.02	0.15	-0.40	[-0.35, 0.18]
DS02	LSTM s1	9	0.25	0.30	0.08	0.08	[-0.03, 0.26]
DS03	LSTM s1	18	0.01	-0.04	0.14	-0.26	[-0.31, 0.29]
DS02	LSTM s2	9	-0.17	0.13	0.10	0.08	[-0.54, -0.14]
DS03	LSTM s2	18	0.04	0.02	0.25	-0.24	[-0.23, 0.30]

Table S22: C0: pooled-threshold healthy FPR (%) by phase within the calibration engines at the 1% target, in-sample and from one calibration engine to the other, where calibration-to-audit mismatch is absent by construction.

Dataset	Run	Evaluation	Climb	Cruise	Descent	Overall	Highest
DS02	PCA	in-sample	0.044	0.470	2.416	1.000	descent
		engine 18 to 20	0.015	0.586	2.309	1.057	descent
		engine 20 to 18	0.067	0.371	2.317	0.874	descent
DS03	PCA	in-sample	0.448	0.493	1.742	1.000	descent
		engine 4 to 8	0.176	0.600	1.732	0.911	descent
		engine 8 to 4	0.803	0.250	1.825	1.137	descent
DS02	IF s0	in-sample	0.155	0.383	2.421	1.000	descent
		engine 18 to 20	0.230	0.494	2.367	1.105	descent
		engine 20 to 18	0.094	0.272	2.271	0.828	descent
DS03	IF s0	in-sample	0.356	0.746	1.617	1.000	descent
		engine 4 to 8	0.356	1.113	2.322	1.362	descent
		engine 8 to 4	0.261	0.388	1.102	0.683	descent
DS02	IF s1	in-sample	0.137	0.370	2.450	1.000	descent
		engine 18 to 20	0.215	0.515	2.592	1.191	descent
		engine 20 to 18	0.075	0.249	2.203	0.792	descent
DS03	IF s1	in-sample	0.347	0.732	1.633	1.000	descent
		engine 4 to 8	0.338	1.130	2.328	1.365	descent
		engine 8 to 4	0.264	0.359	1.174	0.710	descent
DS02	IF s2	in-sample	0.143	0.388	2.425	1.000	descent
		engine 18 to 20	0.206	0.486	2.484	1.138	descent
		engine 20 to 18	0.092	0.293	2.302	0.845	descent
DS03	IF s2	in-sample	0.411	0.804	1.537	1.000	descent
		engine 4 to 8	0.360	1.228	2.219	1.364	descent
		engine 8 to 4	0.390	0.376	1.095	0.713	descent
DS02	LSTM s0	in-sample	0.267	0.719	1.943	1.000	descent
		engine 18 to 20	0.347	1.230	2.155	1.322	descent
		engine 20 to 18	0.233	0.398	1.929	0.815	descent
DS03	LSTM s0	in-sample	0.438	0.971	1.397	1.000	descent
		engine 4 to 8	0.570	1.958	2.146	1.650	descent
		engine 8 to 4	0.310	0.420	0.876	0.600	descent
DS02	LSTM s1	in-sample	0.298	0.636	2.012	1.000	descent
		engine 18 to 20	0.284	0.897	1.988	1.125	descent
		engine 20 to 18	0.325	0.462	2.050	0.904	descent
DS03	LSTM s1	in-sample	0.430	1.102	1.306	1.000	descent
		engine 4 to 8	0.592	1.984	2.127	1.658	descent
		engine 8 to 4	0.278	0.434	0.786	0.553	descent
DS02	LSTM s2	in-sample	0.324	0.696	1.921	1.000	descent
		engine 18 to 20	0.359	1.195	2.116	1.299	descent
		engine 20 to 18	0.305	0.427	1.936	0.850	descent
DS03	LSTM s2	in-sample	0.432	1.154	1.266	1.000	descent
		engine 4 to 8	0.593	2.259	2.223	1.790	cruise
		engine 8 to 4	0.288	0.436	0.698	0.515	descent

Post-plan error decomposition

Table S23: Post-plan descriptive decomposition (deviation Dev-1) of the equal-cell mean squared nominal error over audit engine $\times$ phase cells at the 1% target into overall bias, phase, engine and interaction shares.

Dataset	Run	Arm	RMS (pp)	Bias ² (%)	Phase (%)	Engine (%)	Engine $\times$ phase (%)
DS02	PCA	P	1.80	7	85	4	5
		C	1.54	19	13	35	34
		C'	1.96	18	10	45	27
	IF s0	P	0.96	5	72	17	6
		C	2.55	20	14	39	27
		C'	2.38	21	6	53	19
	IF s1	P	1.20	8	76	15	1
		C	2.25	17	14	39	30
		C'	2.26	21	5	54	19
	IF s2	P	1.04	7	74	16	4
		C	2.53	19	15	38	28

(continued)

Dataset	Run	Arm	RMS (pp)	Bias ² (%)	Phase (%)	Engine (%)	Engine×phase (%)
DS03	LSTM s0	C'	2.42	22	7	52	20
		P	0.79	7	51	24	18
		C	1.53	15	15	36	34
	LSTM s1	C'	2.00	31	1	54	14
		P	0.80	8	45	20	27
		C	1.37	15	13	40	33
	LSTM s2	C'	1.75	26	1	55	18
		P	0.82	11	35	23	31
		C	1.61	13	18	34	35
	PCA	C'	1.99	30	2	53	15
		P	0.66	10	57	19	14
		C	0.80	48	12	21	19
	IF s0	C'	0.84	41	18	22	19
		P	0.83	6	70	9	15
		C	0.77	26	5	36	33
	IF s1	C'	0.58	10	37	20	33
		P	0.83	4	68	12	17
		C	0.71	28	6	37	28
	IF s2	C'	0.57	8	39	19	34
		P	0.84	6	65	12	17
		C	0.81	27	6	38	30
	LSTM s0	C'	0.59	14	34	19	33
		P	1.03	25	29	27	20
		C	0.95	45	4	21	31
	LSTM s1	C'	1.01	58	1	16	24
		P	1.09	23	28	29	20
		C	0.98	43	5	23	30
	LSTM s2	C'	1.05	55	3	17	25
		P	1.17	21	22	34	22
		C	0.99	38	4	26	33
		C'	1.11	56	1	16	27

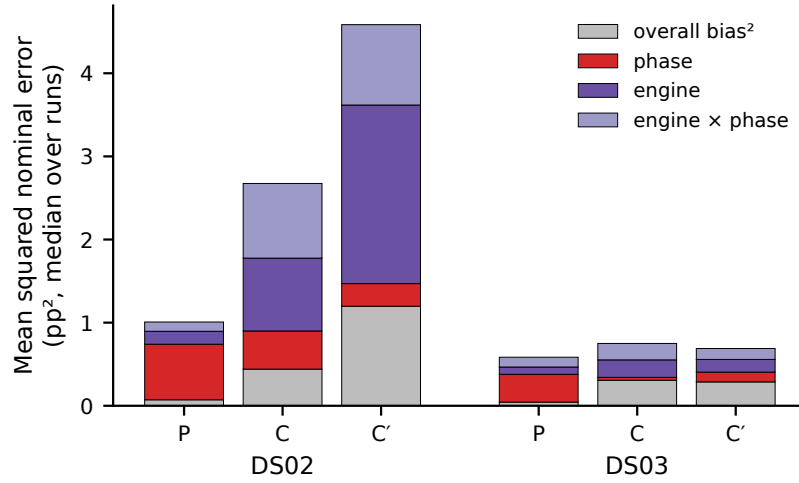


Figure S4: Post-plan descriptive decomposition (Dev-1) of the equal-cell mean squared nominal error at the 1% target (median over the seven detector runs).